

Hippocratic Writings, Volume II

ΙΠΠΟΚΡΑΤΟΥΣ ΣΥΓΓΡΑΜΜΑΤΑ Β΄

Hippocrates

translated by Francis Adams (1849)

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On Injuries of the Head

Head 1

Men's heads are by no means all like to one another, nor are the sutures of the head of all men constructed in the same form. Thus, whoever has a prominence in the anterior part of the head (by prominence is meant the round protuberant part of the bone which projects beyond the rest of it), in him the sutures of the head take the form of the Greek letter *tau*, T; for the head has the shorter line running transverse before the prominence, while the other line runs through the middle of the head, all the way to the neck. But whoever has the prominence in the back part of the head, in him the sutures are constructed in quite the opposite form to the former; for in this case the shorter line runs in front of the prominence, while the longer runs through the middle all along to the forehead. But whoever has a prominence of the head both before and behind, in him the sutures resemble the Greek letter *êta* H; for the long lines of the letter run transverse before each prominence while the short one runs through the middle and terminates in the long lines. But whoever has no prominence on either part he has the sutures of the head resembling the Greek letter *chi*; for the one line comes transverse to the temple while the other passes along the middle of the head. The bone at the middle of the head is double, the hardest and most compact part being the upper portion, where it is connected with the skin, and the lowest, where it is connected with the meninx (*dura mater*); and from the uppermost and lowermost parts the bone gradually becomes softer and less compact, till you come to the *diploe*. The *diploe* is the most porous, the softest, and most cavernous part. But the whole bone of the head, with the exception of a small portion of the uppermost and lowermost portions of it, is like a sponge; and the bone has in it many juicy substances, like caruncles; and if one will rub them with the fingers, some blood will issue from them. There are also in the bone

certain very slender and hollow vessels full of blood. So it is with regard to hardness, softness, and porosity.

Head 2

In respect to thickness and thinness; the thinnest and weakest part of the whole head is the part about the bregma; and the bone there has the smallest and thinnest covering of flesh upon it, and the largest proportion of brain is situated in that region of the head. And hence it happens that from similar or even smaller wounds and instruments, when a person is wounded to the same or a less degree, the bone of the head there is more contused, fractured, and depressed; and that injuries there are more deadly and more difficult to cure; and it is more difficult to save one's life in injuries there than in any other part of the head; that from having sustained a similar or even a less wound a man will die, and that, too, in a shorter space of time than from a wound in any other part of the head. For the brain about the bregma feels more quickly and strongly any mischief that may occur to the flesh or the bone; for the brain about the bregma is in largest quantity, and is covered by the thinnest bone and the least flesh. Of the other portions, the weakest is that about the temples; for it is the conjunction of the lower jaw with the cranium, and there is motion there up and down as at a joint; and the organ of hearing is near it; and further, a hollow and important vein runs along the temple. But the whole bone of the head behind the vertex and the ear is stronger than the whole anterior part, and the bone itself has a larger and deeper covering of flesh upon it. And hence it follows, that when exposed to the same or even greater injuries from instruments of the same or greater size, the bone is less liable to be fractured and depressed than elsewhere; and that in a fatal accident the patient will live longer when the wound is in the posterior part of the head than when elsewhere; and that pus takes longer time to form and penetrate through the bone to the brain, owing to the thickness of the bone; and moreover, as there is less brain in that part of the head, more persons who are wounded in the back part of the head escape than of those

who are wounded in the anterior part. And in fatal cases, a man will survive longer in winter than in summer, whatever be the part of the head in which the wound is situated.

Head 3

As to the *haedrae* (dints or marks?) of sharp and light weapons, when they take place in the bone without fissure, contusion, or depression inwards (and these take place equally in the anterior and posterior part of the head), death, when it does occur, does not properly result from them. A suture appearing in a wound, when the bone is laid bare, on whatever part of the head the wound may have been inflicted, is the weakest point of the head to resist a blow or a weapon, when the weapon happens to be impinged into the suture itself; but more especially when this occurs in the bregma at the weakest part of the head, and the sutures happen to be situated near the wound, and the weapon has hit the sutures themselves.

Head 4

The bone in the head is liable to be wounded in the following modes, and there are many varieties in each of these modes of fracture: When a wounded bone breaks, in the bone comprehending the fissure, contusion necessarily takes place where the bone is broken; for an instrument that breaks the bone occasions a contusion thereof more or less, both at the fracture and in the parts of the bone surrounding the fracture. This is the first mode. But there are all possible varieties of fissures; for some of them are fine, and so very fine that they cannot be discovered, either immediately after the injury, or during the period in which it would be of use to the patient if this could be ascertained. And some of these fissures are thicker and wider, certain of them being very wide. And some of them extend to a greater, and some to a smaller, distance. And some are more straight, nay, completely straight; and some are more curved, and that in a remarkable degree. And some are deep, so as to extend downwards and through the whole bone; and some are less so, and do not penetrate through the whole bone.

Head 5

But a bone may be contused, and yet remain in its natural condition without any fracture in it; this is the second mode. And there are many varieties of contusion; for they occur to a greater and less degree, and to a greater depth, so as sometimes to extend through the whole bone; or to a less depth, so as not to extend through the whole bone; and to a greater and smaller length and breadth. But it is not possible to recognize any of these varieties by the sight, so as to determine their form and extent; neither, indeed, is it visible to the eyes when any mischief of this kind takes place, and immediately after the injury, whether or not the bone has been actually bruised, as is likewise the case with certain fractures at a distance from the seat of injury.

Head 6

And the bone being fractured, is sometimes depressed inwards from its natural level along with the fractures, otherwise there would be no depression; for the depressed portion being fractured and broken off, is pushed inwards, while the rest of the bone remains in its natural position; and in this manner a fracture is combined with the depression. This is the third mode. There are many varieties of depression, for it may comprehend a greater and a smaller extent of bone, and may either be to a greater depth, or less so, and more superficial.

Head 7

When a *hedra*, or dint of a weapon, takes place in a bone, there may be a fracture combined with it; and provided there be a fracture, contusion must necessarily be joined, to a greater or less extent, in the seat of the dint and fracture, and in the bone which comprehends them. This is the fourth mode. And there may be a *hedra*, or indentation of the bone, along with contusion of the surrounding bone, but without any fracture either in the *hedra* or in the contusion inflicted by the weapon. But the indentation of a weapon takes place in a bone, and is called *hedra*, when the bone remaining in its natural

state, the weapon which struck against the bone leaves its impression on the part which it struck. In each of these modes there are many varieties, with regard to the contusion and fracture, if both these be combined with the *hedra*, or if contusion alone, as it has been already stated that there are many varieties of contusion and fracture. And the *hedra*, or dint, of itself may be longer and shorter, crooked, straight, and circular; and there are many varieties of this mode, according to the shape of the weapon; and they may be more or less deep, and narrower or broader, and extremely broad. When a part is cleft, the cleft or notch which occurs in the bone, to whatever length or breadth, is a *hedra*, if the other bones comprehending the cleft remain in their natural position, and be not driven inwards; for in this case it would be a depression, and no longer a *hedra*.

Head 8

A bone may be injured in a different part of the head from that on which the person has received the wound, and the bone has been laid bare. This is the fifth mode. And for this misfortune, when it occurs, there is no remedy; for when this mischief takes place, there is no means of ascertaining by any examination whether or not it has occurred, or on what part of the head.

Head 9

Of these modes of fracture, the following require trepanning: the contusion, whether the bone be laid bare or not; and the fissure, whether apparent or not. And if, when an indentation (*hedra*) by a weapon takes place in a bone it be attended with fracture and contusion, and even if contusion alone, without fracture, be combined with the indentation, it requires trepanning. A bone depressed from its natural position rarely requires trepanning; and those which are most pressed and broken require trepanning the least; neither does an indentation (*hedra*) without fracture and contusion require trepanning; nor does a notch, provided it is large and wide; for a notch and a *hedra* are the same.

Head 10

In the first place, one must examine the wounded person, in what part of the head the wound is situated, whether in the stronger or weaker parts; and ascertain respecting the hairs about the wound, whether they have been cut off by the instrument, and have gone into the wound; and if so, one should declare that the bone runs the risk of being denuded of flesh, and of having sustained some injury from the weapon. These things one should say from a distant inspection, and before laying a hand on the man; but on a close examination one should endeavor to ascertain clearly whether the bone be denuded of flesh or not; and if the denuded bone be visible to the eyes, this will be enough; but otherwise an examination must be made with the sound. And if you find the bone denuded of the flesh, and not safe from the wound, you must first ascertain the state of the bone, and the extent of the mischief, and what assistance it stands in need of. One should also inquire of the wounded person how and in what way he sustained the injury; and if it be not apparent whether the bone has sustained an injury or not, it will be still more necessary, provided the bone be denuded, to make inquiry how the wound occurred, and in what manner; for when contusions and fractures exist in the bone, but are not apparent, we must ascertain, in the first place from the patient's answers, whether or not the bone has sustained any such injuries, and then find out the nature of the case by word and deed, with the exception of sounding. For sounding does not discover to us whether the bone has sustained any of these injuries or not; but sounding discovers to us an indentation inflicted by a weapon, and whether a bone be depressed from its natural position, and whether the bone be strongly fractured; all which may also be ascertained visibly with the eyes.

Head 11

And a bone sustains fractures, either so fine as to escape the sight, or such as are apparent, and contusions which are not apparent, and depression from its natural position, especially when one person is intentionally wounded by another, or when, whether intentionally or

not, a blow or stroke is received from an elevated place, and if the instrument in the hand, whether used in throwing or striking, be of a powerful nature, and if a stronger person wound a weaker. Of those who are wounded in the parts about the bone, or in the bone itself, by a fall, he who falls from a very high place upon a very hard and blunt object is in most danger of sustaining a fracture and contusion of the bone, and of having it depressed from its natural position; whereas he that falls upon more level ground, and upon a softer object, is likely to suffer less injury in the bone, or it may not be injured at all. Of those instruments which, falling upon the head, wound the parts about the bone, or the bone itself, that which falls from a very high place, and the least on a level with the person struck, and which is at the same time very hard, very blunt, and very heavy, and which is the least light, sharp, and soft, such an instrument would occasion a fracture and contusion of the bone. And there is most danger that the bone may sustain these injuries, under such circumstances, when the wound is direct and perpendicular to the bone, whether struck from the hand or from a throw, or when any object falls upon the person, or when he is wounded by falling, or in whatever way the bone sustains a direct wound from this instrument. Those weapons which graze the bone obliquely are less apt to fracture, contuse, or depress the bone, even when the bone is denuded of flesh; for in some of those wounds thus inflicted the bone is not laid bare of the flesh. Those instruments more especially produce fractures in the bone, whether apparent or not, and contusions, and inward depression of the bone, which are rounded, globular, smooth on all sides, blunt, heavy, and hard; and such weapons bruise, compress, and pound the flesh; and the wounds inflicted by such instruments, whether obliquely or circularly, are round, and are more disposed to suppurate, and to have a discharge, and take longer time to become clean; for the flesh which has been bruised and pounded must necessarily suppurate and slough away. But weapons of an oblong form, being, for the most part, slender, sharp, and light, penetrate the flesh rather than bruise it, and the bone in

like manner; and such an instrument may occasion a *hedra* and a cut (for a *hedra* and a cut are same thing); but weapons of this description do not produce contusions, nor fractures, nor depressions inwardly. And in addition to the appearances in the bone, which you can detect by the sight, you should make inquiry as to all these particulars (for they are symptoms of a greater or less injury), whether the wounded person was stunned, and whether darkness was diffused over his eyes, and whether he had vertigo, and fell to the ground.

Head 12

When the bone happens to be denuded of flesh by the weapon, and when the wound occurs upon the sutures, it is difficult to distinguish the indentation (*hedra*) of a weapon which is clearly recognized in other parts of the bone, whether it exist or not, and especially if the *hedra* be seated in the sutures themselves. For the suture being rougher than the rest of the bone occasions confusion, and it is not clear which is the suture, and which the mark inflicted by the instrument, unless the latter (*hedra*) be large. Fracture also for the most part is combined with the indentation when it occurs in the sutures; and this fracture is more difficult to discern when the bone is broken, on this account, that if there be a fracture, it is situated for the most part in the suture. For the bone is liable to be broken and slackened there, owing to the natural weakness of the bone there, and to its porosity, and from the suture being readily ruptured and slackened: but the other bones which surround the suture remain unbroken, because they are stronger than the suture. For the fracture which occurs at the suture is also a slackening of the suture, and it is not easy to detect whether the bone be broken and slackened by the indentation of a weapon occurring in the suture, or from a contusion of the bone at the sutures; but it is still more difficult to detect a fracture connected with contusion. For the sutures, having the appearance of fissures, elude the discernment and sight of the physician, as being rougher than the rest of the bone, unless the bone be strongly cut and slackened, (for a cut and a *hedra* are the same

thing.) But it is necessary, if the wound has occurred at the sutures, and the weapon has impinged on the bone or the parts about it, to pay attention and find out what injury the bone has sustained. For a person wounded to the same, or a much smaller, extent, and by weapons of the same size and quality, and even much less, will sustain a much greater injury, provided he has received the blow at the sutures, than if it was elsewhere. And many of these require trepanning, but you must not apply the trepan to the sutures themselves, but on the adjoining bone.

Head 13

And with regard to the cure of wounds in the head, and the mode of detecting injuries in the bone which are not apparent, the following is my opinion:— In a wound of the head, you must not apply anything liquid, not even wine, but as little as possible, nor a cataplasm, nor conduct the treatment with tents, nor apply a bandage to an ulcer on the head, unless it be situated on the forehead, in the part which is bare of hairs, or about the eyebrow and eye, for wounds occurring there require cataplasms and bandages more than upon any other part of the head. For the rest of the head surrounds the whole forehead, and the wounds wherever situated become inflamed and swelled, owing to an influx of blood from the surrounding parts. And neither must you apply cataplasms and bandages to the forehead at all times; but when the inflammation is stopped and the swelling has subsided, you must give up the cataplasms and bandages. A wound in any other part of the head must not be treated with tents, bandages, or cataplasms, unless it also requires incision. You must perform incision on wounds situated on the head and forehead, whenever the bone is denuded of flesh, and appears to have sustained some injury from the blow, but the wound has not sufficient length and breadth for the inspection of the bone, so that it may be seen whether it has received any mischief from the blow, and of what nature the injury is, and to what extent the flesh has been contused, and whether the bone has sustained any injury, or whether it be uninjured by the blow, and

has suffered no mischief; and with regard to the treatment, what the wound, and the flesh, and the injury of the bone stand in need of. Ulcers of this description stand in need of incision; and, if the bone be denuded of the flesh, and if it be hollow, and extend far obliquely, we cut up the cavity wherever the medicine cannot penetrate readily, whatever medicine it may be; and wounds which are more inclined to be circular and hollow, and for the most part others of the like shape, are cut up by making a double incision in the circle lengthways, according to the figure of the man, so as to make the wound of a long form. Incisions may be practiced with impunity on other parts of the head, with the exception of the temple and the parts above it, where there is a vein that runs across the temple, in which region an incision is not to be made. For convulsions seize on a person who has been thus treated; and if the incision be on the left temple, the convulsions seize on the right side; and if the incision be on the right side, the convulsions take place on the left side.

Head 14

When, then, you lay open a wound in the head on account of the bones having been denuded of the flesh, as wishing to ascertain whether or not the bone has received an injury from the blow, you must make an incision proportionate to the size of the wound, and as much as shall be judged necessary. And in making the incision you must separate the flesh from the bone where it is united to the membrane (*pericranium?*) and to the bone, and then fill the whole wound with a tent, which will expand the wound very wide next day with as little pain as possible; and along with the tents apply a cataplasm, consisting of a mass (*maza*) of fine flour pounded in vinegar, or boiled so as to render it as glutinous as possible. On the next day, when you remove the tent, having examined the bone to see what injury it has sustained, if the wound in the bone be not right seen by you, nor can you discover what mischief the bone itself has sustained, but the instrument seems to have penetrated to the bone so as to have injured it, you must scrape the bone with a raspatory to a

depth and length proportionate to the suture of the patient, and again in a transverse direction, for the sake of the fractures which are not seen, and of the contusions which are not discovered, as not being accompanied with depression of the bone from its natural position. For the scraping discovers the mischief, if the injuries in the bone be not otherwise manifest. And if you perceive an indentation (*hedra*) left in the bone by the blow, you must scrape the dint itself and the surrounding bones, lest, as often happens, there should be a fracture and contusion, or a contusion alone, combined with the dint, and escape observation. And when you scrape the bone with the raspatory, and it appears that the wound in the bone requires the operation, you must not postpone it for three days, but do it during this period, more especially if the weather be hot, and you have had the management of the treatment from the commencement. If you suspect that the bone is broken or contused, or has sustained both these injuries, having formed your judgment from the severity of the wound, and from the information of the patient, as that the person who inflicted the wound, provided it was done by another person, was remarkably strong, and that the weapon by which he was wounded was of a dangerous description, and then that the man had been seized with vertigo, dimness of vision, and stupor, and fell to the ground,—under these circumstances, if you cannot discover whether the bone be broken, contused, or both the one and the other, nor can see the truth of the matter, you must dissolve the jet-black ointment, and fill the wound with it when thus dissolved, and apply a linen rag smeared with oil, and then a cataplasm of the maza with a bandage; and on the next day, having cleaned out the wound, scrape the bone with the raspatory. And if the bone is not sound, but fractured and contused, the rest of it which is scraped will be white; but the fracture and contusion, having imbibed the preparation, will appear black, while the rest of the bone is white. And you must again scrape more deeply the fracture where it appears black; and, if you thus remove the fissure, and cause it to disappear, you may conclude that there has

been a contusion of the bone to a greater or less extent, which has occasioned the fracture that has disappeared under the raspatory; but it is less dangerous, and a matter of less consequence, when the fissure has been effaced. But if the fracture extend deep, and do not seem likely to disappear when scraped, such an accident requires trepanning. But having performed this operation, you must apply the other treatment to the wound.

Head 15

You must be upon your guard lest the bone sustain any injury from the fleshy parts if not properly treated. When the bone has been sawed and otherwise denuded, whether it be actually sound, or only appears to be so, but has sustained some injury from the blow, there may be danger of its suppurating (although it would not otherwise have done so), if the flesh which surrounds the bone be ill cured, and become inflamed and strangled; for it gets into a febrile state, and becomes much inflamed. For the bone acquires heat and inflammation from the surrounding flesh, along with irritation and throbbing, and the other mischiefs which are in the flesh itself, and from these it gets into a state of suppuration. It is a bad thing for the flesh (*granulations?*) in an ulcer to be moist and mouldy, and to require a long time to become clean. But the wound should be made to suppurate as quickly as possible; for, thus the parts surrounding the wound would be the least disposed to inflammation, and would become the soonest clean; for the flesh which has been chopped and bruised by the blow, must necessarily suppurate and slough away. But when cleaned the wound must be dried, for thus the wound will most speedily become whole, when flesh devoid of humors grows up, and thus there will be no fungous flesh in the sore. The same thing applies to the membrane which surrounds the brain: for when, by sawing the bone, and removing it from the meninx, you lay the latter bare, you must make it clean and dry as quickly as possible, lest being in a moist state for a considerable time, it become soaked therewith and swelled; for when these things occur, there is danger of its mortifying.

Head 16

A piece of bone that must separate from the rest of the bone, in consequence of a wound in the head, either from the indentation (*hedra*) of a blow in the bone, or from the bone being otherwise denuded for a long time, separates mostly by becoming exsanguous. For the bone becomes dried up and loses its blood by time and a multiplicity of medicines which are used; and the separation will take place most quickly, if one having cleaned the wound as quickly as possible will next dry it, and the piece of bone, whether larger or smaller. For a piece of bone which is quickly dried and converted, as it were, into a shell, is most readily separated from the rest of the bone which retains its blood and vitality; for, the part having become exsanguous and dry, more readily drops off from that which retains its blood and is alive.

Head 17

Such pieces of bone as are depressed from their natural position, either being broken off or chopped off to a considerable extent, are attended with less danger, provided the membrane be safe; and bones which are broken by numerous and broader fractures are still less dangerous and more easily extracted. And you must not trepan any of them, nor run any risks in attempting to extract the pieces of bone, until they rise up of their own accord, upon the subsidence of the swelling. They rise up when the flesh (*granulations*) grows below, and it grows from the diploe of the bone, and from the sound portion, provided the upper table alone be in a state of necrosis. And the flesh will shoot up and grow below the more quickly, and the pieces of bone ascend, if one will get the wound to suppurate and make it clean as quickly as possible. And when both the tables of the bone are driven in upon the membrane, I mean the upper and lower, the wound, if treated in the same way, will very soon get well, and the depressed bones will quickly rise up.

Head 18

The bones of children are thinner and softer, for this reason, that they contain more blood [than those of adults]; and they are porous and spongy, and neither dense nor hard. And when wounded to a similar or inferior degree by weapons of the same or even of an inferior power, the bone of a young person more readily and quickly suppurates, and that in less time than the bone of an older person; and in accidents, which are to prove fatal, the younger person will die sooner than the elder. But if the bone is laid bare of flesh, one must attend and try to find out, what even is not obvious to the sight, and discover whether the bone be broken and contused, or only contused; and if, when there is an indentation in the bone, whether contusion, or fracture, or both be joined to it; and if the bone has sustained any of these injuries, we must give issue to the blood by perforating the bone with a small trepan, observing the greatest precautions, for the bone of young persons is thinner and more superficial than that of elder persons.

Head 19

When a person has sustained a mortal wound on the head, which cannot be cured, nor his life preserved, you may form an opinion of his approaching dissolution, and foretell what is to happen from the following symptoms which such a person experiences. When a bone is broken, or cleft, or contused, or otherwise injured, and when by mistake it has not been discovered, and neither the raspatory nor trepan has been applied as required, but the case has been neglected as if the bone were sound, fever will generally come on before the fourteenth day if in winter, and in summer the fever usually seizes after seven days. And when this happens, the wound loses its color, and the inflammation dies in it; and it becomes glutinous, and appears like a pickle, being of a tawny and somewhat livid color; and the bone then begins to sphacelate, and turns black where it was white before, and at last becomes pale and blanched. But when suppuration is fairly established in it, small blisters form on the tongue and he dies delirious. And, for the most part, convulsions seize the other side of

the body; for, if the wound be situated on the left side, the convulsions will seize the right side of the body; or if the wound be on the right side of the head, the convulsion attacks the left side of the body. And some become apoplectic. And thus they die before the end of seven days, if in summer; and before fourteen, if in winter. And these symptoms indicate, in the same manner, whether the wound be older or more recent. But if you perceive that fever is coming on, and that any of these symptoms accompany it, you must not put off, but having sawed the bone to the membrane (*meninx*), or scraped it with a raspator, (and it is then easily sawed or scraped,) you must apply the other treatment as may seem proper, attention being paid to circumstances.

Head 20

When in any wound of the head, whether the man has been trepanned or not, but the bone has been laid bare, a red and erysipelatous swelling supervenes in the face, and in both eyes, or in either of them, and if the swelling be painful to the touch, and if fever and rigor come on, and if the wound look well, whether as regards the flesh or the bone, and if the parts surrounding the wound be well, except the swelling in the face, and if the swelling be not connected with any error in the regimen, you must purge the bowels in such a case with a medicine which will evacuate bile; and when thus purged the fever goes off, the swelling subsides, and the patient gets well. In giving the medicine you must pay attention to the strength of the patient.¹

Head 21

With regard to trepanning, when there is a necessity for it, the following particulars should be known. If you have had the management of the case from the first, you must not at once saw the bone down to the *meninx*; for it is not proper that the membrane should be laid bare and exposed to injuries for a length of time, as in the end it may become fungous. And there is another danger if you saw the bone down to the *meninx* and remove it at once, lest in the act of sawing you should wound the *meninx*. But in trepanning, when

only a very little of the bone remains to be sawed through, and the bone can be moved, you must desist from sawing, and leave the bone to fall out of itself.² For to a bone not sawed through, and where a portion is left of the sawing, no mischief can happen; for the portion now left is sufficiently thin. In other respects you must conduct the treatment as may appear suitable to the wound. And in trepanning you must frequently remove the trepan, on account of the heat in the bone, and plunge it in cold water. For the trepan being heated by running round, and heating and drying the bone, burns it and makes a larger piece of bone around the sawing to drop off, than would otherwise do. And if you wish to saw at once down to the membrane, and then remove the bone, you must also, in like manner, frequently take out the trepan and dip it in cold water. But if you have not charge of the treatment from the first, but undertake it from another after a time, you must saw the bone at once down to the meninx with a serrated trepan, and in doing so must frequently take out the trepan and examine with a sound (specillum), and otherwise along the tract of the instrument. For the bone is much sooner sawn through, provided there be matter below it and in it, and it often happens that the bone is more superficial, especially if the wound is situated in that part of the head where the bone is rather thinner than in other parts. But you must take care where you apply the trepan, and see that you do so only where it appears to be particularly thick, and having fixed the instrument there, that you frequently make examinations and endeavor by moving the bone to bring it up. Having removed it, you must apply the other suitable remedies to the wound. And if, when you have the management of the treatment from the first, you wish to saw through the bone at once, and remove it from the membrane, you must, in like manner, examine the tract of the instrument frequently with the sound, and see that it is fixed on the thickest part of the bone, and endeavor to remove the bone by moving it about. But if you use a perforator (*trepan?*), you must not penetrate to the membrane, if you operate on a case which you have had the charge of from the first,

but must leave a thin scale of bone, as described in the process of sawing.

On the Surgery

Surg. 1

It's the business of the physician to know, in the first place, things similar and things dissimilar; those connected with things most important, most easily known, and in anywise known;³ which are to be seen, touched, and heard; which are to be perceived in the sight, and the touch, and the hearing, and the nose, and the tongue, and the understanding; which are to be known by all the means we know other things.

Surg. 2

The things relating to surgery, are—the patient; the operator; the assistants; the instruments; the light, where and how; how many things, and how; where the body, and the instruments; the time; the manner; the place.

Surg. 3

The operator is either sitting or standing, conveniently for himself, for the person operated upon, for the light. There are two kinds of light, the common and the artificial; the common is not at our disposal, the artificial is at our disposal. There are two modes of using each, either to the light, or from the light (to the side?). There is little use of that which is from (*or oblique to the light*), and the degree of it is obvious. As to opposite the light, we must turn the part to be operated upon to that which is most brilliant of present and convenient lights, unless those parts which should be concealed, and which it is a shame to look upon; thus the part that is operated upon should be opposite the light, and the operator opposite the part operated upon, except in so far as he does not stand in his own light; for in this case the operator will indeed see, but the thing operated upon will not be seen. With regard to himself when sitting, his feet

should be raised to a direct line with his knees, and nearly in contact with one another; the knees a little higher than the groins, and at some distance from one another, for the elbows to rest upon them. The robe, in a neat and orderly manner, is to be thrown over the elbows and shoulders equally and proportionally. With regard to the part operated upon; we have to consider how far distant, and how near, above, below, on this side on that side, or in the middle. The measure as to distance and proximity is, that the elbows do not press the knees before, nor the sides behind; that the hands be not raised higher than the breasts, nor lower than so as that when the breast reposes on the knees he may have the hands at right angles with the arm: thus it is as regards the medium; but as concerns this side or that, the operator must not be beyond his seat, but in proportion as he may require turning he must shift the body, or part of the body, that is operated upon. When standing, he must make his inspection, resting firmly and equally on both feet; but he must operate while supporting himself upon either leg, and not the one on the same side with the hand which he makes use of; the knee being raised to the height of the groins as while sitting; and the other measures in like manner. The person operated upon should accommodate the operator with regard to the other parts of his body, either standing, sitting, or lying; so as that he may continue to preserve his figure, avoid sinking down, shrinking from, turning away; and may maintain the figure and position of the part operated upon, during the act of presentation, during the operation, and in the subsequent position.

Surg. 4

The nails should be neither longer nor shorter than the points of the fingers; and the surgeon should practice with the extremities of the fingers, the index-finger being usually turned to the thumb; when using the entire hand, it should be prone; when both hands, they should be opposed to one another. It greatly promotes a dexterous use of the fingers when the space between them is large, and when the thumb is opposed to the index. But it is clearly a disease when the

thumb is impaired from birth, or when, from a habit contracted during the time of nursing, it is impeded in its motions by the fingers. One should practice all sorts of work with either of them, and with both together (for they are both alike), endeavouring to do them well, elegantly, quickly, without trouble, neatly, and promptly.

Surg. 5

The instruments, and when and how they should be prepared, will be treated of afterwards; so that they may not impede the work, and that there may be no difficulty in taking hold of them, with the part of the body which operates. But if another gives them, he must be ready a little beforehand, and do as you direct.

Surg. 6

Those about the patient must present the part to be operated upon as may seem proper, and they must hold the rest of the body steady, in silence, and listening to the commands of the operator.

Surg. 7

There are two views of bandaging: that which regards it while doing, and that which regards it when done. It should be done quickly, without pain, with ease, and with elegance; quickly, by despatching the work; without pain, by being readily done; with ease, by being prepared for everything; and with elegance, so that it may be agreeable to the sight. By what mode of training these accomplishments are to be acquired has been stated. When done, it should fit well and neatly; it is neatly done when with judgment, and when it is equal and unequal, according as the parts are equal or unequal. The forms of it (the bandage?) are the simple, the slightly winding (called *ascia*), the sloping (*sima*), the monocus, the rhombus, and the semi-rhombus. The form of bandage should be suitable to the form and the affection of the part to which it is applied.

Surg. 8

There are two useful purposes to be fulfilled by bandaging: (*first*), strength, which is imparted by the compression and the number of folds. In one case the bandage effects the cure, and in another it

contributes to the cure. For these purposes this is the rule—that the force of the constriction be such as to prevent the adjoining parts from separating, without compressing them much, and so that the parts may be adjusted but not forced together; and that the constriction be small at the extremities, and least of all in the middle. The knot and the thread that is passed through should not be in a downward but in an upward direction, regard being had to the circumstances under which the case is presented; to position, to the bandaging, and to the compression. The commencement of the ligatures is not to be placed at the wound, but where the knot is situated. The knot should not be placed where it will be exposed to friction, nor where it will be in the way, nor where it will be useless. The knot and the thread should be soft, and not large.

Surg. 9

(*Second.*) One ought to be well aware that every bandage has a tendency to fall off towards the part that declines or becomes smaller; as, for example, upwards, in the case of the head, and downwards, in the case of the leg. The turns of the bandage should be made from right to left, and from left to right, except on the head, where it should be in a straight direction. When opposite parts are to be bandaged together, we must use a bandage with two heads; or if we make use of a bandage with one head, we must attach it in like manner at some fixed point: such, for example, as the middle of the head; and so in other cases. Those parts which are much exposed to motion, such as the joints, where there is a flexion, should have few and slight bandages applied to them, as at the ham; but where there is much extension, the bandage should be single and broad, as at the knee-pan; and for the maintenance of the bandage in its proper place, some turns should be carried to those parts which are not much moved, and are lank, such as the parts above and below the knee. In the case of the shoulder; a fold should be carried round by the other armpit; in that of the groin, by the flanks of the opposite side; and of the leg, to above the calf of the leg. When the bandage has a tendency

to escape above, it should be secured below, and *vice versa*; and where there is no means of doing this, as in the case of the head, the turns are to be made mostly on the most level part of the head, and the folds are to be done with as little obliquity as possible, so that the firmest part being last applied may secure the portions which are more movable. When we cannot secure the bandaging by means of folds of the cloth, nor by suspending them from the opposite side, we must have recourse to stitching it with ligatures, either passed circularly or in the form of a seam.

Surg. 10

The bandages should be clean, light, soft, and thin. One should practice rolling with both hands together, and with either separately. One should also choose a suitable one, according to the breadth and thickness of the parts. The heads of the bandages should be hard, smooth, and neatly put on. That sort of bandaging is the worst which quickly falls off; but those are bad bandages which neither compress nor yet come off.

Surg. 11

The following are the object which the upper bandage, the under bandage, or both aim at: The object of the under bandage is either to bring together parts that are separated, or to compress such as are expanded, or to separate what are contracted, or to restore to shape what are distorted, or the contrary. It is necessary to prepare pieces of linen cloth, which are light, thin, soft, clean, having no seams nor protuberances on them, but sound, and able to bear some stretching, or even a little more than required; not dry, but wetted with a juice suitable to the purpose required. We must deal with parts separated (*in a sinus?*) in such wise, that the parts which are raised may touch the bottom without producing pressure; we must begin on the sound part, and terminate at the wound; so that whatever humor is in it may be expelled, and that it may be prevented from collecting more. And straight parts are to be bandaged in a straight direction, and oblique obliquely, in such a position as to create no pain; and so that there

may be no constriction nor falling off on a change of position, either for the purpose of taking hold of anything, or laying the limb; and that muscles, veins, nerves, and bones may be properly placed and adjusted to one another. It should be raised or laid in a natural position, so as not to occasion pain. In those cases in which an abscess is formed, we must act in a contrary way. When our object is to bring together parts which have become expanded, in other respects we must proceed on the same plan; and we must commence the bringing together from some considerable distance; and after their approach, we must apply compression, at first slight, and afterwards stronger, the limit of it being the actual contact of the parts. In order to separate parts which are drawn together, when attended with inflammation, we must proceed on the opposite plan; but when without inflammation, we must use the same preparations, but bandage in the opposite direction. In order to rectify distorted parts, we must proceed otherwise on the same principles; but the parts which are separated must be brought together by an underbandage, by agglutinants, and by suspending it (*the limb?*) in its natural position. And when the deformities are the contrary, this is to be done on the contrary plan.

Surg. 12

In fractures we must attend to the length, breadth, thickness, and number of the compresses. The length should be that of the bandaging; the breadth, three or four fingers; thickness, three or fourfold; number so as to encircle the limb, neither more nor less; those applied for the purpose of rectifying a deformity, should be of such a length as to encircle it; the breadth and thickness being determined by the vacuity, which is not to be filled up at once. The upper bandages are two, the first of which is to be carried from the seat of the injury upwards, and the second from the seat of the injury downwards, and from below upwards; the parts about the seat of the injury being most compressed, the extremities least, and the rest in proportion. The upper bandages should take in a considerable portion

of the sound parts. We must attend to the number, length, and breadth of the bandages; the number must be such as not to be inferior to what the injury requires, nor occasion compression with the splints, nor prove cumbersome, nor occasion any slipping of them, nor render them inefficient. As to length and breadth, they should be three, four, five, or six cubits in length, and as many fingers broad. The folds of the strings (*selvages?*) should be such as not to occasion pressure; they are to be soft and not thick; and all these things are to be proportionate to the length, breadth, and thickness of the part affected. The splints are to be smooth, even, and rounded at the extremities; somewhat less all along than the upper bandaging, and thickest at the part to which fracture inclines. Those parts where there are tuberosities, and which are devoid of flesh, such as the ankles or fingers, we must guard from the splints which are placed over them, either by position, or by their shortness. They are to be secured by the strings in such a manner as not to occasion pressure at first. A soft, consistent, and clean cerate should be rubbed into the folds of the bandage.

Surg. 13

As to the temperature and quantity of the water used, its heat should be just such as the hand can bear, and it ought to be known that a large quantity is best for producing relaxation and attenuation, whereas a moderate quantity is best for incarnating and softening. The limit to the affusion is, to stop when the parts become swelled up, and before the swelling subsides; for the parts swell up at first, and fall afterward.

Surg. 14

The object on which to (*the limb?*) is laid should be soft, smooth, and sloping upwards toward the protuberant parts of the body, such as the heel or hips, so that there may be no projection, nor bending inwards, nor turning aside. The canal (*spout or gutter?*) should rather comprehend the whole limb than the half of it, attention being paid to the injury and to whatever else appears to create inconvenience.

Surg. 15

The presentation of the injured part to the physician, the extension, the arrangement, and so forth, are to be regulated according to nature. What is nature in these operations is to be determined by the accomplishment of the object which we have in view, and for this purpose we must look to the part in the state of rest, in its middle state, and to habit; in regard to the state of rest and relaxation, as in the arm, that it be in a line with the hand; and with regard to the medium between flexion and extension, that the forearm be at right angles to the arm; and with regard to habit, it should be considered that some limbs bear certain positions preferably, as, for example, the thighs extension; for in such attitudes the parts can best bear to be placed for a considerable time without a change of posture. And in the change from the state of distention, the muscles, veins, nerves, and bones, when properly arranged and secured, will preserve their relations to one another while the limb is raised or placed.

Surg. 16

The extension should be most powerful when the largest and thickest bones, or when both are broken; next when the under-bone, and least of all, when the upper. When immoderate, it is injurious, except in the case of children. The limb should be a little elevated. The model by which we judge if the part be properly set is the sound part of the same name, or the part which is its pair.

Surg. 17

Friction can relax, brace, incarnate, attenuate: hard braces, soft relaxes, much attenuates, and moderate thickens.

Surg. 18

The following should be the state of matters on the first application of the bandage. The person to whom it has been applied should say that he feels the compression particularly at the seat of the injury, but very little at the extremities; the parts should be adjusted but not pressed together, and that rather by the number of the bandages than by the force of the constriction; and the tightness should rather be on the

increase during the first day and night; but on the next it should be less, and on the third the bandages should be loose. On the next day a soft swelling should be observed in the extremities; and on the third day, when the bandaging is loosed, the swelling should be found diminished in size, and this should be the case every time the bandages are removed. At the second application of the bandage, it should be ascertained whether the dressing has been properly done, and then greater compression should be made, and with more bandages; and on the third, still greater, and still more. On the seventh day from the first dressing, when the bandages are loosed, the limb should be found slender and the bones mobile. We must then have recourse to the splints, provided the limb be free of swelling, pruritus, and ulceration, and allow them to remain until twenty days after the accident; but if any suspicions arise, the bandages must be loosed in the interval. The splints should be tightened every third day.

Surg. 19

The suspending of a fractured limb in a sling, the disposition of it, and the bandaging, all have for their object to preserve it in position. The principal considerations with regard to the position are the habits and the peculiar nature of each of the limbs: the varieties are shown in running, walking, standing, lying, action, repose.

Surg. 20

It should be kept in mind that exercise strengthens, and inactivity wastes.

Surg. 21

Compression should be produced by the number of bandages, rather than by the force of the constriction.

Surg. 22

In cases of ecchymosis, contusions, sprains, or swellings not attended with inflammations, blood is to be expelled from the wound, in greatest quantity to the upper part, and in smallest to the inferior; neither the arm nor the leg should be placed in a declining position: the head of the bandage should be placed on the wound, and there the

greatest pressure should be made; the least at the extremities, and intermediately in the middle; the last fold of the bandage should be at the upper part of the body. As to binding and compression, these objects are to be attained rather by the number of the bandages than the force of the constriction; and moreover, in these cases the bandages should be thin, light, soft, clean, broad, sound, so that they may effect their purpose, even without splints. And we must use affusions.

Surg. 23

Dislocations, sprains, diastases of bones, violent separation, abruption of the extremities of bones, and distrainings, so as to induce *varus* or *valgus*, in these cases we must apply the bandages so as not to compress the part whence the displacement took place, and that we may render them tight at the side to which the displacement was, and give the limb an inclination in the opposite direction, and that in an excessive degree. We employ bandages, compresses, suspension of the limb in a sling, attitude, extension, friction, rectification; and along with these the affusion of much water.

Surg. 24

In treating parts which are atrophied, we must comprehend a considerable part of the sound limb with the bandage, so that by the influx thereby produced, the wasted part may acquire a supply greater than its loss, and may be thus disposed to growth and restoration of its fleshy parts. It is better also to bandage the parts above, as the thigh in the case of the leg, and also the thigh and leg of the opposite side, so that they may be placed in similar circumstances, and may both equally be deprived of motion; and that the supply of nourishment may be alike curtailed and open to both. The compression should be the effect rather of the number of the bandages than of their tightness. We relax first the part most requiring it, and have recourse to that kind of friction which will promote the growth of flesh, and to affusion. No splints.

Surg. 25

Those things which are for the purpose of giving support and strength to the part, as to the breast, side, head, and so forth, are used in such cases as the following: for pulsations, that there may be no motion in the part; and in separation at the sutures of the skull, in order to give support; and in order to strengthen the chest and head, in coughs, sneezings, and other movements. In all these cases the same measure of bandaging is to be observed, for where the injury is, there the bandage should compress most, and something soft is to be placed below that suits with the complaint; and we must not apply the bandages tighter than just to stop the pulsations from creating disturbance, and that the separated parts at the sutures may be brought into contact, they must not be such as absolutely to stop the coughs and sneezings, but so as to give support, and, without occasioning uneasiness, prevent the parts from being shaken.

On Fractures

Fract. 1

In treating fractures and dislocations, the physician must make the extension as straight as possible, for this is the most natural direction. But if it incline to either side, it should rather turn to that of pronation, for there is thus less harm than if it be toward supination. Those, then, who act in such cases without deliberation, for the most part do not fall into any great mistake, for the person who is to have his arm bound, presents it in the proper position from necessity, but physicians who fancy themselves learned in these matters, are they who commit blunders. There is no necessity for much study, then, in order to set a broken arm, and in a word, any ordinary physician can perform it; but I am under the necessity of giving the longer directions on this subject, because I know physicians who have the reputation of being skilled in giving the proper positions to the arm in binding it up, while in reality they are only showing their own ignorance. But many other things in our art are judged of in this

manner, for people rather admire what is new, although they do not know whether it be proper or not, than what they are accustomed to, and know already to be proper; and what is strange, they prefer to what is obvious. I must now state what the mistakes of medical men are, which I wish to unteach, and what instructions I have to give as to the management of the arm; for what I have to say regarding it, will apply to the other bones in the body.

Fract. 2

The arm, then, for that is the subject we were treating of, was presented in the prone position to be bound, but the physician forced his patient to hold it as the archers do when they project the shoulder, and in this position he bound it up, thinking within himself that he was acting according to Nature, and in proof of this he pointed out that all the bones in the fore-arm were thus in a straight line, and that the integuments both inside and outside, were also in a straight line, and that the flesh and nerves (*tendons*?) were thus put in their natural position, and he appealed to what happens in archery, as a proof of this. And so saying, and so doing, he is looked up to as a sage; and yet he forgets that in all the other arts and performances, whether executed by strength or dexterity, what is reckoned the natural position is not the same, and that in the same piece of work it may happen that the natural position of the right arm is not the same as that of the left. For there is one attitude in throwing the javelin, and another in slinging, another in casting stones, another in boxing, and another in a state of repose. And whatever arts one examines, it will be found that the natural position of the arms is not the same in each, but that in every case the arms are put into the attitude which suits best with the instrument that is used, and the work to be performed. In practicing archery, no doubt this is the best attitude of the left arm, for ginglymoid extremity of the humerus being fixed in the cavity of the ulna, in this position, throws the bones of the forearm and arm into a line, as if they constituted a single bone, and all flexion at the joint is prevented in this position. It is no doubt certain that the

member is thus put into the most unbending and extended position possible, so as not to be overcome or yield when the string is drawn by the right arm, and thus will the archer be enabled to draw the string farthest, and discharge his arrow with the greatest force and rapidity, for arrows thus discharged have the greatest swiftness and force, and are carried to the greatest distances. But there is nothing in common between the binding up of an arm and archery. Moreover, if having thus bound up the arm, the physician direct the patient to keep it thus, he will occasion him greater pain than he had from the wound itself; and thus also, if the physician order him to bend the arm, neither the bones, the nerves, nor the flesh will any longer be in the same condition, but will be arranged differently, having overcome the bandaging. What use, then, is there of the archer's attitude? And these mistakes, the physician, conceited in his knowledge, would probably not have committed if he had allowed the patient himself to present his arm.

Fract. 3

But another physician putting the arm into the state of supination, gives orders to extend the arm thus, and bandages it in this position, reckoning it the one according to nature, judging thus from the skin, and also fancying the bones to be thus in their natural position, because the bone which protrudes at the wrist, where the little finger is, appears to be in a line with the bone from which people measure the bone of the fore-arm. These things he brings forward as proofs that the parts are in their natural state, and he is supposed to speak correctly. But, indeed, if the arm be kept stretched in a supine position, it will become very painful, and this fact any one may ascertain by extending his own arm in this attitude. And also a weaker man grasping with his hands a stronger man whose arm is turned in a supine position, could lead him wherever he chose, and neither, if a man held a sword thus in his hand, could he make any proper use of it, so constrained is this position. And, moreover, if, when a physician has thus bound up the arm, he allow it to remain in the same position, the

patient will endure greater pain if he walk about, but considerable, even if he remain at rest. And thus, too, if he shall bend the arm, the muscles and the bones must necessarily assume a different position. But, in addition to other mischief, he is ignorant of these facts regarding the position, that the bone which protrudes at the wrist, close to the little finger, belongs to the fore-arm, whereas the one at the joint, from which people measure the fore-arm, is the head of the humerus. He fancies that both these belong to the same bone, and many others are of this opinion. The latter, in fact, is the same part as that which is called the elbow, upon which we sometimes rest, and when he holds the arm thus in a supine position, in the first place the bone appears distorted, and in the next place the tendons which extend from the carpus along the inner side and from the fingers become distorted while the arm has a supine position; for these tendons proceed to the bone of the humerus, from which the fore-arm is measured. Such, and so many mistakes and marks of ignorance are committed, regarding the natural construction of the arm. But if one will extend a broken arm as I direct, he will turn the bone, situated at the extremity of the little finger, into the straight line, and also the one at the elbow, and the tendons which stretch from the carpus to the extremity of the humerus will be placed in the straight line; and when the arm is suspended in a sling, it will be in the same attitude as that in which it was bound up, and will give no pain to the patient when he walks about, nor when he lies reclined, and will not become fatigued. The man should be so seated that the prominent part of the bone may be turned to the brightest light which is at hand, so that the operator in making the extension, may be at no loss to discover if it be sufficiently straight. The prominence of a broken bone could not escape being detected by the hand of an experienced person, when applied for this purpose, and, moreover, the projecting part is particularly painful to the touch.

In cases of fracture in either of the bones of the forearm, it is easier to effect a cure if the upper bone be broken, although it be the thicker one, both because the sound bone is situated below, and forms a support to it, and because the deformity is more easily concealed, there being a thick mass of flesh on the upper side, except near to the wrist. But the lower bone is without a covering of flesh, is not easily concealed, and requires stronger extension. If it is not this bone, but the other which is broken, a more feeble extension proves sufficient, but if both be broken, a more powerful extension is required. In the case of a young person I have known the extension made more strong than was necessary, but in general the extension made is less than what is required. And when they are extended, the physician should apply the palms of the hands, and adjust the fractured parts and then having rubbed the parts with cerate, but not in large quantity so that the bandages may not come off, it is to be bound up in this state, care being taken that the hand be not lower than the elbow, but a little higher, so that the blood do not flow toward the extremity, but may be determined to the upper part; and then it is to be secured with the bandage, the head of which is to be placed at the fracture, and the bandage should impart firmness to the parts without occasioning strong compression. When you have carried the bandage twice or thrice round at the seat of the fracture, it is to be carried upward, so that the afflux of blood into it may be stopped, and the bandage should terminate there, and the first bandages ought not to be long. The head of the second bandage is also to be placed upon the seat of the fracture, and a single round of it being made there, it is then to be carried downward, and is not to be applied so tight as the other, and there should be greater distances between the turns, so that the bandage may prove sufficient to revert to the spot where the other terminated. The bandages may be rolled to the left hand or to the right, or to whatever side suits best with the position of the fractured arm, or according to the inclination which it may have. Afterward we must place along the arm, compresses, smeared with a little cerate, for

thus they occasion less uneasiness, and are more easily arranged. And then we must apply the bandages crossways, sometimes to the right hand, and sometimes to the left, for the most part beginning below and terminating above, but sometimes commencing above and ending below. The parts which are thinly covered with flesh should be wrapped round with compresses, and inequalities should be made up, not by a number of folds at once, but by degrees. Some slack turns are also to be made around the wrist, to this side and to that. These two bandages are sufficient at first.

Fract. 5

And these are the signs that the patient has been well treated and properly bandaged: if you ask him if the arm feels tight, and he says it does, but moderately so, and especially about the fracture; and this reply he should make all along, if the bandage be properly applied. And these are symptoms of the bandaging being moderately tight; if for the first day and night he fancies that the tightness does not diminish, but rather increases; and if on the next day there be a soft swelling in the hand, for this is a sign of moderate compression, but at the end of the second day the compression should feel less, and on the third day the bandaging should appear loose. And if any of these symptoms be wanting, you may conclude that the bandaging is slacker than it should be; or if any of these symptoms be in excess, you may infer that the compression is more than moderate; and judging from these, you will apply the next bandages either slacker or tighter. Having removed the bandages on the third day, you must make extension and adjust the fracture, and bind it up again; and if the first bandaging was moderately applied, the second bandaging should be made somewhat tighter. The heads the bandages should be placed on the fractures as in the former case; for, so doing, the humors will be driven to the extremities, whereas if you bandage any other part beforehand, the humors will be forced from it to the seat of the fracture: it is of much importance that this should be properly understood. Thus the bandaging and compression should always

commence at the seat of the fracture, and everything else should be conducted on the same principle, so that the farther you proceed from the fracture, the compression should always be the less. The bandages should never be actually loose, but should be smoothly put on. At each dressing the number of bandages should be increased; and the patient, if asked, should answer, that he feels the bandages somewhat tighter than on the former occasion, especially about the fracture, and everything else in proportion; and with respect to the swelling, the pain, and recovery, everything should proceed as after the former dressing. But on the third day the outer bandaging should appear looser. Then having removed the bandages, you should bind it up again, somewhat tighter than before, and with all the bandages which will be required on the occasion, and afterwards one ought to experience the same train of symptoms as at the former periods of bandaging.

Fract. 6

When the third day arrives, that is to say, the seventh from the first dressing, if properly done, the swelling in the hand should be not very great; and the part which has been bandaged should be found more slender and less swelled at each time, and on the seventh day the swelling should be quite gone, and the broken bones should be more readily moved, and admit of being easily adjusted. And if these things be so, you should, after setting the fracture, apply the bandages so as to suit the splints, and a little more tight than formerly, unless there be more pain from the swelling in the hand. When you have applied the bandages, you must adjust the splints all around the limb, and secure them with strings so loose as just to keep them in their place, without the application of the splints contributing at all to the compression of the arm. After this the pain and recovery should proceed as in the preceding periods of the bandaging. But if, on the third day, the patient say that the bandaging is loose, you must then fasten the splints, especially at the fracture, but also elsewhere, wherever the bandaging is rather loose than tight. The splint should

be thickest where the fracture protrudes, but it should not be much more so than elsewhere. Particular attention should be paid to the line of the arm corresponding to the thumb, so that no splint be laid on it, but upon each side of it, nor in the line of the little finger where the bone is prominent at the wrist, but on each side of it. And if it be found necessary that splints should be applied in these directions at the seat of the fracture, they should be made shorter than the others, so as that they may not reach the bones which are prominent at the wrist, for otherwise there is danger of ulceration, and of the tendons being laid bare. The splints should be adjusted anew every third day, in a very gentle manner, always keeping in mind that the object of the splints is to maintain the lower bandages in their place, and that they are not needed in order to contribute to the compression.

Fract. 7

If, then, you see that the bones are properly adjusted by the first dressings, and that there is no troublesome pruritus in the part, nor any reason to suspect ulceration, you may allow the arm to remain bandaged in the splints until after the lapse of more than twenty days. The bones of the fore-arm generally get consolidated in thirty days altogether; but there is nothing precise in this matter, for one constitution differs from another, and one period of life from another. When you remove the bandages, you must pour hot water on the arm and bind it up again, but somewhat slacker, and with fewer bandages than formerly: and again on the third day you undo the bandages, and bind it still more loosely, and with still fewer bandages. And if, while the arm is bound up in the splints, you should at any time suspect that the bones do not lie properly, or if anything about the bandages annoys the patient, you should loose them at the middle of the time, or a little earlier, and apply them again. A diet slightly restricted will be sufficient in those cases in which there was no external wound at first, or when the bone does not protrude; but one should live rather sparingly until the tenth day, as being now deprived of exercise; and tender articles of food should be used, such as moderately loosen the

bowels; but one should abstain altogether from flesh and wine, and then by degrees resume a more nourishing diet. This diet. may be laid down as a just rule in the treatment of fractures, both as to how they should be treated, and what will be the results of a proper plan of treatment; so that one may know, that if things do not turn out thus, there has been some defect or excess in the treatment. And in this simple plan of treatment it is necessary to attend also to the following directions, which some physicians pay little attention to, although, when improperly executed, they are capable of marring the whole process of bandaging: for if both the bones be broken, or the lower one only, and the patient who has got his arm bandaged keep it slung in a shawl, and that the shawl is particularly loose at the fracture, so that the arm is not properly suspended at this end or that, in this case the bone must necessarily be found distorted upwards; whereas, when both bones are thus broken, if the arm recline in the shawl at the wrist and elbow, but the rest of it be not kept up, the bone in this case will be distorted to the lower side. The greater part of the arm and the wrist of the hand should therefore be equally suspended in a broad soft shawl.

Fract. 8

When the arm is broken, if one stretch the fore-arm and adjust it while in this position, the muscle of the arm will be bound while extended; but when the dressing is over, and the patient bends his arm at the elbow, the muscle of the arm will assume a different shape. The following, then, is the most natural plan of setting the arm: having got a piece of wood a cubit or somewhat less in length, like the handles of spades, suspend it by means of a chain fastened to its extremities at both ends; and having seated the man on some high object, the arm is to be brought over, so that the armpit may rest on the piece of wood, and the man can scarcely touch the seat, being almost suspended; then having brought another seat, and placed one or more leather pillows under the arm, so as to keep it a moderate height while it is bent at a right angle, the best plan is to put round

the arm a broad and soft skin, or broad shawl, and to hang some great weight to it, so as to produce moderate extension; or otherwise, while the arm is in the position I have described, a strong man is to take hold of it at the elbow and pull it downward. But the physician standing erect, must perform the proper manipulation, having the one foot on some pretty high object, and adjusting the bone with the palms of his hands; and it will readily be adjusted, for the extension is good if properly applied. Then let him bind the arm, commencing at the fracture, and do otherwise as directed above; let him put the same questions and avail himself of the same signs to ascertain whether the arm be moderately tight or not; and every third day let him bind it anew and make it tighter; and on the seventh or ninth day let him bind it up with splints, and leave it so until after the lapse of more than thirty days. And if he suspect that the bone is not lying properly, let him remove the bandages in the interval, and having adjusted the arm, let him bind it up again. The bone of the arm is generally consolidated in forty days. When these are past, the dressing is to be removed, and fewer and slacker bandages applied instead of it. The patient is to be kept on a stricter diet, and for a longer space of time than in the former case; and we must form our judgment of it from the swelling in the hand, looking also to the strength of the patient. This also should be known, that the arm is naturally inclined outward; to this side, therefore, the distortion usually takes place, if not properly treated; but indeed, all the other bones are usually distorted during treatment for fracture to that side to which they naturally incline. When, therefore, anything of this kind is suspected, the arm is to be encircled in a broad shawl, which is to be carried round the breast, and when the patient goes to rest, a compress of many folds, or some such thing, is to be folded and placed between the elbow and the side, for thus the bending of the bone will be rectified, but care must be taken lest it be inclined too much inwards.

The human foot is composed of several small bones like the hand. These bones therefore are scarcely ever broken, unless the skin at the same time be wounded by some sharp and heavy body. The treatment of such injuries, therefore, will be delivered under the head of wounds. But if any bone be moved from its place, or a joint of the toes be luxated, or any of the bones of the part called the tarsus be displaced, it must be forced back again to its place as described with regard to the hand; and is to be treated with cerate, compresses, and bandages, like the fractures, with the exception of the splints; and is to be secured tightly in the same way, and the bandages renewed on the third day; and the patient thus bandaged should return the same answers as in fractures, as to the bandages feeling tight or slack. All these bones recover perfectly in twenty days, except those that are connected with the bones of the leg, and are in a line with them. It is advantageous to lie in bed during the whole of this time; but the patients, thinking light of the complaint, have not perseverance to do this, and they walk about before they get well; wherefore many of these do not make a perfect recovery. And often the pain puts them in mind of the injury; and deservedly, for the feet sustain the weight of the whole body. When, therefore, they walk about before they are whole, the joints which have been luxated are cured incompletely; and, on that account, while walking about, they have pains in the leg from time to time.

Fract. 10

But those bones which are connected with the bones of the leg are larger than the others, and the cure of them when luxated is more protracted. The mode of treatment then is the same; but we must use more bandages and more splints, and the bandage is to be carried round to this side and to that, and pressure is to be made as in the other cases, particularly at the seat of the luxation, and the first circles of the bandages are to be made there. And at each time the bandages are taken off, much hot water is to be used, for in all injuries at joints the affusion of hot water in large quantity is to be had

recourse to. And the same symptoms of compression and relaxation should manifest themselves in the same times, as in the cases formerly treated of, and the subsequent bandagings should be conducted in like manner. These cases get completely well for the most part in forty days, if the patients have resolution to keep their bed; but if not, they are subjected to the complaints formerly described, or still worse.

Fract. 11

In persons who jumping from any high object pitch upon their heel with great force, the bones are separated, and the veins pour forth their contents, owing to the contusion of the flesh surrounding the bone, and hence a swelling and much pain supervene. For this bone (*os calcis*) is not a small one, protrudes beyond the line of the leg, and is connected with important veins and tendons; for the back tendon of the leg is inserted into this bone. Such cases are to be treated with cerate, and with compresses and bandages; and hot water is to be used in large quantity; and they require many bandages, which ought to be particularly good and appropriate. And if the patient happen to have a tender skin about the heel, nothing is to be done to it; but if, as some have it, the skin be thick and hardened, it is to be pared down smoothly and thinned, but without wounding it. It is not everybody who can apply the bandage properly in such cases; for if one shall bind the parts, as in other accidents about the ankle, sometimes bringing a fold round the foot and sometimes round the tendon, these turns leave out the heel, which is the seat of the contusion, and thus there is danger that the *os calcis* may sphacelate; and if this should take place, the impediment may endure for life and also in all the other cases of sphacelus, not proceeding from such a cause as this; as when, from being carelessly allowed to lie in a certain position during confinement to bed, the heel becomes black, or when a serious wound has occurred in the leg and it is long of healing, and is connected with the heel, or when the same thing happens in the thigh, or when in any disease a protracted decubitus takes place on the back, in all such

cases the sores are inveterate, troublesome, and frequently break out again, unless particular attention be paid to the cure, along with much rest, as in all the cases attended with sphacelus. And cases of sphacelus connected with this cause, in addition to other inconveniences, are attended with great danger to the whole body. For they are apt to be attended with very acute fevers, of the continual type, accompanied with tremblings, hiccup, aberration of intellect, and which prove fatal within a few days: and there may be lividities of bloody veins⁴, with nausea, and gangrene from pressure; these diseases may occur, besides the sphacelus.⁵ Those which have been described are the most violent contusion; but in general the contusions are mild, and no great care is required with regard to the treatment, and yet it must be conducted properly. But when the contusion appears to be severe, we must do as described above, making many turns of the bandage around the heel, sometimes carrying it to the extremity of the foot, sometimes to the middle, and sometimes around the leg; and, in addition, all the surrounding parts are to be bandaged in this direction and that, as formerly described; and the compression should not be made strong, but we should make use of many bandages, and it is better also to administer hellebore the same day or on the morrow; and the bandages should be removed on the third day and reapplied. And these are the symptoms by which we discover whether the case will get worse or not: when the extravasated blood, the lividities, and the surrounding parts become red and hard, there is danger of an exacerbation. But if there be no fever, we must give emetics, as has been said, and administer the other remedies which are applicable when the fever is not of a continual type; but if continual fever be present, we must not give strong medicines, but enjoin abstinence from solid food and soups, and give water for drink, and not allow wine but *oxyglyky* (a composition from vinegar and honey?). But if the case be not going to get worse, the ecchymosed and livid parts, and those surrounding them become greenish and not hard; for this is a satisfactory proof in all cases of

ecchymosis, that they are not to get worse; but when lividity is complicated with hardness, there is danger that the part may become blackened. And we must so manage the foot as that it may be generally raised a little higher than the rest of the body. Such a patient will get well in sixty days if he keep quiet.

Fract. 12

The leg consists of two bones, of which the one is much more slender than the other at one part, but not much more slender at another. These are connected together at the foot, and form a common epiphysis, but they are not united together along the line of the leg; and at the thigh they are united together and form an epiphysis, and this epiphysis has a diaphysis; but the other bone in a line with the little toe is a little longer. Such is the nature of the bones of the leg.

Fract. 13

Sometimes the bones connected with the foot are displaced, sometimes both bones with their epiphysis; sometimes the whole epiphysis is slightly moved, and sometimes the other bone. These cases are less troublesome than the same accidents at the wrist, if the patients will have resolution to give them rest. The mode of treatment is the same as that of the other, for the reduction is to be made, as of the other, by means of extension, but greater force is required, as the parts of the body concerned are stronger in this case. But, for the most part, two men will be sufficient, by making extension in opposite directions, but, not withstanding, if they are not sufficiently strong, it is easy to make more powerful extension in the following way: having fixed in the ground either the nave of a wheel, or any such object, something soft is to be bound round the foot, and then some broad thongs of ox-skin being brought round it, the heads of the thongs are to be fastened to a pestle or any other piece of wood, the end of which is to be inserted into the nave, and it, the pestle, is to be pulled away, while other persons make counter-extension by grasping the shoulders and the ham. It is also sometimes necessary to secure the upper extremity otherwise; this if you desire to effect, fasten deeply in

the ground a round, smooth piece of wood, and place the upper extremity of the piece of wood at the perineum, so that it may prevent the body from yielding to the pulling at the foot, and, moreover, to prevent the leg while stretched, from inclining downward; some person seated at his side should push back the hip, so that the body may not turn round with the pulling, and for this purpose, if you think fit, pieces of wood may be fastened about the armpits on each side, and they are to be stretched by the hands, and thus secured, while another person takes hold of the limb at the knee, and aids in thus making counter-extension. Or thus, if you prefer it: having bound other thongs of leather about the limb, either at the knee, or around the thigh, and having fastened another nave of a wheel in the ground above the head, and adjusted the thongs piece of wood adapted to the nave, extension may thus be made in the opposite direction to the feet. Or if you choose, it may be done thus: instead of the naves, lay a moderate-sized beam under the couch, and then having fastened pieces of wood in this beam, both before and behind the head, make counter-extension by means of thongs, or place windlasses at this extremity and that, and make extension by means of them. There are many other methods of making extension. But the best thing is, for any physician who practices in a large city, to have prepared a proper wooden machine, with all the mechanical powers applicable in cases of fractures and dislocation, either for making extension, or acting as a lever. For this purpose it will be sufficient to possess a board in length, breadth, and thickness, resembling the quadrangular threshing-boards made of oak.

Fract. 14

When you have made proper extension, it is easy reduce the joint, for the displaced bone is thus raised into a line with the other. And the bones are to be adjusted with the palms of the hands, pressing upon the projecting bone with the one, and making counter-pressure below the ankle with the other. When you have replaced the bones, you must apply the bandages while the parts are upon the stretch, if you

possibly can; but if prevented by the thongs, you must loose them, and make counter-extension until you get the bandages applied. The bandage is to be applied in the manner formerly described, the heads of the bandages being placed on the projecting part, and the first turns made in like manner, and so also with regard to the number of compresses and the compression; and turns of the bandages are to be brought frequently round on this and on that side of the ankle. But this joint must be bound more tight at the first dressing than in the case of the hand. But when you have applied the bandage, you must place the bandaged part somewhat higher than the rest of the body, and in such a position that the foot may hang as little as possible. The attenuation of the body is to be made proportionate to the magnitude of the luxation, for one luxation is to be a small, and another to a great extent. But in general we must reduce more, and for a longer time, in injuries about the legs, than in those about the hands; for the former parts are larger and thicker than the latter, and it is necessary that the body should be kept in a state of rest, and in a recumbent position. There is nothing to prevent or require the limb to be bandaged anew on the third day. And all the treatment otherwise is to be conducted in like manner, as in the preceding cases. And if the patient have resolution to lie quiet, forty days will be sufficient for this purpose, if only the bones be properly reduced, but if he will not lie quiet, he will not be able to use the limb with ease, and he will find it necessary to wear a bandage for a long time. When the bones are not properly replaced, but there has been some defect in this respect, the hip, the thigh, and the leg become wasted, and if the dislocation be inward, the external part of the thigh is wasted, and *vice versa*. But for the most part the dislocation is inward.

Fract. 15

And when both bones of the leg are broken without a wound of the skin, stronger extension is required. We may make extension by some of the methods formerly described, provided the bones ride over one another to a considerable degree. But extension by men is also

sufficient, and for the most part two strong men will suffice, by making extension and counterextension. Extension must naturally be made straight in a line with the leg and thigh, whether on account of a fracture of the bones of the leg or of the thigh. And in both cases they are to be bandaged while in a state of extension, for the same position does not suit with the leg and the arm. For when the fractured bones of the arm or fore-arm are bandaged, the fore-arm is suspended in a sling, and if you bind them up while extended, the figures of the fleshy parts will be changed in bending the arm at the elbow, for the elbow cannot be kept long extended, since persons are not in the custom of keeping the joint long in this form, but in a bent position, and persons who have been wounded in the arm, and are still able to walk about, require to have the arm bent at the elbow-joint. But the leg, both in walking and standing, is habitually extended, either completely or nearly so, and is usually in a depending position from its construction, and in order that it may bear the weight of the rest of the body. Wherefore it readily bears to be extended when necessary, and even when in bed the limb is often in this position. And when wounded, necessity subdues the understanding, since the patients become incapable of raising themselves up, so that they neither think of bending the limb nor of getting up erect, but remain lying in the same position. For these reasons, neither the same position nor the same mode of bandaging applies to the arm and to the leg. If, then, extension by means of men be sufficient, we should not have recourse to any useless contrivances, for it is absurd to employ mechanical means when not required; but if extension by men be not sufficient, you may use any of the mechanical powers which is suitable. When sufficiently extended, it will be easy to adjust the bones and bring them into their natural position, by straightening and arranging them with the palms of the hand.

Fract. 16

When the parts are adjusted, you should apply the bandages while the limb is in a stretched position, making the first turns to the right or to

the left, as may be most suitable; and the end of the bandage should be placed over the fracture, and the first turns made at that place; and then the bandage should be carried up the leg, as described with regard to the other fractures. But the bandages should be broader and longer, and more numerous, in the case of the leg than in that of the arm. And when it is bandaged it should be laid upon some smooth and soft object, so that it may not be distorted to the one side or the other, and that there may be no protrusion of the bones either forward or backward; for this purpose nothing is more convenient than a cushion, or something similar, either of linen or wool, and not hard; it is to be made hollow along its middle, and placed below the limb. With regard to the canals (*gutters*?) usually placed below fractured legs, I am at a loss whether to advise that they should be used or not. For they certainly are beneficial, but not to the extent which those who use them suppose. For the canals do not preserve the leg at rest as they suppose; nor, when the rest of the body is turned to the one side or the other, does the canal prevent the leg from following, unless the patient himself pay attention; neither does the canal prevent the limb from being moved without the body to the one side or the other. And a board is an uncomfortable thing to have the limb laid upon, unless something soft be placed above it. But it is a very useful thing in making any subsequent arrangements of the bed and in going to stool. A limb then may be well or ill arranged with or without the canal. But the common people have more confidence, and the surgeon is more likely to escape blame, when the canal is placed under the limb, although it is not *secundum artem*. For the limb should by all means lie straight upon some level and soft object, since the bandaging must necessarily be overcome by any distortion in the placing of the leg, whenever or to whatever extent it may be inclined. The patient, when bandaged, should return the same answers as formerly stated, for the bandaging should be the same, and the same swellings should arise in the extremities, and the slackening of the bandages in like manner, and the new bandaging on the third day; and the bandaged part

should be found reduced in swelling; and the new bandagings should be more tightly put on, and more pieces of cloth should be used; and the bandages should be carried loosely about the foot, unless the wound be near the knee. Extension should be made and the bones adjusted at every new bandaging; for, if properly treated, and if the swelling progress in a suitable manner, the bandaged limb will have become more slender and attenuated, and the bones will be more mobile, and yield more readily to extension. On the seventh, the ninth, or the eleventh day, the splints should be applied as described in treating of the other fractures. Attention should be paid to the position of the splints about the ankles and along the tendon of the foot which runs up the leg. The bones of the leg get consolidated in forty days, if properly treated. But if you suspect that anything is wanting to the proper arrangement of the limb, or dread any ulceration, you should loose the bandages in the interval, and having put everything right, apply them again.

Fract. 17

But if the other bone (*fibula?*) of the leg be broken, less powerful extension is required, and yet it must not be neglected, nor be performed slovenly, more especially at the first bandaging. For in all cases of fracture this object should be attained then as quickly as possible. For when the bandage is applied tight while the bones are not properly arranged, the properly arranged, the part becomes more painful. The treatment otherwise is the same.

Fract. 18

Of the bones of the leg, the inner one, called the tibia, is the more troublesome to manage, and requires the greater extension; and if the broken bones are not properly arranged, it is impossible to conceal the distortion, for the bone is exposed and wholly uncovered with flesh; and it is much longer before patients can walk on the leg when this bone is broken. But if the outer bone be broken, it causes much less trouble, and the deformity, when the bones are not properly set, is much more easily concealed, the bone being well covered with flesh;

and the patients speedily get on foot, for it is the inner bone of the leg which supports the most of the weight of the body. For along with the thigh, as being in a line with weight thrown upon the thigh, the inner bone has more work to sustain; inasmuch as it is the head of the thigh-bone which sustains the upper part of the body, and it is on the inner and not on the outer side of the thigh, being in a line with the tibia; and the other half of the body approximates more to this line than to the external one; and at the same time the inner bone is larger than the outer, as in the fore-arm the bone in the line of the little finger is the slenderer and longer. But in the joint of the inferior extremity, the disposition of the longer bone is not alike, for the elbow and the ham are bent differently. For these reasons when the external bone is broken, the patients can soon walk about; but in fractures of the inner, it is a long time before they can walk.

Fract. 19

When the thigh-bone is broken, particular pains should be taken with regard to the extension that it may not be insufficient, for when excessive, no great harm results from it. For, if one should bandage a limb while the extremities of the bone are separated to a distance from one another by the force of the extension, the bandaging will not keep them separate, and so the bones will come together again as soon as the persons stretching it let go their hold; for the fleshy parts (*muscles?*) being thick and strong, are more powerful than the bandaging, instead of being less so. In the case then which we are now treating of, nothing should be omitted in order that the parts may be properly distended and put in a straight line; for it is a great disgrace and an injury to exhibit a shortened thigh. For the arm, when shortened, might be concealed, and the mistake would not be great; but a shortened thigh-bone would exhibit the man maimed. For when the sound limb is placed beside it, being longer than the other, it exposes the mistake, and therefore it would be to the advantage of a person who would be improperly treated that both his legs should be broken, rather than either of them; for in this case the one would be of

the same length as the other. When, then, proper extension has been made, you must adjust the parts with the palms of the hands, and bandage the limb in the manner formerly described, placing the hands of the bandages as was directed, and making the turns upward. And the patient should return the same answers to the same questions as formerly, should be pained and recover in like manner, and should have the bandaging renewed in the same way; and the application of the splints should be the same. The thigh-bone is consolidated in forty days.

Fract. 20

But this also should be known, that the thigh-bone is curved rather to the outside than to the inside, and rather forward than backward; when not properly treated, then, the distortions are in these directions; and the bone is least covered with flesh at the same parts, so that the distortion cannot be concealed. If, therefore, you suspect anything of this kind, you should have recourse to the mechanical contrivances recommended in distortion of the arm. And a few turns of the bandage should be brought round by the hip and the loins, so that the groin and the articulation near the perineum may be included in the bandage; and moreover, it is expedient that the extremities of the splints should not do mischief by being placed on parts not covered with the bandages. The splints, in fact, should be carefully kept off the naked parts at both ends; and the arrangement of them should be so managed, as that they may not be placed on the natural protuberances of the bone at the knee-joint, nor on the tendon which is situated there.

Fract. 21

The swellings which arise in the ham, at the foot, or in any other part from the pressure, should be well wrapped in unsoured and carded wool, washed with wine and oil, and anointed with cerate, before bandaging; and if the splints give pain they should be slackened. You may sooner reduce the swellings, by laying aside the splints, and applying plenty of bandages to them, beginning from below and

rolling upward; for thus the swellings will be most speedily reduced, and the humors be propelled to the parts above the former bandages. But this form of bandaging must not be used unless there be danger of vesications or blackening in the swelling, and nothing of the kind occurs unless the fracture be bound too tight, or unless the limb be allowed to hang, or it be rubbed with the hand, or some other thing of an irritant nature be applied to the skin.

Fract. 22

More injury than good results from placing below the thigh a canal which does not pass farther down than the ham, for it neither prevents the body nor the leg from being moved without the thigh. And it creates uneasiness by being brought down to the ham, and has a tendency to produce what of all things should be avoided, namely, flexion at the knee, for this completely disturbs the bandages; and when the thigh and leg are bandaged, if one bend the limb at the knee, the muscles necessarily assume another shape, and the broken bones are also necessarily moved. Every endeavor then should be made to keep the ham extended. But it appears to me, that a canal which embraces the limb from the nates to the foot is of use. And moreover, a shawl should be put loosely round at the ham, along with the canal, as children are swathed in bed; and then, if the thigh-bone gets displaced either upward or to the side, it can be more easily kept in position by this means along with the canal. The canal then should be made so as to extend all along the limb or not used at all.

Fract. 23

The extremity of the heel should be particularly attended to, so that it may be properly laid, both in fractures of the leg and of the thigh. For if the foot be placed in a dependent position, while the rest of the body is supported, the limb must present a curved appearance at the forepart of the leg; and if the heel be placed higher than is proper, and if the rest of the leg be rather too low, the bone at the forepart of the leg must present a hollow, more especially if the heel of the patient be naturally large. But all the bones get consolidated more slowly, if not

laid properly, and if not kept steady in the same position, and in this case the callus is more feeble.

Fract. 24

These things relate to cases in which there is fracture of the bones without protrusion of the same or wound of any other kind. In those cases in which the bones are simply broken across, and are not comminuted, but protrude, if reduced the same day or next, and secured in their place, and if there be no reason to anticipate that any splintered bones will come away; and in those in which the broken bones do not protrude, nor is the mode of fracture such that there is reason to expect the splinters will come out, some physicians heal the sores in a way which neither does much good nor harm, by means of a cleansing application, applying pitch ointment, or some of the dressings for fresh wounds, or anything else which they are accustomed to do, and binding above them compresses wetted with wine, or greasy wool, or something else of the like nature. And when the wounds become clean and are now healed, they endeavor to bind up the limb with plenty of bandages, and keep it straight with treatment does some good, and never much harm. The bones, however, can never be equally well restored to their place, but the part is a little more swelled than it should be; and the limb will be somewhat shortened, provided both bones either of the leg or forearm have been fractured.

Fract. 25

There are others who treat such cases at first with bandages, applying them on both sides of the seat of the injury, but omit them there, and leave the wound uncovered, and afterward they apply to the wound some cleansing medicine, and complete the dressing with compresses dipped in wine and greasy wool. This plan of treatment is bad, and it is clear that those who adopt this mode of practice are guilty of great mistakes in other cases of fracture as well as these. For it is a most important consideration to know in what manner the head of the bandage should be placed and at what part the greatest pressure

should be, and what benefits would result from applying the end of the bandage and the pressure at the proper place, and what mischiefs would result from applying the head of the bandage and the pressure otherwise than at the proper place. Wherefore it has been stated in the preceding part of the work what are the results of either; and the practice of medicine bears witness to the truth of it, for in a person thus bandaged, a swelling must necessarily arise on the wound. For, if even a sound piece of skin were bandaged on either side, and a part were left in the middle, the part thus left unbandaged would become most swelled, and would assume a bad color; how then could it be that a wound would not suffer in like manner? The wound then must necessarily become discolored and its lips everted, the discharge will be ichorous and without pus, and the bones, which should not have got into a state of necrosis, exfoliate; and the wound gets into a throbbing and inflamed condition. And they are obliged to apply a cataplasm on account of the swelling, but this is an unsuitable application to parts which are bandaged on both sides, for a useless load is added to the throbbing which formerly existed in it. At last they loose the bandages when matters get very serious, and conduct the rest of the treatment without bandaging; and notwithstanding, if they meet with another case of the same description, they treat it in the same manner, for they do not think that the application of the bandages on both sides, and the exposure of the wound are the cause of what happened, but some other untoward circumstance. Wherefore I would not have written so much on this subject, if I had not well known that this mode of bandaging is unsuitable, and yet that many conduct the treatment in this way, whose mistake it is of vital importance to correct, while what is here said is a proof, that what was formerly written as to the circumstances under which bandages should be tightly applied to fractures or otherwise has been correctly written.

As a general rule it may be said, that in those cases in which a separation of bone is not expected, the same treatment should be applied as when the fractures are not complicated with an external wound; for the extension, adjustment of the bones, and the bandaging, are to be conducted in the same manner. To the wound itself a cerate mixed with pitch is to be applied, a thin folded compress is to be bound upon it, and the parts around are to be anointed with white cerate. The cloths for bandages and the other things should be torn broader than in cases in which there is no wound, and the first turn of the bandage should be a good deal broader than the wound. For a narrower bandage than the wound binds the wound like a girdle, which is not proper, or the first turn should comprehend the whole wound, and the bandaging should extend beyond it on both sides. The bandage then should be put on in the direction of the wound, and should be not quite so tight as when there is no wound, but the bandage should be otherwise applied in the manner described above. The bandages should be of a soft consistence, and more especially so in such cases than in those not complicated with a wound. The number of bandages should not be smaller, but rather greater than those formerly described. When applied, the patient should have the feeling of the parts being properly secured, but not too tight, and in particular he should be able to say that they are firm about the wound. And the intervals of time during which the parts seem to be properly adjusted, and those in which they get loose, should be the same as those formerly described. The bandages should be renewed on the third day, and the after treatment conducted in the same manner as formerly described, except that in the latter case the compression should be somewhat less than in the former. And if matters go on properly, the parts about the wound should be found at every dressing always more and more free of swelling, and the swelling should have subsided on the whole part comprehended by the bandages. And the suppurations will take place more speedily than in the case of wounds treated otherwise; and the pieces of flesh

in the wound which have become black and dead, will sooner separate and fall off under this plan of treatment than any other, and the sore will come more quickly to cicatrization when thus treated than otherwise. The reason of all this is, that the parts in which the wound is situated, and the surrounding parts, are kept free of swelling. In all other respects the treatment is to be conducted as in cases of fracture without a wound of the integuments. Splints should not be applied. On this account the bandages should be more numerous than in the former case, both because they must be put on less tight, and because the splints are later of being applied. But if you do apply the splints, they should not be applied along the wound, and they are to be put on in a loose manner, especial care being taken that there may be no great compression from the splints. This direction has been formerly given. And the diet should be more restricted, and for a longer period, in those cases in which there is a wound at the commencement, and when the bones protrude through the skin; and, in a word, the greater the wound, the more severe and protracted should the regimen be.

Fract. 27

The treatment of the sores is the same in those cases of fracture in which there was no wound of the skin at first, but one has formed in the course of treatment, owing to the pressure of the splints occasioned by the bandages, or from any other cause. In such cases it is ascertained that there is an ulcer, by the pain and the throbbing; and the swelling in the extremities becomes harder than usual, and if you apply your finger the redness disappears, but speedily returns. If you suspect anything of the kind you must loose the dressing, if there be any itching below the under-bandages, or in any other part that is bandaged, and used a pitched cerate instead of the other. If there be nothing of that, but if the ulcer be found in an irritable state, being very black and foul, and the fleshy parts about to suppurate, and the tendons to slough away, in these cases no part is to be exposed to the air, nor is anything to be apprehended from these suppurations, but the treatment is to be conducted in the same manner as in those cases

in which there was an external wound at first. You must begin to apply the bandages loosely at the swelling in the extremities, and then gradually proceed upward with the bandaging, so that it may be tight at no place, but particularly firm at the sore, and less so elsewhere. The first bandages should be clean and not narrow, and the number of bandages should be as great as in those cases in which the splints were used, or somewhat fewer. To the sore itself a compress, anointed with white cerate, will be sufficient, for if a piece of flesh or nerve (*tendon?*) become black, it will fall off; for such sores are not to be treated with acrid, but with emollient applications, like burns. The bandages are to be renewed every third day, and no splints are to be applied, but rest is to be more rigidly maintained than in the former cases, along with a restricted diet. It should be known, that if any piece of flesh or tendon be to come away, the mischief will spread much less, and the parts will much more speedily drop off, and the swelling in the surrounding parts will much more completely subside, under this treatment, than if any of the cleansing applications be put upon the sore. And if any part that is to come away shall fall off, the part will incarnate sooner when thus treated than otherwise, and will more speedily cicatrize. Such are the good effects of knowing how a bandage can be well and moderately applied. But a proper position, the other parts of the regimen, and suitable bandages cooperate.

Fract. 28

If you are deceived with regard to a recent wound, supposing there will be no exfoliation of the bones, while they are on the eve of coming out of the sore, you must not hesitate to adopt this mode of treatment; for no great mischief will result, provided you have the necessary dexterity to apply the bandages well and without doing any harm. And this is a symptom of an exfoliation of bone being about to take place under this mode of treatment; pus runs copiously from the sore, and appears striving to make its escape. The bandage must be renewed more frequently on account of the discharge, since otherwise fevers come on; if the sore and surrounding parts be compressed by

the bandages they become wasted. Cases complicated with the exfoliation of very small bones, do not require any change of treatment, only the bandages should be put on more loosely, so that the discharge of pus may not be intercepted, but left free, and the dressings are to be frequently renewed until the bone exfoliate, and the splints should not be applied until then.

Fract. 29

Those cases in which the exfoliation of a larger piece of bone is expected, whether you discover this at the commencement, or perceive subsequently that it is to happen, no longer require the same mode of treatment, only that the extension and arrangement of the parts are to be performed in a manner that has been described; but having formed double compresses, not less than half a fathom in breadth (being guided in this by the nature of the wound), and considerably shorter than what would be required to go twice round the part that is wounded, but considerably longer than to go once round, and in number what will be sufficient, these are to be dipped in a black austere wine; and beginning at the middle, as is done in applying the double-headed bandage, you are to wrap the part around and proceed crossing the heads in the form of the bandage called “ascia.” These things are to be done at the wound, and on both sides of it; and there must be no compression, but they are to be laid on so as to give support to the wound. And on the wound itself is to be applied the pitched cerate, or one of the applications to recent wounds, or any other medicine which will suit with the embrocation. And if it be the summer season, the compresses are to be frequently damped with wine; but if the winter season, plenty of greasy wool, moistened with wine and oil, should be applied. And a goat’s skin should be spread below, so as to carry off the fluids which run from the wound; these must be guarded against, and it should be kept in mind, that parts which remain long in the same position are subject to excoriations which are difficult to cure.

Fract. 30

In such cases as do not admit of bandaging according to any of the methods which have been described, or which will be described, great pains should be taken that the fractured part of the body be laid in a right position, and attention should be paid that it may incline upward rather than downward. But if one would wish to do the thing well and dexterously, it is proper to have recourse to some mechanical contrivance, in order that the fractured part of the body may undergo proper and not violent extension; and this means is particularly applicable in fractures of the leg. There are certain physicians who, in all fractures of the leg, whether bandages be applied or not, fasten the sole of the foot to the couch, or to some other piece of wood which they have fixed in the ground near the couch. These persons thus do all sorts of mischief but no good; for it contributes nothing to the extension that the foot is thus bound, as the rest of the body will no less sink down to the foot, and thus the limb will no longer be stretched, neither will it do any good toward keeping the limb in a proper position, but will do harm, for when the rest of the body is turned to this side or that, the bandaging will not prevent the foot and the bones belonging to it from following the rest of the body. For if it had not been bound it would have been less distorted, as it would have been the less prevented from following the motion of the rest of the body. But one should sew two balls of Egyptian leather, such as are worn by persons confined for a length of time in large shackles, and the balls should have coats on each side, deeper toward the wound, but shorter toward the joints; and the balls should be well stuffed and soft, and fit well, the one above the ankles, and the other below the knee. Sideways it should have below two appendages, either of a single or double thong, and short, like loops, the one set being placed on either side of the ankle, and the other on the knee. And the other upper ball should have others of the same kind in the same line. Then taking four rods, made of the cornel tree, of equal length, and of the thickness of a finger, and of such length that when bent they will admit of being adjusted to the appendages, care should be taken that

the extremities of the rods bear not upon the skin, but on the extremities of the balls. There should be three sets of rods, or more, one set a little longer than another, and another a little shorter and smaller, so that they may produce greater or less distention, if required. Either of these sets of rods should be placed on this side and that of the ankles. If these things be properly contrived, they should occasion a proper and equable extension in a straight line, without giving any pain to the wound; for the pressure, if there is any, should be thrown at the foot and the thigh. And the rods are commodiously arranged on either side of the ankles, so as not to interfere with the position of the limb; and the wound is easily examined and easily arranged. And, if thought proper, there is nothing to prevent the two upper rods from being fastened to one another; and if any light covering be thrown over the limb, it will thus be kept off from the wound. If, then, the balls be well made, handsome, soft, and newly stitched, and if the extension by the rods be properly managed, as has been already described, this is an excellent contrivance; but if any of them do not fit properly, it does more harm than good. And all other mechanical contrivances should either be properly done, or not be had recourse to at all, for it is a disgraceful and awkward thing to use mechanical means in an unmechanical way.

Fract. 31

Moreover, the greater part of physicians treat fractures, both with and without an external wound, during the first days, by means of unwashed wool, and there does not appear to be anything improper in this. It is very excusable for those who are called upon to treat newly-received accidents of this kind, and who have no cloth for bandages at hand, to do them up with wool; for, except cloth for bandages, one could not have anything better than wool in such cases; but a good deal should be used for this purpose, and it should be well carded and not rough, for in small quantity and of a bad quality it has little power. But those who approve of binding up the limb with wool for a day or two, and on the third and fourth apply bandages, and make the

greatest compression and extension at that period, such persons show themselves to be ignorant of the most important principles of medicine; for, in a word, at no time is it so little proper to disturb all kinds of wounds as on the third and fourth day; and all sort of probing should be avoided on these days in whatever other injuries are attended with irritation. For, generally, the third and fourth day in most cases of wounds, are those which give rise to exacerbations, whether the tendency be to inflammation, to a foul condition of the sore, or to fevers. And if any piece of information be particularly valuable this is; to which of the most important cases in medicine does it not apply? and that not only in wounds but in many other diseases, unless one should call all other diseases wounds. And this doctrine is not devoid of a certain degree of plausibility, for they are allied to one another in many respects. But those who maintain that wool should be used until after the first seven days, and then that the parts should be extended and adjusted, and secured with bandages, would appear not to be equally devoid of proper judgment, for the proper judgment, for the most dangerous season for inflammation is then past, and the bones being loose can be easily set after the lapse of these days. But still this mode of treatment is far inferior to that with bandages from the commencement; for, the latter method exhibits the patient on the seventh day free from inflammation, and ready for complete bandaging with splints; while the former method is far behind in this respect, and is attended with many other bad effects which it would be tedious to describe.

Fract. 32

In those cases of fracture in which the bones protrude and cannot be restored to their place, the following mode of reduction may be practiced:— Some small pieces of iron are to be prepared like the levers which the cutters of stone make use of, one being rather broader and another narrower; and there should be three of them at least, and still more, so that you may use those that suit best; and then, along with extension, we must use these as levers, applying the

under surface of the piece of iron to the under fragment of the bone, and the upper surface to the upper bone; and, in a word, we must operate powerfully with the lever as we would do upon a stone or a piece of wood. The pieces of iron should be as strong as possible, so that they may not bend. This is a powerful assistance, provided the pieces of iron be suitable, and one use them properly as levers. Of all the mechanical instruments used by men, the most powerful are these three, the axis in peritrochio, the lever, and the wedge. Without these, one or all, men could not perform any of their works which require great force. Wherefore, reduction with the lever is not to be despised, for the bones will be reduced in this way, or not at all. But if the upper fragment which rides over the other does not furnish a suitable point of support a suitable point of support for the lever, but the protruding part is sharp, you must scoop out of the bone what will furnish a proper place for the lever to rest on. The lever, along with extension, may be had recourse to on the day of the accident, or next day, but by no means on the third, the fourth, and the fifth. For if the limb is disturbed on these days, and yet the fractured bones not reduced, inflammation will be excited, and this no less if they are reduced; for convulsions are more apt to occur if reduction take place, than if the attempt should fail. These facts should be well known, for if convulsions should come on when reduction is effected, there is little hope of recovery; but it is of use to displace the bones again if this can be done with out trouble. For it is not at the time when the parts are in a particularly relaxed condition that convulsions and tetanus are apt to supervene, but when they are more than usually tense. In the case we are now treating of, we should not disturb the limb on the aforesaid days, but strive to keep the wound as free from inflammation as possible, and especially encourage suppuration in it. But when seven days have elapsed, or rather more, if there be no fever, and if the wound be not inflamed, then there will be less to prevent an attempt at reduction, if you hope to succeed; but otherwise you need not take and give trouble in vain.

Fract. 33

When you have reduced the bones to their place, the modes of treatment, whether you expect the bones to exfoliate or not, have been already described. All those cases in which an exfoliation of bone is expected, should be treated by the method of bandaging with cloths, beginning for the most part at the middle of the bandage, as is done with the double-headed bandage; but particular attention should be paid to the shape of the wound, so that its lips may gape or be distorted as little as possible under the bandage. Sometimes the turns of the bandage have to be made to the right, and sometimes to the left, and sometimes a double-headed bandage is to be used.

Fract. 34

It should be known that bones, which it has been found impossible to reduce, as well as those which are wholly denuded of flesh, will become detached. In some cases the upper part of the bone is laid bare, and in others the flesh dies all around; and, from a sore of long standing, certain of the bones become carious, and some not, some more, and some less; and in some the small, and in others the large bones. From what has been said it will be seen, that it is impossible to tell in one word when the bones will separate. Some come away more quickly, owing to their smallness, and some from being merely fixed at the point; and some, from pieces not separating, but merely exfoliating, become dried up and putrid; and besides, different modes of treatment have different effects. For the most part, the bones separate most quickly in those cases in which suppuration takes place most quickly, and when new flesh is most quickly formed, and is particularly sound, for the flesh which grows up below in the wound generally elevates the pieces of bone. It will be well if the whole circle of the bone separate in forty days; for in some cases it is protracted to sixty days, and in some to more; for the more porous pieces of bone separate more quickly, but the more solid come away more slowly; but the other smaller splinters in much less time, and others otherwise. A portion of bone which protrudes should be sawn off for the following

reasons: if it cannot be reduced, and if it appears that only a small piece is required in order that it may get back into its place; and if it be such that it can be taken out, and if it occasions inconvenience and irritates any part of the flesh, and prevents the limb from being properly laid, and if, moreover, it be denuded of flesh, such a piece of bone should be taken off. With regard to the others, it is not of much consequence whether they be sawed off or not. For it should be known for certain, that such bones as are completely deprived of flesh, and have become dried, all separate completely. Those which are about to exfoliate should not be sawn off. Those that will separate completely must be judged of from the symptoms that have been laid down.

Fract. 35

Such cases are to be treated with compresses and vinous applications, as formerly laid down regarding bones which will separate. We must avoid wetting it at the beginning with anything cold; for there is danger of febrile rigors, and also of convulsions; for convulsions are induced by cold things, and also sometimes by wounds. It is proper to know that the members are necessarily shortened in those cases in which the bones have been broken, and have healed the one across the other, and in those cases in which the whole circle of the bone has become detached.

Fract. 36

Those cases in which the bone of the thigh, or of the arm, protrudes, do not easily recover. For the bones are large, and contain much marrow; and many important nerves, muscles, and veins are wounded at the same time. And if you reduce them, convulsions usually supervene; and, if not reduced, acute bilious fevers come on, with singultus and mortification. The chances of recovery are not fewer in those cases in which the parts have not been reduced, nor any attempts made at reduction. Still more recover in those cases in which the lower, than those in which the upper part of the bone protrudes; and some will recover when reduction has been made, but very rarely indeed. For modes of treatment and peculiarity of constitution make a

great difference as to the capability of enduring such an injury. And it makes a great difference if the bones of the arm and of the thigh protrude to the inside; for there are many and important vessels situated there, some of which, if wounded, will prove fatal; there are such also on the outside, but of less importance. In wounds of this sort, then, one ought not to be ignorant of the dangers, and should prognosticate them in due time. But if you are compelled to have recourse to reduction, and hope to succeed, and if the bones do not cross one another much, and if the muscles are not contracted (for they usually are contracted), the lever in such cases may be advantageously employed.

Fract. 37

Having effected the reduction, you must give an emollient draught of hellebore the same day, provided it has been reduced on the day of the accident, but otherwise it should not be attempted. The wound should be treated with the same things as are used in fractures of the bones of the head, and nothing cold should be applied; the patient should be restricted from food altogether, and if naturally of a bilious constitution, he should have for a diet a little fragrant *oxyglyky* sprinkled on water; but if he is not bilious, he should have water for drink; and if fever of the continual type come on, he is to be confined to this regimen for fourteen days at least, but if he be free of fever, for only seven days, and then you must bring him back by degrees to a common diet. To those cases in which the bones have not been reduced, a similar course of medicine should be administered, along with the same treatment of the sores and regimen; and in like manner the suspended part of the body should not be stretched, but should rather be contracted, so as to relax the parts about the wound. The separation of the bones is protracted, as also was formerly stated. But one should try to escape from such cases, provided one can do so honourably, for the hopes of recovery are small, and the dangers many; and if the physician do not reduce the fractured bones he will

be looked upon as unskillful, while by reducing them he will bring the patient nearer to death than to recovery.

Fract. 38

Luxations and subluxations at the knee are much milder accidents than subluxations and luxations at the elbow. For the knee-joint, in proportion to its size, is more compact than that of the arm, and has a more even conformation, and is rounded, while the joint of the arm is large, and has many cavities. And in addition, the bones of the leg are nearly of the same length, for the external one overtops the other to so small an extent as hardly to deserve being mentioned, and therefore affords no great resistance, although the external nerve (*ligament?*) at the ham arises from it; but the bones of the fore-arm are unequal, and the shorter is considerably thicker than the other, and the more slender (*ulna?*) protrudes, and passes up above the joint, and to it (the *olecranon?*) are attached the nerves (*ligaments?*) which go downward to the junction of the bones; and the slender bone (*ulna?*) has more to do with the insertion of the ligaments in the arm than the thick bone (*radius?*). The configuration then of the articulations, and of the bones of the elbow, is such as I have described. Owing to their configuration, the bones at the knee are indeed frequently dislocated, but they are easily reduced, for no great inflammation follows, nor any constriction of the joint. They are displaced for the most part to the inside, sometimes to the outside, and occasionally into the ham. The reduction in all these cases is not difficult, but in the dislocations inward and outward, the patient should be placed on a low seat, and the thigh should be elevated, but not much. Moderate extension for the most part sufficeth, extension being made at the leg, and counter-extension at the thigh.

Fract. 39

Dislocations at the elbow are more troublesome than those at the knee, and, owing to the inflammation which comes on, and the configuration of the joint, are more difficult to reduce if the bones are not immediately replaced. For the bones at the elbow are less subject

to dislocation than those of the knee, but are more difficult to reduce and keep in their position, and are more apt to become inflamed and ankylosed.

Fract. 40

For the most part the displacements of these bones are small, sometimes toward the ribs, and sometimes to the outside; and the whole articulation is not displaced, but that part of the humerus remains in place which is articulated with the cavity of the bone of the forearm that has a protuberance (*ulna?*). Such dislocations, to whatever side, are easily reduced, and the extension is to be made in the line of the arm, one person making extension at the wrist, and another grasping the armpit, while a third, applying the palm of his hand to the part of the joint which is displaced, pushes it inward, and at the same time makes counterpressure on the opposite side near the joint with the other hand.

The end of the humerus at the elbow gets displaced (*subluxated?*) by leaving the cavity of the ulna. Such luxations readily yield to reduction, if applied before the parts get inflamed. The displacement for the most part is to the inside, but sometimes to the outside, and they are readily recognized by the shape of the limb. And often such luxations are reduced without any powerful extension. In dislocations inward, the joint is to be pushed into its place, while the fore-arm is brought round to a state of pronation. Such are most of the dislocations at the elbow.

Fract. 41

But if the articular extremity of the humerus be carried to either side above the bone of the fore-arm, which is prominent, into the hollow of the arm (?), this rarely happens; but if it does happen, extension in the straight line is not so proper under such circumstances; for in such a mode of extension, the process of the ulna (*olecranon?*) prevents the bone of the arm (*humerus?*) from passing over it. In dislocations of this kind, extension should be made in the manner described when treating of the bandaging of fractured bones of the arm, extension

being made upward at the armpit, while the parts at the elbow are pushed downward, for in this manner can the humerus be most readily raised above its cavity; and when so raised, the reduction is easy with the palms of the hand, the one being applied so as to make pressure on the protuberant part of the arm, and the other making counter-pressure, so as to push the bone of the fore-arm into the joint. This method answers with both cases. And perhaps this is the most suitable mode of reduction in such a case of dislocation. The parts may be reduced by extension in a straight line, but less readily than thus.

Fract. 42

If the arm be dislocated forward—this rarely happens, indeed, but what would a sudden shock not displace? for many other things are removed from their proper place, notwithstanding a great obstacle,—in such a violent displacement the part (*olecranon?*) which passes above the prominent part of the bones is large, and the stretching of the nerves (*ligaments?*) is intense; and yet the parts have been so dislocated in certain cases. The following is the symptom of such a displacement: the arm cannot be bent in the least degree at the elbow, and upon feeling the joint the nature of the accident becomes obvious. If, then, it is not speedily reduced, strong and violent inflammation, attended with fever, will come on, but if one happen to be on the spot at the time it is easily reduced. A piece of hard linen cloth (or a piece of hard linen, not very large, rolled up in a ball, will be sufficient) is to be placed across the bend of the elbow, and the arm is then to be suddenly bent at the elbow, and the hand brought up to the shoulder. This mode of reduction is sufficient in such displacements; and extension in the straight line can rectify this manner of dislocation, but we must use at the same time the palms of the hands, applying the one to the projecting part of the humerus at the bend of the arm for the purpose of pushing it back, and applying the other below to the sharp extremity of the elbow, to make counter-pressure, and incline the parts into the straight line. And one may use with advantage in

this form of dislocation the method of extension formerly described, for the application of the bandages in the case of fracture of the arm; but when extension is made, the parts are to be adjusted, as has been also described above.

Fract. 43

But if the arm be dislocated backward (but this very rarely happens, and it is the most painful of all, and the most subject to bilious fevers of the continual type, which prove fatal in the course of a few days), in such a case the patient cannot extend the arm. If you are quickly present, by forcible extension the parts may return to their place of their own accord; but if fever have previously come on, you must no longer attempt reduction, for the pain will be rendered more intense by any such violent attempt. In a word, no joint whatever should be reduced during the prevalence of fever, and least of all the elbow-joint.

Fract. 44

There are also other troublesome injuries connected with the elbow-joint; for example, the thicker bone (*radius?*) is sometime partially displaced from the other, and the patient can neither perform extension nor flexion properly. This accident becomes obvious upon examination with the hand at the bend of the arm near the division of the vein that runs up the muscle. In such a case it is not easy to reduce the parts to their natural state, nor is it easy, in the separation of any two bones united by symphysis, to restore them to their natural state, for there will necessarily be a swelling at the seat of the diastasis. The method of bandaging a joint has been already described in treating of the application of bandages to the ankle.

Fract. 45

In certain cases the process of the ulna (*olecranon?*) behind the humerus is broken; sometimes its cartilaginous part, which gives origin to the posterior tendon of the arm, and sometimes its fore part, at the base of the anterior coronoid process; and when this displacement takes place, it is apt to be attended with malignant fever.

The joint, however, remains in place, for its whole base protrudes at that point. But when the displacement takes place where its head overtops the arm, the joint becomes looser if the bone be fairly broken across. To speak in general terms, all cases of fractured bones are less dangerous than those in which the bones are not broken, but the veins and important nerves (*tendons*?) situated in these places are contused; for the risk of death is more immediate in the latter class of cases than in the former, if continual fever come on. But fractures of this nature seldom occur.

Fract. 46

It sometimes happens that the head of the humerus is fractured at its epiphysis; and this, although it may appear to be a much more troublesome accident, is in fact a much milder one than the other injuries at the joint.

Fract. 47

The treatment especially befitting each particular dislocation has been described; and it has been laid down as a rule, that immediate reduction is of the utmost advantage, owing to the rapid manner in which inflammation of the tendons supervenes. For even when the luxated parts are immediately reduced, the tendons usually become stiffened, and for a considerable time prevent extension and flexion from being performed to the ordinary extent. All these cases are to be treated in a similar way, whether the extremity of the articulating bone be snapped off, whether the bones be separated, or whether they be dislocated; for they are all to be treated with plenty of bandages, compresses, and cerate, like other fractures. The position of the joint in all these cases should be the same, as when a fractured arm or forearm has been bound up. For this is the most common position in all dislocations, displacements, and fractures; and it is the most convenient for the subsequent movements, whether of extension or flexion, as being the intermediate stage between both. And this is the position in which the patient can most conveniently carry or suspend his arm in a sling. And besides, if the joint is to be stiffened by callus,

it were better that this should not take place when the arm is extended, for this position will be a great impediment and little advantage; if the arm be wholly bent, it will be more useful; but it will be much more convenient to have the joint in the intermediate position when it becomes ankylosed. So much with regard to position.

Fract. 48

In bandaging, the head of the first bandage should be placed at the seat of the injury, whether it be a case of fracture, of dislocation, or of diastasis (*separation?*), and the first turns should be made there, and the bandages should be applied most firmly at that place, and less so on either side. The bandaging should comprehend both the arm and the fore-arm, and on both should be to a much greater extent than most physicians apply it, so that the swelling may be expelled from the seat of the injury to either side. And point of the fore-arm should be comprehended in the bandaging, whether the injury be in that place or not, in order that the swelling may not collect there. In applying bandages, we must avoid as much as possible accumulating many turns of the bandage at the bend of the arm. For the principal compression should be at the seat of the injury, and the same rules are to be observed, and at the same periods, with regard to compression and relaxation, as formerly described respecting the treatment of broken bones; and the bandages should be renewed every third day; and they should appear loose on the third day, as in the other case. And splints should be applied at the proper time (for there is nothing unsuitable in them, whether the bones be fractured or not, provided there is no fever); they should be particularly loose, whether applied to the arm or the forearm, but they must not be thick. It is necessary that they should be of unequal size, and that the one should ride over the other, whenever from the flexion it is judged proper. And the application of the compresses should be regulated in the same manner as has been stated with regard to the splints; and they should be put on in a somewhat more bulky form at the seat of the injury. The

periods are to be estimated from the inflammation, and from what has been written on them above.

On the Articulations

Artic. 1

I am acquainted with one form in which the shoulder-joint is dislocated, namely, that into the armpit; I have never seen it take place upward nor outward; and yet I do not positively affirm whether it might be dislocated in these directions or not, although I have something which I might say on this subject. But neither have I ever seen what I considered to be a dislocation forward. Physicians, indeed, fancy that dislocation is very apt to occur forward, and they are more particularly deceived in those persons who have the fleshy parts about the joint and arm much emaciated; for, in all such cases, the head of the arm appears to protrude forward. And I in one case of this kind having said that there was no dislocation, exposed myself to censure from certain physicians and common people on that account, for they fancied that I alone was ignorant of what everybody else was acquainted with, and I could not convince them but with difficulty, that the matter was so. But if one will strip the point of the shoulder of the fleshy parts, and where the muscle (*deltoid?*) extends, and also lay bare the tendon that goes from the armpit and clavicle to the breast (*pectoral muscle?*), the head of the humerus will appear to protrude strongly forward, although not dislocated, for the head of the humerus naturally inclines forward, but the rest of the bone is turned outward. The humerus is connected obliquely with the cavity of the scapula, when the arm is stretched along the sides; but when the whole arm is stretched forward, then the head of the humerus is in a line with the cavity of the humerus, and no longer appears to protrude forward. And with regard to the variety we are now treating of, I have never seen a case of dislocation forward; and yet I do not speak decidedly respecting it, whether such a dislocation may take

place or not. When, then, a dislocation into the armpit takes place, seeing it is of frequent occurrence, many persons know how to reduce it, for it is an easy thing to teach all the methods by which physicians effect the reductions, and the best manner of applying them. The strongest of those methods should be used when the difficulty of reduction is particularly great. The strongest is the method to be last described.

Artic. 2

Those who are subject to frequent dislocations at the shoulder-joint, are for the most part competent to effect the reduction themselves; for, having introduced the knuckles of the other hand into the armpit, they force the joint upward, and bring the elbow toward the breast. The physician might reduce it in the same manner, if having introduced his fingers into the armpit on the inside of the dislocated joint, he would force it from the ribs, pushing his own head against the acromion, in order to make counter-pressure, and with his knees applied to the patient's elbow pushing the arm to the sides. It will be of advantage if the operator has strong hands, or the physician may do as directed with his head and hands, while another person brings the elbow toward the breast. Reduction of the shoulder may also be effected by carrying the fore-arm backward to the spine, and then with the one hand grasping it at the elbow, to bend the arm upward, and with the other to support it behind at the articulation. This mode of reduction, and the one formerly described, are not natural, and yet by rotating the bone of the joint, they force it to return.

Artic. 3

Those who attempt to perform reduction with the heel, operate in a manner which is an approach to the natural. The patient must lie on the ground upon his back, while the person who is to effect the reduction is seated on the ground upon the side of the dislocation; then the operator, seizing with his hand the affected arm, is to pull it, while with his heel in the armpit he pushes in the contrary direction, the right heel being placed in the right armpit, and the left heel in the

left armpit. But a round ball of a suitable size must be placed in the hollow of the armpit; the most convenient are very small and hard balls, formed from several pieces of leather sewed together. For without something of the kind the heel cannot reach to the head of the humerus, since, when the arm is stretched, the armpit becomes hollow, the tendons on both sides of the armpit making counter-contraction so as to oppose the reduction. But another person should be seated on the other side of the patient to hold the sound shoulder, so that the body may not be dragged along when the arm of the affected side is pulled; and then, when the ball is placed in the armpit, a supple piece of thong sufficiently broad is to be placed round it, and some person taking hold of its two ends is to seat himself above the patient's head to make counter-extension, while at the same time he pushes with his foot against the bone at the top of the shoulder. The ball should be placed as much on the inside as possible, upon the ribs, and not upon the head of the humerus.

Artic. 4

There is another method of reduction performed by the shoulder of a person standing. The person operating in this way, who should be taller than the patient, is to take hold of his arm and place the sharp point of his own shoulder in the patient's armpit, and push it in so that it may lodge there, and having for his object that the patient may be suspended at his back by the armpit, he must raise himself higher on this shoulder than the other; and he must bring the arm of the suspended patient as quickly as possible to his own breast. In this position he should shake the patient when he raises him up, in order that the rest of the body may be a counterpoise to the arm which is thus held. But if the patient be very light, a light child should be suspended behind along with him. These methods of reduction are all of easy application in the palestra, as they can all be performed without instruments, but they may also be used elsewhere.

Artic. 5

Those who accomplish the reduction by forcibly bending it round a pestle, operate in a manner which is nearly natural. But the pestle should be wrapped in a soft shawl (for thus it will be less slippery), and it should be forced between the ribs and the head of the humerus. And if the pestle be short, the patient should be seated upon something, that his arm can with difficulty pass above the pestle. But for the most part the pestle should be longer, so that the patient when standing may be almost suspended upon the piece of wood. And then the arm and forearm should be stretched along the pestle, whilst some person secures the opposite side of the body by throwing his arms round the neck, near the clavicle.

Artic. 6

But the method with a ladder is another of the same kind, and still better, since by it the body can be more safely counterpoised on this side; and that, while in the method which the piece of wood resembling a pestle, there is danger of the body tumbling to either side. But some round thing should be tied upon the step of the ladder which may be fitted to the armpit, whereby the head of the bone may be forced into its natural place.

Artic. 7

The following, however, is the strongest of all the methods of reduction. We must get a piece of wood, five, or at least four inches broad, two inches in thickness, or still thinner, and two cubits in length, or a little less; and its extremity at one end should be rounded, and made very narrow and very slender there, and it should have a slightly projecting edge (*ambe*) on its round extremity, not on the part that is to be applied to the side, but to the head of the humerus, so that it may be adjusted in the armpit at the sides under the head of the humerus; and a piece of soft shawl or cloth should be glued to the end of the piece of wood, so as to give the less pain upon pressure. Then having pushed the head of this piece of wood as far inward as possible between the ribs and the head of the humerus, the whole arm is to be stretched along this piece of wood, and is to be bound round at

the arm, the fore-arm, and the wrist, so that it may be particularly well secured; but great pains should be taken that the extremity of this piece of wood should be introduced as far as possible into the armpit, and that it is carried past the head of the humerus. Then a cross-beam is to be securely fastened between two pillars, and afterward the arm with the piece of wood attached to it is to be brought over this cross-beam, so that the arm may be on the one side of it and the body on the other, and the cross-beam in the armpit; and then the arm with the piece of wood is to be forced down on the one side of the cross-beam, and the rest of the body on the other. The cross-beam is to be bound so high that the rest of the body may be raised upon tip-toes. This is by far the most powerful method of effecting reduction of the shoulder; for one thus operates with the lever upon the most correct principles, provided only the piece of wood be placed as much as possible within the head of the humerus, and thus also the counter-balancing weights will be most properly adjusted, and safely applied to the bone of the arm. Wherefore recent cases in this way may be reduced more quickly than could be believed, before even extension would appear to be applied; and this is the only mode of reduction capable of replacing old dislocations, and this it will effect, unless flesh has already filled up the (glenoid) cavity, and the head of the humerus has formed a socket for itself in the place to which it has been displaced; and even in such an old case of dislocation, it appears to me that we could effect reduction (for what object would a lever power properly applied not it move?), but it would not remain in its place, but would be again displaced as formerly. The same thing may be effected by means of the ladder, by preparing it in the same manner. If the dislocation be recent, a large Thessalian chair may be sufficient to accomplish this purpose; the wood, however, should be dressed up as described before; but the patient should be seated sideways on the chair, and then the arm, with the piece of wood attached to it, is to be brought over the back of the chair, and force is to be applied to the arm, with the wood on the one

side, and the body on the other side. The same means may be applied with a double door. One should always use what happens to be at hand.

Artic. 8

Wherefore it should be known that one constitution differs much from another as to the facility with which dislocations in them may be reduced, and one articular cavity differs much from another, the one being so constructed that the bone readily leaps out and another less so; but the greatest difference regards the binding together of the parts by the nerves (*ligaments?*) which are slack in some and tight in others. For the humidity in the joints of men is connected with the state of the ligaments, when they are slack and yielding; for you may see many people who are so humid (*flabby?*) that when they choose they can disarticulate their joints without pain, and reduce them in like manner. The habit of the body also occasions a certain difference, for in those who are in a state of embonpoint and fleshy the joint is rarely dislocated, but is more difficult to reduce; but when they are more attenuated and leaner than usual, then they are subject to dislocations which are more easily reduced. And the following observation is a proof that matters are so; for in cattle the thighs are most apt to be dislocated at the hip-joint, when they are most particularly lean, which they are at the end of winter, at which time then they are particularly subject to dislocations (if I may be allowed to make such an observation while treating of a medical subject); and therefore Homer has well remarked, that of all beasts oxen suffer the most at that season, and especially those employed at the plow as being worked in the winter season. In them, therefore, dislocations happen most frequently, as being at that time most particularly reduced in flesh. And other cattle can crop the grass when it is short, but the ox cannot do so until it becomes long; for, in the others, the projection of the lip is slender, and so is the upper lip, but in the ox the projection of the lip is thick, and the upper jaw is thick and obtuse, and therefore they are incapable of seizing short herbs. But the

solidungula as having prominent teeth in both their front jaws, can crop the grass and grasp it with their teeth while short, and delight more in short grass than in rank; for, in general, short grass is better and more substantial than rank, as having not yet given out its fructification. Wherefore the poet has the following line: “As when to horned cattle dear the vernal season comes,”⁶ because rank grass appears to be most sought after by them. But otherwise in the ox, this joint is slacker than in other animals, and, therefore, this animal drags his foot in walking more than any other, and especially when lank and old. For all these reasons the ox is most particularly subject to dislocations; and I have made the more observations respecting him, as they confirm all that was said before on this subject. With regard, then, to the matter on hand, I say that dislocations occur more readily, and are more speedily reduced in those who are lean than in those who are fleshy; and in those who are humid and lank there is less inflammation than in such as are dry and fleshy, and they are less compactly knit hereafter, and there is more mucosity than usual in cases not attended with inflammation, and hence the joints are more liable to luxations; for, in the main, the articulations are more subject to mucosities in those who are lean than in those who are fleshy; and the flesh of lean persons who have not been reduced by a proper course of discipline abounds more with mucosity than that of fat persons. But in those cases in which the mucosity is accompanied with inflammation, the inflammation binds (*braces?*) the joint, and hence those who have small collections of mucosities are not very subject to dislocations, which they would be if the mucosity had not been accompanied with more or less inflammation.

Artic. 9

In cases of dislocation those persons who are not attacked with inflammation of the surrounding parts, can use the shoulder immediately without pain, and do not think it necessary to take any precautions with themselves; it is therefore the business of the physician to warn them beforehand that dislocation is more likely to

return in such cases than when the tendons have been inflamed. This remark applies to all the articulations, but particularly to those of the shoulder and knee, for these are the joints most subject to luxations. But those who have inflammation of the ligaments cannot use the shoulder, for the pain and the tension induced by the inflammation prevent them. Such cases are to be treated with cerate, compresses, and plenty of bandages; but a ball of soft clean wool is to be introduced into the armpit, to fill up the hollow of it, that it may be a support to the bandaging, and maintain the joint *in situ*. The arm, in general, should be inclined upward as much as possible, for thus it will be kept at the greatest possible distance from the place at which the head of the humerus escaped. And when you bandage the shoulder you must fasten the arms to the sides with a band, which is to be carried round the body. The shoulder should be rubbed gently and softly. The physician ought to be acquainted with many things, and among others with friction; for from the same name the same results are not always obtained; for friction could brace a joint when unseasonably relaxed, and relax it when unseasonably hard; but we will define what we know respecting friction in another place. The shoulder, then, in such a state, should be rubbed with soft hands; and, moreover, in a gentle manner, and the joint should be moved about, but not roughly, so as to excite pain. Things get restored sometimes in a greater space of time, and sometimes in a smaller.

Artic. 10

A dislocation may be recognized by the following symptoms:—Since the parts of a man's body are proportionate to one another, as the arms and the legs, the sound should always be compared with the unsound, and the unsound with the sound, not paying regard to the joints of other individuals (for one person's joints are more prominent than another's), but looking to those of the patient, to ascertain whether the sound joint be unlike the unsound. This is a proper rule, and yet it may lead to much error; and on this account it is not sufficient to know this art in theory, but also by actual practice; for

many persons from pain, or from any other cause, when their joints are not dislocated, cannot put the parts into the same positions as the sound body can be put into; one ought therefore to know and be acquainted beforehand with such an attitude. But in a dislocated joint the head of the humerus appears lying much more in the armpit than it is in the sound joint; and also, above, at the top of the shoulder, the part appears hollow, and the acromion is prominent, owing to the bone of the joint having sunk into the part below; there is a source of error in this case also, as will be described afterward, for it deserves to be described; and also, the elbow of the dislocated arm is farther removed from the ribs than that of the other; but by using force it may be approximated, though with considerable pain; and also they cannot, with the elbow extended, raise the arm to the ear, as they can the sound arm, nor move it about as formerly in this direction and that. These, then, are the symptoms of dislocation at the shoulder. The methods of reduction and the treatment are as described.

Artic. 11

It deserves to be known how a shoulder which is subject to frequent dislocations should be treated. For many persons owing to this accident have been obliged to abandon gymnastic exercises, though otherwise well qualified for them; and from the same misfortune have become inept in warlike practices, and have thus perished. And this subject deserves to be noticed, because I have never known any physician treat the case properly; some abandon the attempt altogether, and others hold opinions and practice the very what is proper. For physicians have burned the shoulders subject to dislocation, at the top of the shoulder, at the anterior part where the head of the humerus protrudes, and a little behind the top of the shoulder; these burnings, if the dislocation of the arm were upward, or forward, or backward, would have been properly performed; but now, when the dislocation is downward, they rather promote than prevent dislocations, for they shut out the head of the humerus from the free space above. The cautery should be applied thus: taking hold

with the hands of the skin at the armpit, it is to be drawn into the line, in which the head of the humerus is dislocated; and then the skin thus drawn aside is to be burnt to the opposite side. The burnings should be performed with irons, which are not thick nor much rounded, but of an oblong form (for thus they pass the more readily through), and they are to be pushed forward with the hand; the cauteries should be red-hot, that they may pass through as quickly as possible; for such as are thick pass through slowly, and occasion eschars of a greater breadth than convenient, and there is danger that the cicatrices may break into one another; which, although nothing very bad, is most unseemly, or awkward. When you have burnt through, it will be sufficient, in most cases, to make eschars only in the lower part; but if there is no danger of the ulcers passing into one another, and there is a considerable piece of skin between them, a thin spatula is to be pushed through these holes which have been burned, while, at the same time, the skin is stretched, for otherwise the instrument could not pass through; but when you have passed it through you must let go the skin, and then between the two eschars you should form another eschar with a slender iron, and burn through until you come in contact with the spatula. The following directions enable you to determine how much of the skin of the armpit should be grasped; all men have glands in the armpit greater or smaller, and also in many other parts of the body. But I will treat in another work of the whole constitution of the glands, and explain what they are, what they signify, and what are their offices. The glands, then, are not to be taken hold of, nor the parts internal to the glands; for this would be attended with great danger, as they are adjacent to the most important nerves. But the greater part of the substances external to the glands are to be grasped, for there is no danger from them. And this, also, it is proper to know, that if you raise the arm much, you will not be able to grasp any quantity of skin worth mentioning, for it is all taken up with the stretching; and also the nerves, which by all means you must avoid wounding, become exposed and stretched in this

position; but if you only raise the arm a little, you can grasp a large quantity of skin, and the nerves which you ought to guard against are left within, and at a distance from the operation. Should not, then, the utmost pains be taken in the whole practice of the art to find out the proper attitude in every case? So much regarding the armpit, and these contractions will be sufficient, provided the eschars be properly placed. Without the armpit there are only two places where one might place the eschars to obviate this affection; the one before and between the head of the humerus and the tendon at the armpit; and then the skin may be fairly burned through, but not to any great depth, for there is a large vein adjacent, and also nerves, neither of which must be touched with the heat. But externally, one may form another eschar considerably above the tendon at the armpit, but a little below the head of the humerus; and the skin must be burned fairly through, but it must not be made very deep, for fire is inimical to the nerves. Through the whole treatment the sores are to be so treated, as to avoid all strong extension of the arm, and this is to be done moderately, and only as far as the dressing requires; for thus they will be less cooled (for it is of importance to cover up all sorts of burns if one would treat them mildly), and then the lips of them will be less turned aside; there will be less hemorrhage and fear of convulsions. But when the sores have become clean, and are going on to cicatrization, then by all means the arm is to be bound to the side night and day; and even when the ulcers are completely healed, the arm must still be bound to the side for a long time; for thus more especially will cicatrization take place, and the wide space into which the humerus used to escape will become contracted.

Artic. 12

When attempts to reduce a dislocated shoulder have failed, if the patient be still growing, the bone of the affected arm will not increase like the sound one, for although it does increase in so far it becomes shorter than the other; and those persons called *weasel-armed*, become so from two accidents, either from having met with this dislocation in

utero, or from another accident, which will be described afterward. But those who while they were children have had deep-seated suppurations about the head of the bone, all become weasel-armed; and this, it should be well known, will be the issue, whether the abscess be opened by an incision or cautery, or whether it break spontaneously. Those who are thus affected from birth are quite able to use the arm yet neither can they raise the arm to the ear, by extending the elbow, but they do this much less efficiently than with the sound arm. But in those who have had the shoulder dislocated after they were grown up, and when it has not been reduced, the top of the shoulder becomes much less fleshy, and the habit of body at that part is attenuated; but when they cease to have pain, whatever they attempt to perform by raising the elbow from the sides obliquely, they can no longer accomplish as formerly; but whatever acts are performed by carrying the arm around by the sides, either backward or forward, all those they can perform; for they can work with an auger or a saw, or with a hatchet, and can dig, by not raising the elbow too much, and do all other kinds of work which are done in similar attitudes.

Artic. 13

In those cases where the acromion has been torn off, the bone which is thus separated appears prominent. The bone is the bond of connection between the clavicle and scapula, for in this respect the constitution of man is different from that of other animals; physicians are particularly liable to be deceived in this accident (for as the separated bone protrudes, the top of the shoulder appears low and hollow), so that they make preparations as if for dislocation of the shoulder; for I have known many physicians, otherwise not inexpert at the art, who have done much mischief by attempting to reduce such shoulders, thus supposing it a case of dislocation; and they did not desist until they gave over mistake of supposing that they had reduced the shoulder. The treatment, in these cases, is similar to that which is applicable in others of a like kind, namely, cerate, compresses, and

suitable bandaging with linen cloths. The projecting part must be pushed down, and the greater number of compresses are to be placed on it, and most compression is to be applied at that part, and the arm being fastened to the side is to be kept elevated; for thus the parts which had been torn asunder are brought into closest proximity with one another. All this should be well known, and if you choose you may prognosticate safely that no impediment, small or great, will result from such an injury at the shoulder, only there will be a deformity in the place, for the bone cannot be properly restored to its natural situation, but there must necessarily be more or less tumefaction in the upper part. For neither can any other bone be made exactly as it was, which having become incorporated with another bone, and having grown to it as an apophysis, has been torn from its natural situation. If properly bandaged, the acromion becomes free of pain in a few days.

Artic. 14

When a fractured clavicle is fairly broken across it is more easily treated, but when broken obliquely it is more difficult to manage. Matters are different in these cases from what one would have supposed; for a bone fairly broken across can be more easily restored to its natural state, and with proper care the upper part may be brought down by means of suitable position and proper bandaging, and even if not properly set, the projecting part of the bone is not very sharp. But in oblique fractures the case is similar to that of bones which have been torn away, as formerly described; for they do not admit of being restored to their place, and the prominence of the bone is very sharp. For the most part, then, it should be known, no harm results to the shoulder or to the rest of the body from fracture of the clavicle, unless it sphacelate, and this rarely happens. A deformity, however, may arise from fracture of the clavicle, and in these cases it is very great at first, but by and by it becomes less. A fractured clavicle, like all other spongy bones, gets speedily united; for all such bones form callus in a short time. When, then, a fracture has recently

taken place, the patients attach much importance to it, as supposing the mischief greater than it really is, and the physicians bestow great pains in order that it may be properly bandaged; but in a little time the patients, having no pain, nor finding any impediment to their walking or eating, become negligent; and the physicians finding they cannot make the parts look well, take themselves off, and are not sorry at the neglect of the patients, and in the meantime the callus is quickly formed.⁷ The method of dressing which is most appropriate, is similar to that used in ordinary cases, consisting of cerate, compresses, and bandages; and it should be most especially known in this operation, that most compresses should be placed on the projecting bone, and that the greatest pressure should be made there. There are certain physicians who make a show of superior skill by binding a heavy piece of lead on the part in order to depress the projecting bone; but this mode of treatment does not apply to the clavicle, for it is impossible to depress the projecting part to any extent worth mentioning. There are others who, knowing the fact that the bandages are apt to slip off, and that they do not keep the projecting parts in their place, apply compresses and bandages like the others, and then having girt the patient with a girdle, where it is usually applied with most effect, they make a heap of the compresses upon the projecting bone when they apply them, and having fastened the head of the bandage to the girdle in front, they apply it so as to bring the turns of it into the line of the clavicle, carrying them to the back, and then bringing them around the girdle they carry them to the fore part and again backward. There are others who do not apply the bandage round the girdle, but carry the rounds of it by the perineum and anus, and along the spine, so as to compress the fracture. To an inexperienced person these methods will appear not far from natural, but when tried, they will be found of no service; for they do not remain firm any length of time, even if the patient keep his bed, although in this position they answer best; and yet even when lying in bed, should he bend his leg, or should his trunk be bent, all

the will be displaced; and, moreover, the bandaging is inconvenient, in as much as the anus is comprehended by it, and many turns of the bandage are crowded there in a narrow space. And in the method with the girdle, the girdle cannot be so firmly girt around, but that the turns of the bandage force the girdle to ascend, and hence of necessity all the other bandages must be slackened. He would seem to me to come nearest his purpose, although after all he effects but little, who would take a few turns round the girdle, but would use the bandage principally to secure the former bandaging; for in this manner the bandages would be most secure, and would mutually assist one another. Every thing now almost has been said which applies to fracture of the clavicle. But this also should be known, that in fractures of the clavicle, it is the part attached to the breast which is uppermost, and that the piece attached to the acromion is the lowermost. The cause of this is, that for the most part the breast can neither be depressed nor raised, there being but a slight movement of the joint at the breast, for the sternum is connected together on both sides with the spine. The clavicle admits of most motion at the joint of the shoulder, and this arises from its connection with the acromion. And, moreover, when broken, the part which is connected with the sternum flies upward, and is not easily forced downward; for it is naturally light, and there is more room for it above than below. But the shoulder, the arm, and the parts connected with them, are easily moved from the sides and breast, and, on that account, they admit of being considerably elevated and depressed. When, therefore, the clavicle is broken, the fragment attached to the shoulder inclines downward, for it inclines much more readily with the shoulder and arm downward than upward. Matters being as I have stated, they act imprudently who think to depress the projecting end of the bone. But it is clear that the under part ought to be brought to the upper, for the former is the movable part, and that which has been displaced from its natural position. It is obvious, therefore, that there is no other way of applying force to it (for the bandages no more force it to than they

force it from); but if one will push the arm when at the sides as much as possible upward, so that the shoulder may appear as sharp as possible, it is clear that in this way it will be adjusted to the fragment of the bone connected with the breast from which it was torn. If one then will apply a bandage, *secundum artem*, for the purpose of promoting a speedy cure, and will reckon everything else of no value, except the position as described, he will form a correct opinion of the case, and will effect a cure in the speediest and most appropriate manner. It is of great importance, however, that the patient should lie in a recumbent posture. Fourteen days will be sufficient if he keep quiet, and twenty at most.

Artic. 15

But if the clavicle be fractured in the opposite manner (which does not readily happen), so that the fragment of bone connected with the breast is depressed, while the piece connected with the acromion is raised up and rides over other, this case does not require much management, for if the shoulder and arm be let go, the fragments of the bone will be adjusted to one another, and an ordinary bandage will suffice, and the callus will be formed in the course of a few days.

Artic. 16

If the fracture be not thus, but if it incline either forward or backward, it may be restored to its natural position, by raising the shoulder with the arm as formerly described, and brought back to its natural place, when the cure will be speedily accomplished. Most of the varieties of displacement may be rectified by raising the arm upward. When the upper bone is displaced laterally or downward, it would favor the adaptation of the parts if the patient would lie on his back, and if some elevated substance were placed between the shoulder-blades, so that the breast may be depressed as much as possible upon the two sides; and if, while another person raised the arm extended along the sides, the physician, applying the palm of the one hand to the head of the bone, would push it away, and with the other would adjust the broken bones, he would thus reduce the parts most readily to their natural

position. But, as formerly stated, the upper bone (*sternal fragment?*) is rarely depressed downward. In most cases, after the bandages have been applied, that position is beneficial in which the elbow is fixed to the same side, and the shoulder is kept elevated; but in certain cases, the shoulder is to be raised, as has been directed, and the elbow is to be brought forward to the breast, and the hand laid on the acromion of the sound side. If the patient has the resolution to lie in bed, something should be placed so as to support the shoulder, and keep it as much elevated as possible. But if he walk about, the arm should be slung in a shawl, which embraces the point of the elbow, and is passed round the neck.

Artic. 17

When the elbow-joint is displaced or dislocated to the side or outward, while its sharp point (*olecranon?*) remains in the cavity of the humerus, extension is to be made in a straight line, and the projecting part is to be pushed backward and to the side.

Artic. 18

In complete dislocations toward either side, extension is to be made as in bandaging fracture of the arm; for thus the rounded part of the elbow will not form an obstacle to it. Dislocation, for the most part, takes place toward the sides (*inwardly?*). Reduction is to be effected by separating (the bones) as much as possible, so that the end (of the humerus) may not come in contact with the olecranon, and it is to be carried up, and turned round, and not forced in a straight line, and, at the same time, the opposite sides are to be pushed together, and propelled into their proper place. It will further assist if rotation of the fore-arm be made at the elbow, sometimes turning it into a supine position, and sometimes into a prone. The position for the treatment consists in keeping the hand a little higher than the elbow, and the arm at the sides; then it may either be suspended or laid at rest, for either position will answer; and nature and the usage of common means will accomplish the cure, if the callus does not form improperly: it is formed quickly. The treatment is to be conducted

with bandages according to the rule for bandaging articulations, and the point of the elbow is to be included in the bandage.

Artic. 19

Dislocations at the elbow give rise to the most serious consequences, such as fevers, pain, nausea, vomitings of pure bile, and more especially when the humerus is displaced backward from pressure on the nerve, which occasions numbness; next to it is the dislocation forward; the treatment is the same; reduction in dislocation backward is by extension and adaptation; the symptom of this variety-loss of the power of extension; of dislocation forward-loss of the power of flexion, and in this case reduction is to be accomplished by placing a hard ball (in the bend of the elbow), and bending the fore-arm about it, along with sudden extension.

Artic. 20

Diastasis of the bones may be recognized by examining the part where the vein that runs along the arm divides.

Artic. 21

In those cases callus is quickly formed. In congenital dislocations the bones below the seat of the injury are shorter than natural, and, mostly, those nearest to the place; namely, the bones of the fore-arm, next those of the hand; and, third, those of the fingers. The arm and shoulder are stronger, owing to the nourishment which they receive, and the other arm, from the additional work which it has to perform, is still more strong. Wasting of the flesh takes place on the inside if the dislocation be on the outside; or otherwise, on the side opposite the dislocation.

Artic. 22

When the elbow is dislocated either inward or outward, extension is to be made with the fore-arm at a right angle to the arm; the arm, suspended by means of a shawl passed through the armpit, and a weight attached to the extremity of the elbow; or force may be applied with the hands; when the articular extremity has been cleared, the displaced parts are to be rectified with the palms of the hand, as in

dislocations of the hands. It is to be bandaged, suspended in a sling, and placed while in this attitude.

Artic. 23

Dislocations backward are to be rectified by the palms of the hands, along with sudden extension; the two acts are to be performed together, as in other cases of the kind. But in dislocation forward the arm is to be bent around a ball of cloth of proper size, and at the same time replaced.

Artic. 24

But if the displacement be on the other side, both these operations are to be performed in effecting the adjustment. For conducting the treatment, the position and bandaging are the same as in the other cases. But all these cases may be reduced by ordinary distention.

Artic. 25

Of the methods of reduction, some operate by raising up the part, some by extension, and some by rotation: the last consists in rapidly turning the fore-arm to this side and that.

Artic. 26

The joint of the hand is dislocated either inward or outward, most frequently inward. The symptoms are easily recognized: if inward, the patient cannot at all bend his fingers; and if outward, he cannot extend them. With regard to the reduction,—by placing the fingers above a table, extension and counter-extension are to be made by other persons, while with the palm or heel of the hand on the projecting bone one pushes forward, and another from behind on the other bone; some soft substance is to be applied to it, and the arm is to be turned to the prone position if the dislocation was forward, but to the supine, if backward. The treatment is to be conducted with bandages.

Artic. 27

The whole hand is dislocated either inward or outward, or to this side or that, but more especially inward; and sometimes the epiphysis is displaced, and sometimes the other of these bones is separated. In

these cases strong extension is to be applied, and pressure is to be made on the projecting bone, and counter-pressure on the opposite side, both at the same time, behind and at the side, with the hands upon a table, or with the heel. These accidents give rise to serious consequences and deformities; but in the course of time the part gets strong, and admits of being used. The cure is with bandages, which ought to embrace both the hand and fore-arm; and splints are to be applied as far as the fingers; and when they are used they should be more frequently unloosed than infractures, and more copious affusions of water should be used.

Artic. 28

In congenital dislocations (at the wrist) the hand becomes shortened, and the atrophy of the flesh occurs, for the most part, on the side opposite to the dislocation. In an adult the bones remain of their natural size.

Artic. 29

Dislocation at the joint of a finger is easily recognized. Reduction is to be effected by making extension in a straight line, and applying pressure on the projecting bone, and counter-pressure on the opposite side of the other. The treatment is with bandages. When not reduced, callus is formed outside of the joint. When the dislocation takes place at birth, during adolescence the bones below the dislocation are shortened, and the flesh is wasted rather on the opposite than on the same side with the dislocation. When it occurs in an adult the bones remain of their proper size.

Artic. 30

The jaw-bone, in few cases, is completely dislocated, for the zygomatic process formed from the upper jaw-bone (*malar?*) and the bone behind the ear (*temporal?*) shuts up the heads of the under jaw, being above the one (*condyloid process?*), and below the other (*coronoid process?*). Of these extremities of the lower jaw, the one, from its length, is not much exposed to accidents, while the other, the coronoid, is more prominent than the zygoma, and from both these heads nervous

tendons arise, with which the muscles called temporal and masseter are connected; they have got these names from their actions and connections; for in eating, speaking, and the other functional uses of the mouth, the upper jaw is at rest, as being connected with the head by synarthrosis, and not by diarthrosis (*enarthrosis?*): but the lower jaw has motion, for it is connected with the upper jaw and the head by enarthrosis. Wherefore, in convulsions and tetanus, the first symptom manifested is rigidity of the lower jaw; and the reason why wounds in the temporal region are fatal and induce coma, will be stated in another place. These are the reasons why complete dislocation does not readily take place, and this is another reason, because there is seldom a necessity for swallowing so large pieces of food as would make a man gape more than he easily can, and dislocation could not take place in any other position than in great gaping, by which the jaw is displaced to either side. This circumstance, however, contributes to dislocation there; of nerves (*ligaments?*) and muscles around joints, or connected with joints, such as are frequently moved in using the member are the most yielding to extension, in the same manner as well-dressed hides yield the most. With regard, then, to the matter on hand, the jaw-bone is rarely dislocated, but is frequently slackened (*partially displaced?*) in gaping, in the same manner as many other derangements of muscles and tendons arise. Dislocation is particularly recognized by these symptoms: the lower jaw protrudes forward, there is displacement to the opposite side, the coronoid process appears more prominent than natural on the upper jaw, and the patient cannot shut his lower jaw but with difficulty. The mode of reduction which will apply in such cases is obvious: one person must secure the patient's head, and another, taking hold of the lower jaw with his fingers within and without at the chin, while the patient gapes as much as he can, first moves the lower jaw about for a time, pushing it to this side and that with the hand, and directing the patient himself to relax the jaw, to move it about, and yield as much as possible; then all of a sudden the operator must open the mouth,

while he attends at the same time to three positions: for the lower jaw is to be moved from the place to which it is dislocated to its natural position; it is to be pushed backward, and along with these the jaws are to be brought together and kept shut. This is the method of reduction, and it cannot be performed in any other way. A short treatment suffices, a waxed compress is to be laid on, and bound with a loose bandage. It is safer to operate with the patient laid on his back, and his head supported on a leather cushion well filled, so that it may yield as little as possible, but some person must hold the patient's head.

Artic. 31

When the jaw is dislocated on both sides, the treatment is the same. The patients are less able to shut the mouth than in the former variety; and the jaw protrudes farther in this case, but is not distorted; the absence of distortion may be recognized by comparing the corresponding rows of the teeth in the upper and lower jaws. In such cases reduction should be performed as quickly as possible; the method of reduction has been described above. If not reduced, the patient's life will be in danger from continual fevers, coma attended with stupor (for these muscles, when disordered and stretched preternaturally, induce coma); and there is usually diarrhea attended with bilious, unmixed, and scanty dejections; and the vomitings, if any, consist of pure bile, and the patients commonly die on the tenth day.

Artic. 32

In fracture of the lower jaw, when the bone is not fairly broken across, and is still partially retained, but displaced, it should be adjusted by introducing the fingers at the side of the tongue, and making suitable counter-pressure on the outside; and if the teeth at the wound be distorted and loosened, when the bone is adjusted, they should be connected together, not only two, but more of them, with a gold thread, if possible, but otherwise, with a linen thread, until the bone be consolidated, and then the part is to be dressed with cerate, a few

compresses, and a few bandages, which should not be very tight, but rather loose. For it should be well known that in fracture of the jaw, dressing with bandages, if properly performed, is of little advantage, but occasions great mischief if improperly done. Frequent examinations should be made about the tongue, and prolonged pressure should be applied with the fingers, in order to rectify the displaced bone. It would be best if one could do so constantly, but that is impossible.

Artic. 33

But if the bone be fairly broken across (this, however, rarely happens), it is to be set in the manner now described. When adjusted, the teeth are to be fastened together as formerly described, for this will contribute much toward keeping the parts at rest, especially if properly fastened, and the ends of the thread secured with knots. But it is not easy to describe exactly in writing the whole manipulation of the case; but the reader must figure the thing to himself from the description given. Then one must take a piece of Carthaginian leather; if the patient be a younger person, it will be sufficient to use the outer skin, but if an adult the whole thickness of the hide will be required; it is to be cut to the breadth of about three inches, or as much as will be required, and having smeared the jaw with a little gum (for thus it sticks more pleasantly), the end of the skin is to be fastened with the glue near the fractured part of the jaw, at the distance of an inch or a little more, from the wound. This piece is to be applied below the jaw; but the thong should have a cut in it, in the direction of the chin, so that it may go over the sharp point of the chin. Another piece of thong like this, or somewhat broader, is to be glued to the upper part of the jaw, at about the same distance from the wound as the other thong; this thong should be so cut as to encircle the ear. The thongs should be sharp-pointed at the part where they unite, and in gluing them, the flesh of the thong should be turned to the patient's skin, for in this way it will be more tenacious; then we must stretch this thong, but still more so the one at the chin, in order to prevent the fragments of

the jaw from riding over each other, and the thongs are to be fastened at the vertex, and then a bandage is to be bound round the forehead, and a proper apparatus is to be put over all, to prevent the bandages from being displaced. The patient should lie upon the sound side of the jaw, not resting upon the jaw, but upon the head. He is to be kept on a spare diet for ten days, and then nourished without delay. If there be no inflammation during the first days, the jaw is consolidated in twenty days; for callus quickly forms in this, as in all the other porous bones, provided there be no sphacelus (*exfoliation?*). But much remains to be said on the sphacelus of bones in another place. This method of distention with glued substances is mild, of easy application, and is useful for many dislocations in many parts of the body. Those physicians who have not judgment combined with their dexterity, expose themselves in fractures of the jaws, as in other cases, for they apply a variety of bandages to a fractured jaw-bone, sometimes properly, and sometimes improperly. For all such bandaging of a fractured jawbone has a tendency rather to derange the bones connected with the fracture, than to bring them into their natural position.

Artic. 34

But if the lower jaw be disjoined at its symphysis in the chin (there is but one symphysis in the lower jaw, but there are several in the upper; but I am unwilling to digress from the subject, as these matters will have to be touched upon in other kinds of disease)-if, then, the symphysis be separated at the chin, it is the work which anybody can perform, to rectify it; for the part which protrudes is to be pushed inward by pressure with the fingers, and the part that inclines inward is to be forced outward by pushing with the fingers from within. It is after having applied extension to separate the fragments that this is to be done, for they will thus be more easily restored to their natural position, than if one should bring them together by using force. This is proper to be known as applying to all such cases. When you have set the parts, you must fasten the teeth on both sides to one another, as

formerly directed. The treatment is to be accomplished with cerate, a few compresses, and bandages. This part, in particular, requires a short but complex (?) bandaging, for it is nearly cylindrical, though not exactly so; but the turn of the bandage is to be made, if the right jaw was dislocated, to the right hand (that is said to be to the right hand when the right hand conducts the bandaging); but if the other jaw be the seat of the dislocation, the bandaging is to be made in the other direction. And if matters be properly adjusted, and the patient keep quiet, there will be a speedy recovery, and the teeth will be uninjured; but if not, the recovery will be more protracted, the teeth will be distorted, will give trouble, and become useless.

Artic. 35

Of fractures of the nose there are more than one variety, but those who, without judgment, delight in fine bandagings, do much mischief, most especially in injuries about the nose. For this is the most complex of all the forms of bandaging, having most of the turns of the bandage called “ascia,” and rhomboidal intervals and uncovered spaces of the skin. As has been said, those who practice manipulation without judgment are fond of meeting with a case of fractured nose, that they may apply the bandage. For a day or two, then, the physician glories in his performance, and the patient who has been bandaged is well pleased, but speedily the patient complains of the incumbrance of the bandage, and the physician is satisfied, because he has had an opportunity of showing his skill in applying a complex bandage to the nose. Such a bandaging does everything the very reverse of what is proper; for, in the first place, those who have their nose flattened by the fracture, will clearly have the part rendered still more flat, if pressure above be applied to it; and further, those cases in which the nose is distorted to either side, whether at the cartilage or higher up, will evidently derive no benefit from bandaging above it, but will rather be injured; for it will not admit of having compresses properly arranged on either side of the nose, and indeed, persons applying this bandage do not seek to do this.

Artic. 36

This bandaging would appear to me to answer best when the skin surrounding the bone is contused on its ridge near the middle, or if the bone itself have sustained some injury, but not a great one, in such cases, redundant callus forms in the nose, and the part becomes a little too prominent; and yet, even in these cases, the bandaging need not require much trouble, if, indeed, any bandage be applied at all; for it is enough if one lay a waxed compress on the contusion, and then apply the double-headed bandage, thus taking one turn with it. The best application to such accidents is a small cataplasim of wheaten flour, washed, and mixed up into a viscid mass. If the flour be made from good wheat, and if it be glutinous, it should be used alone for all such cases, but if it be not very glutinous, a little of the manna of frankincense, well pulverized, is to be moistened with water, and the flour is to be mixed up with it, or a very little gum may be mixed in like manner.

Artic. 37

In those cases in which the fractured portions are depressed and flattened, if it is depressed in front at the cartilage, something may be introduced into the nostrils to rectify the parts. If not, all such deformities may be restored by introducing the fingers into the nostrils, if this can be managed, but if not, a thick spatula is to be introduced with the fingers, not to the fore part of the nose, but to the depressed portion, and the physician is to take hold of the nose externally on both sides, and at the same time raise it up. And if the fracture be much in the fore part one may introduce into the nostrils as already stated, either caddis scraped from a linen towel, or something such wrapped up in a piece of cloth, or rather stitched in Carthaginian leather, and moulded into a shape suitable to the place into which it is to be introduced. But if the fracture be at a greater distance, it is not possible to introduce anything within, for if it was irksome to bear anything of the kind in the fore part, how is it not to be so when introduced farther in? At first, then, by rectifying the

parts from within, and sparing no pains upon them from without, they are to be brought to their natural position, and set. A fractured nose may be readily restored to shape, especially on the day of the accident, or even a little later, but the physicians act irresolutely, and touch it more delicately at first than they should; for the fingers should be applied on both sides along the natural line of the nose, and it is to be pushed downward, and thus, with pressure from within, the displacement is to be rectified. But for these purposes no physician is equal to the index-fingers of the patient himself, if he will pay attention and has resolution, for they are the most natural means. Either of the fingers is to be placed firmly along the whole nose, and thus it is to be gently held, and steadily, if possible until it become firm, but if not, he himself is to hold it for as long a time as possible, or if he cannot, a child or woman should do it, for the hands ought to be soft. Thus may a fracture of the nose, attended with depression, and not with displacement to the side, but in a straight line, be most properly treated. I have never seen a case of fractured nose which could not be rectified when attempted, before callus is formed, provided the treatment be properly applied. But although men would give a great price to escape being deformed, yet at the same time they do not know how to take care, nor have resolution, if they do not experience pain, nor fear death, although the formation of callus in the nose speedily place, for the most part is consolidated in ten days, provided sphacelus do not take place.

Artic. 38

When the fractured bone is displaced laterally, the treatment is the same, but it is obvious that the reduction is to be made, not by applying equal force on both sides, but by pushing the displaced portion into its natural position, and pressing on it from without, and introducing something into the nostrils, and boldly rectifying the fragments which incline inward, until the whole be properly adjusted, well knowing that if you do not restore the parts at once, it is impossible but that the nose must be distorted. But when you restore

the parts to their natural position, either the patient himself, or some other person, is to apply one finger or more to the part which protrudes, and keep it in position until the fracture be consolidated; but the little finger is, from time to time, to be pushed into the nostril, to rectify the parts which incline inward. When any inflammation supervenes, dough must be used, but attention must still be equally paid to the application of the fingers, although the dough be on the part. But if the fracture be in the cartilage, with lateral displacement, the end of the nose must necessarily be distorted. In such cases some of the aforementioned means of reduction, or whatever suits, is to be introduced into the nostril; but there are many convenient things to be found which have no smell, and are appropriate in other respects; thus, on one occasion, I introduced a slice of sheep's lung, as it happened to be at hand; for sponges, if introduced, imbibe humidities. Then the outer skin of Carthaginian leather is to be taken, and a piece of the size of the thumb, or what will answer, is to be cut off and glued to the outside of the nostril which is turned aside, and then this piece of thong is to be stretched to the proper degree, or rather a little more than what will be sufficient to make the nose straight and regular. Then (for the thong must be long) it is to be brought below the ear and round the head, and the end of the thong may either be glued to the forehead, or a still longer one may be carried all round the head, and secured. This is a natural mode of setting the nose, is of easy application, and is calculated to enable the counter-extension on the nose to be made greater or less, as you may incline. In a case where the fractured nose is turned to the side, the treatment is to be conducted otherwise, as already described; and in most of them the thong ought to be glued to the end of the nose, in order to make extension in the opposite direction.

Artic. 39

When the fracture is complicated with a wound, one need not be troubled on that account, but pitch-cerate or any of the applications for fresh wounds is to be applied to the sores; for, in general, they

admit of easy cure, even when there is reason to apprehend that pieces of bone will come out. The parts, at first, are to be adjusted fearlessly, taking care that nothing is omitted, and, subsequently, they are also to be adjusted with the fingers; more softly, indeed, but still it must be done; and of all parts of the body the nose is modeled with the greatest ease. And there is nothing to prevent us from having recourse to the practice of gluing on the thongs, and drawing the nose to the opposite side, even if there be a wound or the parts be inflamed, for these thongs give no pain.

Artic. 40

In fractures of the ear all sorts of bandages do harm. For one would not think of applying it quite loose, and if applied more tightly, it only does the more harm, for even the sound ear, when confined with a bandage, becomes painful, throbs, and gets into a febrile state. With regard to cataplasms, the heaviest, on the whole, are the worst; but almost all kinds are bad, form abscesses, occasion an increase of humors, and afterward troublesome suppurations; and a fractured ear stands in less need of such applications than any other part; the most ready, if required, is the paste of meal, but neither should it have weight. It should touch as little as possible; for it is a good sometimes to apply nothing at all, both to the ear and to many other cases. Attention must be paid to the patient's position during sleep. And the body must be reduced, more especially if there be danger lest the ear suppurate; it will also be better to open the bowels, and if the patient can be readily made to vomit, this may be accomplished by means of the *syrmaism*. If the part come to suppuration, it should not be hastily opened; for often when matter appears to be formed it is absorbed again, even when no cataplasm is applied. But if forced to open it, the part will get soonest well if transfixed with a cautery, and yet it should be well understood that the ear gets maimed, and is less than the other if burned through. If not burned through, an incision, and not a very small one, should be made on the upper side; for the pus is found to be surrounded with a thicker covering than one would have

supposed; and it may be said, in general, that all parts of a mucous nature and which form mucus, as being all viscid, when touched, slip from below the fingers to either side; and on that account the physician, in such cases, finds that he has to pass his instrument through a thicker substance than he supposed; and in certain ganglionic cases, when the skin is flabby and mucous, many physicians open them, expecting to find a collection in them; here the physician forms a wrong judgment, but by such a procedure no great harm results to the patient from having had the part opened. But with regard to watery parts, and such as are filled with mucus, and which are situated in regions where every one of the parts, if opened, will occasion death or some other injury, these will be treated of in another work. When, therefore, incision is made in the ear, all sorts of cataplasms and pledges should be avoided, and it is to be treated either with applications for recent wounds, or anything else which is neither heavy nor will occasion pain, for if the cartilage be laid bare and abscesses form, the case will be troublesome; this happens from such modes of treatment. In all aggravated cases, the most effectual remedy is the transfixing of the part with a hot iron.

Artic. 41

The vertebrae of the spine when contracted into a hump behind from disease, for the most part cannot be remedied, more especially when the gibbosity is above the attachment of the diaphragm to the spine. Certain of those below the diaphragm are carried off by varices in the legs, more especially by such as occur in the vein at the ham; and in those cases where the gibbosities are removed, the varices take place also in the groin; and some have been carried off by a dysentery when it becomes chronic. And when the gibbosity occurs in youth before the body has attained its full growth, in these cases the body does not usually grow along the spine, but the legs and the arms are fully developed, whilst the parts (about the back) are arrested in their development. And in those cases where the gibbosity is above the diaphragm, the ribs do not usually expand properly in width, but

forward, and the chest becomes sharp-pointed and not broad, and they become affected with difficulty of breathing and hoarseness; for the cavities which inspire and expire the breath do not attain their proper capacity. And they are under the necessity of keeping the neck bent forward at the great vertebra, in order that their head may not hang downward; this, therefore, occasions great contraction of the pharynx by its inclination inward; for, even in those who are erect in stature, dyspnoea is induced by this bone inclining inward, until it be restored to its place. From this frame of body, such persons appear to have more prominent necks than persons in good health, and they generally have hard and unconcocted tubercles in the lungs, for the gibbosity and the distension are produced mostly by such tubercles, with which the neighboring nerves communicate. When the gibbosity is below the diaphragm, in some of these cases nephritic diseases and affections of the bladder supervene, but abscesses of a chronic nature, and difficult to cure, occur in the loins and groins, and neither of these carries off the gibbosity; and in these cases the hips are more emaciated than when the gibbosity is seated higher up; but the whole spine is more elongated in them than in those who have the gibbosity seated higher up, the hair of the pubes and chin is of slower growth and less developed, and they are less capable of generation than those who have the gibbosity higher up. When the gibbosity seizes persons who have already attained their full growth, it usually occasions a crisis of the then existing disease, but in the course of time some of them attack, as in the case of younger persons, to a greater or less degree; but, notwithstanding, for the most part, all these diseases are less malignant. And yet many have borne the affection well, and have enjoyed good health until old age, more especially those persons whose body is inclined to be plump and fat; and a few of them have lived to beyond sixty years of age, but the most of them are more short-lived. In some cases the curvature of the spine is lateral, that is to say, either to the one side or the other; the most of such cases are connected with tubercles (*abscesses?*) within the spine; and in some,

the positions in which they have been accustomed to lie cooperate with the disease. But these will be treated of among the chronic affections of the lungs; for these the most suitable prognostics of what will happen in these cases are given.

Artic. 42

When the spine protrudes backward, in consequence of a fall, it seldom happens that one succeeds in straightening it. Wherefore succussion on a ladder has never straightened anybody, as far as I know, but it is principally practiced by those physicians who seek to astonish the mob—for to such persons these things appear wonderful, for example, if they see a man suspended or thrown down, or the like; and they always extol such practices, and never give themselves any concern whatever may result from the experiment, whether bad or good. But the physicians who follow such practices, as far as I have known them, are all stupid. The device, however, is an old one, and I give great praise to him who first invented this, and any other mechanical contrivance which is according to nature. For neither would I despair, but that if succussion were properly gone about, the spine, in certain cases, might be thereby rectified. But, indeed, for my own part, I have been ashamed to treat all such cases in this way, because such modes of procedure are generally practiced by charlatans.

Artic. 43

Those cases in which the gibbosity is near the neck, are less likely to be benefited by these succussions with the head downward, for the weight of the head, and tops of the shoulders, when allowed to hang down, is but small; and such cases are more likely to be made straight by succussion applied with the feet hanging down, since the inclination downward is greater in this way. When the hump is lower down, it is more likely in this case that succussion with the head downward should do good. If one, then, should think of trying succussion, it may be applied in the following manner:—The ladder is to be padded with leather lined cushions, laid across, and well secured

to one another, to a somewhat greater extent, both in length and breadth, than the space which the man's body will occupy; he is then to be laid on the ladder upon his back, and the feet, at the ankles, are to be fastened, at no great distance from one another, to the ladder, with some firm but soft band; and he is further to be secured, in like manner, both above and below the knee, and also at the nates; and at the groins and chest loose shawls are to be put round in such a fashion as not to interfere with the effect of the succussion; and his arms are to be fastened along his sides to his own body, and not to the ladder. When you have arranged these matters thus, you must hoist up the ladder, either to a high tower or to the gable-end of a house; but the place where you make the succussion should be firm, and those who perform the extension should be well instructed, so that they may let go their hold equally to the same extent, and suddenly, and that the ladder may neither tumble to the ground on either side, nor they themselves fall forward. But, if the ladder be let go from a tower, or the mast of a ship, fastened into the ground with its cordage, it will be better, so that the ropes run upon a pulley or axle-tree. But it is disagreeable even to enlarge upon these matters; and yet, by the contrivances now described, the proper succussion may be made.

Artic. 44

But if the hump be situated very high up, and if succussion be by all means to be used, it will be better to do it with the feet downward, as has been said, for the force downward will be the greater in this case. The patient is to be well fastened to the ladder by cords at the breast, at the neck by means of a very loose shawl so as merely to keep the part properly on the ladder, and the head is to be fastened to the ladder at the forehead, the arms are to be stretched along and attached to the patient's body, and not to the ladder, and the rest of the body is not to be bound, except so as to keep it in place by means of a loose shawl wrapped round it and the ladder; attention, moreover, should be paid that these ligatures do not interfere with the force of the succussion, and the legs are not to be fastened to the ladder, but

should be placed near one another, so as to be in line with the spine. These matters should be thus arranged, if recourse is to be had at all to succussion on a ladder; for it is disgraceful in every art, and more especially in medicine, after much trouble, much display, and much talk, to do no good after all.

Artic. 45

In the first place, the structure of the spine should be known, for this knowledge is requisite in many diseases. Wherefore, on the side turned to the belly (*the anterior?*) the vertebrae are in a regular line, and are united together by a pulpy and nervous band of connection, originating from the cartilages, and extending to the spinal marrow. There are certain other nervous cords which decussate, are attached (*to the vertebrae?*), and are extended from both sides of them. But we will describe in another work the connections of the veins and arteries, their numbers, their qualities, their origin, their functional offices in particular parts, in what sort of sheaths the spinal marrow is inclosed, where they arise, where they terminate, how they communicate, and what their uses. On the opposite side (*behind?*) the vertebrae are connected together by a ginglymoid articulation. Common cords (*nerves?*) are extended to all parts, both those within and without. There is an osseous process from the posterior part of all and each of the vertebra, whether greater or smaller; and upon these processes there are cartilaginous epiphyses, and from them arise nervous productions (*ligaments?*), akin to the external nerves (*tonoi*). The ribs are united to them, having their heads inclined rather to the inside than the out, and every one of them is articulated with the vertebrae; and the ribs in man are very curved, and, as it were, arched. The space between the ribs and the processes of the vertebrae is filled on both sides by muscles, which arise from the neck and extend to the loins (?). The spine, longitudinally, is a straight line slightly curved; from the os sacrum to the great vertebra which is connected with the articulation of the femur, the spine inclines backward, for the bladder, the organs of generation, and the loose portion of the rectum, are

situated there. From this, to the attachment of the diaphragm, the spine inclines inward, and this portion alone, from the internal parts, gives origin to muscles, which are called *psoae*. From this to the great vertebra (*seventh cervical?*) which is above the tops of the shoulders, it is convex behind lengthways; but it is more in appearance than it really is, for the spinous processes are highest in the middle, and less so above and below. The region of the neck is convex before.

Artic. 46

In cases of displacement backward along the vertebrae, it does not often happen, in fact, it is very rare, that one or more vertebrae are torn from one another and displaced. For such injuries do not readily occur, as the spine could not easily be displaced backward but by a severe injury on the fore part through the belly (which would prove fatal), or if a person falling from a height should pitch on the nates, or shoulders (and even in this case he would die, but not immediately); and it also would not readily happen that such a displacement could take place forward, unless some very heavy weight should fall upon it behind; for each of the posterior spinal processes is so constructed, that it would sooner be broken than undergo any great inclination forward from a force which would have to overcome the ligaments and the articulations mutually connecting them. And the spinal marrow would suffer, if from the displacement of a vertebra it were to be bent even to a small extent; for the displaced vertebra would compress the spinal marrow, if it did not break it; and if compressed and strangled, it would induce insensibility of many great and important parts, so that the physician need not give himself any concern about rectifying the displacement of the vertebra, accompanied, as it is, by many other ill consequences of a serious nature. It is evident, then, that such a case could not be reduced either by succussion or by any other method, unless one were to cut open the patient, and then, having introduced the hand into one of the great cavities, were to push outward from within, which one might do on the dead body, but not at all on the living. Wherefore, then, do I

write all this? Because certain persons fancy that they have cured patients in whom the vertebra had undergone complete dislocation forward. Some, indeed, suppose that this is the easiest of all these dislocations to be recovered from, and that such cases do not stand in need of reduction, but get well spontaneously. Many are ignorant, and profit by their ignorance, for they obtain credit from those about them. These are deceived in this way, for they suppose the spinous processes to be the vertebrae themselves, because every one of them appears round to the touch, not knowing that these bones are processes from the vertebrae, as formerly stated; but the vertebrae are at a considerable distance before them; for of all animals, man, in proportion to his bulk, has the belly (*internal cavity?*) the narrowest from behind to before, especially at the breast. When, therefore, any of these processes are severely fractured, whether one or more, the part there appears lower than on either side, and for that reason they are deceived, supposing that the vertebrae are displaced inward. And the attitudes of the patient contribute also to deceive them; for if they attempt to put themselves into a bent position, they are pained, from the skin being stretched at the seat of the injury, and at the same time the fragments of the bones wound the skin still more; but if they bend forward, they feel easier, for the skin at the wound is thus relaxed, and the bones are less disposed to hurt them; and if touched, they shrink and bend forward, and the part which is touched appears empty and soft. All the circumstances now mentioned contribute to deceive the physician. Such patients speedily get well without any bad effects, for callus readily forms in all such bones as are porous.

Artic. 47

There are many varieties of curvature of the spine even in persons who are in good health; for it takes place from natural conformation and from habit, and the spine is liable to be bent from old age, and from pains. Gibbosities (or projections backward) from falls generally take place when one pitches on the nates, or falls on the shoulders. In this case some one of the vertebrae must necessarily appear higher

than natural, and those on either side to a less degree; but yet no one generally has started out of the line of the others, but every one has yielded a little, so that a considerable extent of them is curved. On this account the spinal marrow easily bears such distortions, because they are of a circular shape, and not angular. The apparatus for the reduction in this case must be managed in the following manner: a strong and broad board, having an oblong furrow in it, is to be fastened in the ground, or, in place of the board, we may scoop out an oblong furrow in the wall, about a cubit above the floor, or at any suitable height, and then something like an oaken bench, of a quadrangular shape, is to be laid along (the wall?) at a distance from the wall, which will admit of persons to pass round if necessary, and the bench is to be covered with robes, or anything else which is soft, but does not yield much; and the patient is to be stoved with vapor, if necessary, or bathed with much hot water, and then he is to be stretched along the board on his face, with his arms laid along and bound to his body; the middle, then, of a thong which is soft, sufficiently broad and long, and composed of two cross straps of leather, is to be twice carried along the middle of the patient's breast, as near the armpits as possible, then what is over of the thongs at the armpits is to be carried round the shoulders, and afterward the ends of the thong are to be fastened to a piece of wood resembling a pestle; they are to be adapted to the length of the bench laid below the patient, and so that the pestle-like piece of wood resting against this bench may make extension. Another such band is to be applied above the knees and the ankles, and the ends of the thongs fastened to a similar piece of wood; and another thong, broad, soft, and strong, in the form of a swathe, having breadth and length sufficient, is to be bound tightly round the loins, as near the hips as possible; and then what remains of this swathe-like thong, with the ends of the thongs, must be fastened to the piece of wood placed at the patient's feet, and extension in this fashion is to be made upward and downward, equally and at the same time, in a straight line. For extension thus made could

do no harm, if properly performed, unless one sought to do mischief purposely. But the physicians, or some person who is strong, and not uninstructed, should apply the palm of one hand to the hump, and then, having laid the other hand upon the former, he should make pressure, attending whether this force should be applied directly downward, or toward the head, or toward the hips. This method of applying force is particularly safe; and it is also safe for a person to sit upon the hump while extension is made, and raising himself up, to let himself fall again upon the patient. And there is nothing to prevent a person from placing a foot on the hump, and supporting his weight on it, and making gentle pressure; one of the men who is practiced in the palestra would be a proper person for doing this in a suitable manner. But the most powerful of the mechanical means is this: if the hole in the wall, or in the piece of wood fastened into the ground, be made as much below the man's back as may be judged proper, and if a board, made of lime-tree, or any other wood, and not too narrow, be put into the hole, then a rag, folded several times or a small leather cushion, should be laid on the hump; nothing large, however, should be laid on the back, but just as much as may prevent the board from giving unnecessary pain by its hardness; but the hump should be as much as possible on a line with the hole made in the wall, so that the board introduced into it may make pressure more especially at that spot. When matters are thus adjusted, one person, or two if necessary, must press down the end of the board, whilst others at the same time make extension and counter-extension as along the body, as formerly described. Extension may also be made with axles, which may either be fastened in the ground beside the bench, or the post of the axles may be attached to the bench itself, if you will make them perpendicular and overtopping (*the bench?*) a little at both ends, or at either end of the bench. These powers are easily regulated, so as to be made stronger or weaker, and they are of such force, that if one were to have recourse to them for a mischievous purpose, and not as a remedy, they would operate strongly in this way also; for by making

merely extension and counter-extension longitudinally, without any additional force, one might make sufficient extension; and if, without making extension at all, one were only to press down properly with the board, sufficient force might be applied in this way. Such powers, then, are excellent which admit of being so regulated, that they can be made weaker and stronger as required. And the forces are applied in the natural way; for the pressure above forces the displaced parts into their place. Natural extension restores parts which have come too near one another to their natural position. I, then, am acquainted with no powers which are better or more appropriate than these; for extension along the spine downward has no proper hold at the bone called the os sacrum; and extension upward, along the neck and head, has indeed a hold; but extension thus made is unseemly to behold, and, besides, if increased, may occasion much mischief otherwise. I once made trial of the following plan. Having placed the patient on his back, I put below the hump a bladder, not inflated, and afterward introduced air into the bladder by means of a brass pipe connected with it. But the experiment did not succeed; for, when the man was fairly extended, the bladder yielded, and the air could not be forced into it; and, besides, the hump of the patient was apt to slip off the distended bladder when they were pressed together. But when I did not extend the man strongly, the bladder was swelled up by the air, and the man became more bent forward than proper. I have written this expressly; for it is a valuable piece of knowledge to learn what things have been tried and have proved ineffectual, and wherefore they did not succeed.

Artic. 48

In curvatures forward of the vertebrae from a fall, or from some heavy body falling upon them, in general no one of them is displaced far beyond the others, but if one or more be so displaced, the case proves fatal; but, notwithstanding, as formerly stated, the displacement is circular, and not angular. In such cases, then, the urine and faeces are more apt to be retained than in displacement outward, the feet and

the whole inferior extremities are colder, and the symptoms are more fatal than in the former case; and if they do survive, they are more subject to retention of the urine, and to loss of strength, and to torpor in their legs. But if the displacement be in the upper part, they experience loss of strength and torpor of the whole body. I know no mechanical contrivance by which such a displacement could be reduced, unless that one might be benefited by succussion on a bladder, or any other similar plan of treatment, such as extension, as formerly described. I am not aware of any mode of pressure which might be applied along with the extension, like that of the board in displacement backward; for how could one apply pressure from before through the belly? (*internal cavity?*) The thing is impossible. But neither coughing nor sneezing has any power so as to cooperate with the extension, nor would the injection of air into the bowels have any effect. And to apply large cupping instruments with the view of drawing back the vertebrae which have protruded forward, shows a great error of judgment; for they rather propel than attract, and those who apply them are not aware even of this fact, for the greater will be the inclination forward the greater the instrument applied, the skin being forcibly drawn into the cupping-instrument. I could tell of other modes of succussion than those formerly described, which one might fancy would be more applicable in such an affection; but I have no great confidence in them, and therefore I do not describe them. On the main, it should be known, respecting the accidents which I have briefly described, that displacements forward are of a fatal and injurious nature; but that displacements backward, for the most part, do not prove fatal, nor occasion retention of urine nor torpor of the limbs, for they do not stretch the ducts leading toward the intestines, nor occasion obstruction of the same; but displacements forward produce both these bad effects, and many others in addition. And truly they are more apt to lose the power of their legs and arms, to have torpor of the body, and retention of urine, who experience no displacement either forward or backward, but merely a violent

concussion along the spine, while those who have displacement backward are least subject to these symptoms.

Artic. 49

And one might observe many other instances in medicine, of considerable injuries not proving serious, but producing a crisis in some affection, while less considerable injuries prove more serious, give rise to chronic diseases, and extend their effects to the whole system. Now something similar may happen in fracture of the ribs; for in fracture of one or more ribs, in general, if the fractured bones are not driven inward, nor are laid bare, fever rarely supervenes, neither does it often happen that there is haemoptysis, empyema, and suppurating sores, which require treatment with pledgets, nor necrosis of the bones; and in these cases the ordinary regimen is sufficient. For, unless they be seized with continual fever, a strict diet does more harm than good, by inducing inanition, and increasing the pain, fever, and cough; for moderate fullness of the intestines has a tendency to replace the ribs, while evacuation leads to suspension of the ribs, and suspension induces pain. Ordinary bandaging, externally, is sufficient in such cases; the bandages should be applied moderately tight, along with cerate and compresses, or a pad of wool may be applied. The rib is consolidated in twenty days, for callus soon forms in such bones.

Artic. 50

But when there is contusion of the flesh about the ribs, either from a blow, or a fall, or a bruise, or any like cause, there is often copious vomiting of blood, for there are canals stretched along the vacuity of each rib (*intercostal space?*), and nerves proceeding from the most important parts of the body have their origin there. Many of these, therefore, are troubled with coughs, tubercles, empyema, external suppurations, and sphacelus of the ribs. And even when no such symptoms supervene from contusion of the skin about the ribs, still in such cases there is, generally, more combined pain than in fractures of the ribs, and relapses of pain in the seat of the injury are more apt to

occur. Wherefore some physicians pay much less attention to such injuries, than where the rib is fractured, whereas, if they were wise, they would treat such cases with far greater care than the other; for it is proper that the diet should be restricted, that the patients should remain at rest as much as possible, and abstain from venery, from fat articles of food, from such as excite cough, and from everything strong; they should be bled in the arm, speak as little as possible, should have the contused part bound round with folded compresses, plenty of bandages, broader than the contusion, and which should be smeared with cerate; in applying the bandages, broad and soft shawls should be used, and they should be put on moderately firm, so that the patient will say that they are neither too tight nor loose, and the bandaging should commence at the seat of the injury, and be made more particularly tight there, and the bandaging should be conducted as is done with a double-headed roller, so that the skin about the ribs may not be ruffled, but may lie smooth, and the bandaging should be renewed every day, or every alternate day. It is better also to open the bowels with some gentle medicine, so as just to produce an evacuation of the food, and the diet is to be restricted for ten days, and then the body is to be recruited and filled up; while you are upon the reducing system, the bandaging should be tighter, but when you are making him up again, it must be looser; and, if he spit blood from the commencement, the treatment and bandaging should be continued for forty days; but if there be no haemoptysis, treatment for twenty days will generally be sufficient; but the length of time must be regulated by the magnitude of the injury. When such contusions are neglected, if no greater mischief result there from, at all events the bruised part has its flesh more pulpy than it had formerly. When, therefore, any such thing is left behind, and is not properly dissipated by the treatment, it will be worse if the mucosity be lodged near the bone, for the flesh no longer adheres to the bone as formerly, the bone becomes diseased, and chronic sloughings of the bone in many cases arise from such causes. But if the mischief be not upon the bone, but it

is the flesh itself which is pulpy, relapses and pains will return from time to time, if there happen to be any disorder in the body; wherefore proper bandaging, and for a considerable time, must be had recourse to, until the extravasated blood forming in the bruise be dried up and absorbed, and the part be made up with sound flesh, and the flesh adhere to the bone. The best cure is the cautery in those cases which, from neglect, have become chronic, and the place turns painful, and the flesh is pulpy. And when the flesh itself is pulpy, the burning should be carried as far as the bone, but the bone itself should not be heated; but if it be in the intercostal space, you need not make the burning so superficial, only you must take care not to burn quite through. But if the contusion appear to be at the bone, if it be still recent, and the bone has not yet become necrosed, if it be very small, it is to be burned as has been described; but if the rising along the bone be oblong, several eschars are to be burned over it. Necrosis of the rib will be described along with the treatment of suppurating sores.

Artic. 51

There are four modes of dislocation at the hip-joint: of which modes, dislocation inward takes place most frequently, outward, the most frequently of all the other modes; and it sometimes takes place backward and forward, but seldom. When, therefore, dislocation takes place inward, the leg appears longer than natural, when compared with the other leg, for two reasons truly; for the bone which articulates with the hip-joint is carried from above down to the ischium where it rises up to the pubes, upon it, then, the head of the femur rests, and the neck of the femur is lodged in the cotyloid foramen (*foramen thyroideum?*). The buttock appears hollow externally, from the head of the thigh-bone having shifted inward, and the extremity of the femur at the knee is turned outward, and the leg and foot in like manner. The foot then being turned outward, physicians, from ignorance, bring the sound leg to it and not it to the sound leg; on this account, the injured limb appears to be much longer than the

sound one, and in many other cases similar circumstances lead to error in judgment. Neither does the limb at the groin admit of flexion as in the sound limb, and the head of the bone is felt at the perineum too prominent. These, then, are the symptoms attending dislocation of the thigh inward.

Artic. 52

When, then, a dislocation has not been reduced, but has been misunderstood or neglected, the leg, in walking, is rolled about as is the case with oxen, and the weight of the body is mostly supported on the sound leg, and the limb at the flank, and the joint where the dislocation has occurred is necessarily hollow and bent, while on the sound side the buttock is necessarily rounded. For if one should walk with the foot of the sound leg turned outward, the weight of the body would be thrown upon the injured limb, but the injured limb could not carry it, for how could it? One, then, is forced in walking to turn the leg inward, and not outward, for thus the sound leg best supports its own half of the body, and also that of the injured side. But being hollow at the flank and the hip-joint, they appear small in stature, and are forced to rest on a staff at the side of the sound leg. For they require the support of a staff there, since the nates inclines to this side, and the weight of the body is carried to it. They are forced also to stoop, for they are obliged to rest the hand on the side of the thigh against the affected limb; for the limb which is injured cannot support the body in changing the legs, unless it be held when it is applied to the ground. They who have got an unreduced dislocation inward are forced to put themselves into these attitudes, and this from no premeditation on their part how they should assume the easiest position, but the impediment itself teaches them to choose that which is most conformable to their present circumstances. For persons who have a sore on the foot, or leg, and cannot rest upon the limb, all, even children, walk in this way; for they turn the injured limb outward in walking, and they derive two advantages therefrom, to supply two wants; the weight of the body is not equally thrown upon the limb

turned outward, as upon the one turned inward, for neither is the weight in a line with it, but is much more thrown upon the one under the body; for the weight is in a straight line with it, both in walking and in the shifting of the legs. In this position one can most quickly turn the sound limb under the body, by walking with the unsound limb outward, and the sound inward. In the case we are now treating of, it is well that the body finds out the attitudes which are the easiest for itself. Those persons, then, who have not attained their growth at the time when they met with a dislocation which is not reduced, become maimed in the thigh, the leg, and the foot, for neither do the bones grow properly, but become shortened, and especially the bone of the thigh; and the whole limb is emaciated, loses its muscularity, and becomes enervated and thinner, both from the impediment at the joint, and because the patient cannot use the limb, as it does not lie in its natural position, for a certain amount of exercise will relieve excessive enervation, and it will remedy in so far the deficiency of growth in length. Those persons, then, are most maimed who have experienced the dislocation *in utero*, next those who have met with it in infancy, and least of all, those who are full grown. The mode of walking adopted by adults has been already described; but those who are children when this accident befalls them, generally lose the erect position of the body, and crawl about miserably on the sound leg, supporting themselves with the hand of the sound side resting on the ground. Some, also, who had attained manhood before they met with this accident, have also lost the faculty of walking erect. Those who were children when they met with the accident, and have been properly instructed, stand erect upon the sound leg, but carry about a staff, which they apply under the armpit of the sound side, and some use a staff in both arms; the unsound limb they bear up, and the smaller the unsound limb, the greater facility have they in walking, and their sound leg is no less strong than when both are sound. The fleshy parts of the limb are enervated in all such cases, but those who

have dislocation inward are more subject to this loss of strength than, for the most part, those who have it outward.

Artic. 53

Some tell a story how the Amazonian women dislocate the joints of their male children while mere infants, some at the knee, and others at the hip-joint, that they may be maimed, and that the male sex may not conspire against the female, and that they use them as artisans to perform any sedentary work, such as that of a shoemaker or brazier. Whether these things be true or not I do not know, but this I know, that matters would be such as is represented, provided their children, while infants, were to have their joints dislocated. The consequences of dislocation inward at the hip-joint are much greater than of dislocation outward at the hip-joint, but at the knee, although there be some difference, it is less; but the mode of either impediment is peculiar, their legs are more bandied when the dislocation is outward, but those who have dislocation inward stand erect on their feet with less freedom. In like manner, when the dislocation is at the ankle-joint, if outward they become *vari* (*their toes are turned inward?*), but they can stand; but if the dislocation be inward they become *valgi* (*their toes are turned outward?*), but they have less freedom of standing. The proportional growth of their bones is as follows: in those cases in which the bone of the leg is dislocated, the bones of the feet grow very little, as being very near the injury, but the bones of the leg increase in size, and with very little defect, but the fleshy parts (*muscles?*) are wasted. But when the ankle-joint is in its natural state, but the knee is dislocated, in these cases the bones of the leg do not grow in like manner, but become shortened, as being nearest the seat of the injury, and the bones of the feet also are atrophied, but not in the same proportion; because, as was said a little while ago, the ankle-joint is safe, and if they could use it, as in the case of club-foot, the bones of the foot would be still less atrophied. When the dislocation takes place at the hip-joint, the bone of the thigh, in this case, does not generally grow in like manner, as being the one nearest the seat of the injury,

but becomes shorter than the sound one; but the growth of the bones of the leg is not arrested in like manner; nor of those of the feet, for this reason, that there is no displacement between the bones of the thigh and leg, nor between those of the leg and foot; in those cases, however, the fleshy parts of the whole limb are atrophied; but if they could make use of the limb, the growth of the bones would be still more developed, as formerly stated, only the thigh, although its flesh would be much less wasted, would still be by no means so fleshy as the sound limb. The following observations are a proof of this: those persons who are weasel-armed (*galiancones*) from birth, owing to dislocation of the humerus, or when the accident has happened to them before they have attained their full growth, such persons have the bone of the arm shortened, but those of the fore-arm and hand are little inferior in size to the sound, for the reasons which have been stated, because the humerus is the bone nearest to the joint affected, and, on that account, it is shorter than natural; but the fore-arm is not equally affected by the accident, because the joint at which the bones of the arm and forearm are articulated remains in its natural condition, and the hand is still further distant than the fore-arm from the seat of the injury. Such are the reasons why certain of the bones in this case increase in growth, and certain do not. The laborious office of the hand contributes much to the development of the flesh in the fore-arm and hand, for whatever work is done by the hand, these weasel-armed persons strive to do no less effectually with the other hand than with the sound; for the arms do not support the weight of the body like the legs, and the work performed by them is light. From exercise, then, the fleshy parts on the hand and fore-arm are not atrophied in weasel-armed persons, and by these means the arm, too, gains flesh. But in dislocation inward at the hip-joint, whether from birth or from childhood, the fleshy parts, on that account, are much more atrophied than those of the hand, because the patients cannot exercise the leg. Another proof will be given in the observations which

will be presently stated, that these things are such as I things are such as I have represented.

Artic. 54

When the head of the femur is dislocated outward, the limb in these cases, when compared with the other, appears shortened, and this is natural, for the head of the femur no longer rests on a bone as in dislocation inward, but along the side of a bone which naturally inclines to the side, and it is lodged in flesh of a pulpy and yielding nature, and on that account it appears more shortened. Inwardly, the thigh about the perineum appears more hollow and flabby, but externally the buttock is more rounded, from the head of the thigh having slipped outward, but the nates appear to be raised up, owing to the flesh there having yielded to the head of the thigh-bone; but the extremity of the thigh-bone, at the knee, appears to be turned inward, and the leg and foot in like manner, neither does it admit of flexion like the sound limb. These, then, are the symptoms of dislocation outward.

Artic. 55

When such a dislocation is not reduced in adults, the whole limb appears to be shortened, and in walking they cannot reach the ground with the heel, but they walk with the ball of the foot on the ground, and the points of their toes incline a little inward. But the injured limb, in this case, can support the body much better than in dislocation inward, both because the head of the femur and the neck of its articular extremity, being naturally oblique, have formed a bed under a considerable portion of the hip, and because the extremity of the foot is not forcibly turned outward, but is nearly in a line with the body, and is even inclined more inwardly. When, then, the articular extremity of the femur has worn out a socket for itself in the flesh where it was lodged, and the flesh is lubricated, it ceases to be painful in the course of time, and when it becomes free from pain, they can walk without a staff, if so inclined, and they can support the body on the injured limb. From usage then, in such cases, the fleshy parts are

less enervated than in those which have been mentioned a little before, still, however, they lose their strength more or less; but in general there is more enervation when the dislocation is inward than when it is outward. Some of them, then, cannot wear their shoes, owing to the unbending state of their leg, and some of them can. But when this dislocation takes place *in utero*, and when the dislocation having occurred at any time before manhood, from violence, has not been replaced, or when from disease the articular extremity has started from its socket, and is displaced (for many such cases occur, and from some of them, if the femur become necrosed, obstinate suppurations requiring the use of tents are formed, and in certain of them the bone is laid bare), whether the bone become necrosed or not, the bone of the thigh is much shortened, and does not usually grow like the sound one, the bones, too, of the leg, become shorter than those of the other, but in a small degree, for the same reasons that were formerly stated; such persons can walk, some of them in the same fashion as adults having an unreduced dislocation, and some of them walk with the whole foot on the ground, but limp in walking, being obliged to do so by the shortness of the limb. Such is the result, even though they be carefully and properly trained in the attitudes before they have strength for walking, and in like manner also, after they have acquired the necessary strength; but those persons require the most care who were very young when they met with the accident, for, if neglected while children, the limb becomes entirely useless and atrophied. The fleshy parts of the entire limb are more wasted than those of the sound limb, but this is much less apt to happen in their case than in dislocation inward, owing to usage and exercise, as they are speedily able to make use of the limb, as was stated a little before with regard to the weasel-armed (*galiancones*).

Artic. 56

There are persons who, from birth or from disease, have dislocations outward of both the thighs; in them, then, the bones are affected in like manner, but the fleshy parts in their case lose their strength less;

the legs, too, are plump and fleshy, except that there is some little deficiency at the inside, and they are plump because they have the equal use of both their legs, for in walking they totter equally to this side that. Their nates appear very prominent, from the displacement of the bones of the joint. But if in their case the bones do not sphacelate (*become carious?*) and if they do not become bent above the hip-joint, if nothing of this kind happen to them, they become otherwise sufficiently healthy, but the growth of all the rest of the body, with the exception of the head, is arrested.

Artic. 57

In dislocations of the head of the femur backward, which rarely occur, the patient cannot extend the leg, either at the dislocated joint, or at the ham, to any extent, and of all the dislocations, this is the variety in which the patients have the least power of making extension at the groin and the ham. But, moreover, this also should be known (for it is a valuable piece of knowledge, and of much importance, and yet most yet most people are ignorant of it), that persons in health cannot extend the joint at the ham, if they do not extend the joint at the groin at the same time, unless they raise the foot very high, for in this way they could do it; neither also could they bend the joint at the ham, but with much greater difficulty, if they do not bend the joint at the groin at the same time. There are many other things in the body which have similar connections, both with regard to the contractions of nerves (*ligaments?*), and the positions of muscles, and many of them more worthy of being known than is generally supposed, and with regard to the nature of the intestine and that of the whole internal cavity, and with regard to the displacements and contractions of the uterus; but all these things will be treated of elsewhere, in a work akin to the present one. But with regard to the matter on hand, they cannot make extension, as has been already stated; and the limb appears shortened, for two reasons—first, because it cannot be extended, and also because the bone has slipped into the flesh of the nates; for the head and neck of the femur, in this dislocation, are

carried downward from their natural situation, to the outside of the nates. But yet they can bend the limb, unless prevented by pain, and the leg and foot appear pretty straight, and not much inclined toward either side, but at the groin the flesh, when felt, appears looser, from the bone of the joint having slipped to the other side, but at the nates the head of the femur may be felt to be more prominent than natural. Such are the symptoms accompanying dislocation of the thigh backward.

Artic. 58

When this dislocation occurs in an adult, and is not reduced, he can walk, indeed, after a time, and when the pain has abated, and when he has been accustomed to rotate the articular bone in the flesh; he finds it necessary, however, to make strong flexion at the groin in walking, for two reasons, both because the limb, for the causes already stated, becomes much shorter, and he is far from touching the ground with his heel, and he can barely reach it with the ball of his foot, and not even thus, unless he bend himself at the groins, and also bend with the other leg at the ham. And in this case, he is under the necessity of supporting the upper part of the thigh with his hand at each step: this also contributes, in a certain degree, to make him bend the body at the groins; for, during the shifting of the feet in walking, the body cannot be supported on the unsound be supported on the unsound limb, unless it be pressed to the ground by the hand,—the end of the femur not being placed properly under the body, but having slipped backward to the nates; and if he should try to rest the weight of his body for a little, upon the foot, without any other support, he would fall backward, for there would be a great inclination in this direction, from the hips having protruded backward far beyond the line of the foot, and the spine inclining toward the hips. Such persons can walk, indeed, without a staff, if so accustomed, for because the sole of the foot is in its old line, and is not inclined outward, they do not require anything to balance them. Such, however, as, instead of grasping the thigh, prefer resting their weight upon a staff introduced into the

armpit of the affected side, these, if they use a longer staff, will walk, indeed, more erect, but will not be able to reach the ground with the foot, or if they wish to rest upon the foot, they must take a shorter staff, and will require to bend the body at the groins. The wasting of the fleshy parts is analogous to what happens in the cases formerly described, for the wasting is greatest in those cases in which the patients keep the limb up, and do not exercise it, whilst those who practice walking, have the least atrophy. The sound leg, however, is not benefited, but is rather rendered more deformed, if the injured limb be applied to the ground, for it is forced to cooperate with the other, being protruded at the hip, and bent at the ham. But if the patient does not use the injured limb by applying it to the ground, but carries it up, and rests upon a staff, the sound leg thereby gains strength, for it is employed in its natural position, and further, the exercise gives it strength. But it may be said, these things are foreign to medicine; for what is the use of enlarging upon cases which are already past remedy? This is far from being the case, for it belongs to the knowledge of medicine to be acquainted also with these, and they cannot possibly be separated from one another; for to such as are curable, means are to be used to prevent them from becoming incurable, studying how they may best be prevented from getting into an incurable state. And incurable cases should be known, that they may not be aggravated by useless applications, and splendid and creditable prognostics are made by knowing where, how, and when every case will terminate, and whether it will be converted into a curable or an incurable disease. When then, from birth, or during one's youth, this dislocation backward occurs, and is not reduced, whether it be connected with violence or disease (for many such dislocations occur in diseases, but the nature of the diseases in which dislocations take place, will be described afterward); if, then, the dislocated limb be not reduced, the bone of the thigh becomes shortened, the whole limb is impaired, is arrested in its growth, and loses its flesh from want of use; the articulation at the ham is also

impaired, for the nerves (*ligaments?*) become stretched, from cases formerly stated, wherefore those who have this dislocation, cannot make extension at the knee-joint. In a word, all parts of the body which were made for active use, if moderately used and exercised at the labor to which they are habituated, become healthy, increase in bulk, and bear their age well, but when not used, and when left without exercise, they become diseased, their growth is arrested, and they soon become old. Among these parts the joints and nerves (*ligaments?*), if not used, are not the least liable to be so affected; they are impaired, then, for the reasons we have stated, more in this variety of dislocation than in the others, for the whole limb is wasted, both in its bones and in its fleshy parts. Such persons, then, when they attain their full growth, keep the limb raised and flexed, rest the weight of the body on the other leg, and support themselves with a staff, some with one, and others with two.

Artic. 59

In dislocations of the head of the thigh-bone forward (they are of rare occurrence), the patients cannot extend the leg completely, but least of all can they bend it at the groin; they are pained, also, if forced to bend the limb at the ham. The length of the leg, if compared at the heel, is the same as that of the other; but the extremity of the foot inclines less to project forward. But the whole limb has its natural direction, and inclines neither to this side nor to that. These cases are particularly attended with severe pain, and they are more apt to be accompanied with retention of urine at first than any of the other dislocations; for the head of the thigh-bone is lodged very near to important nerves. And the region of the groin appears swelled out and stretched, while that of the nates is more wrinkled and flabby. The symptoms now stated are those which attend this dislocation of the thigh-bone.

Artic. 60

When persons have attained their full growth before meeting with this dislocation, and when it has not been reduced, upon the

subsidence of the pain, and when the bone of the joint has been accustomed to be rotated in the place where it is lodged, these persons can walk almost erect without a staff, and with the injured leg almost quite straight, as it does not admit of easy flexion at the groin and the ham; owing, then, to this want of flexion at the groin, they keep the limb more straight in walking than they do the sound one. And sometimes they drag the foot along the ground, as not being able to bend the upper part of the limb, and they walk with the whole foot on the ground; for in walking they rest no less on the heel than on the fore part of the foot; and if they could take great steps, they would rest entirely on the heel in walking; for persons whose limbs are sound, the greater the steps they take in walking, rest so much the more on the heel, while they are putting down the one foot and raising the opposite. In this form of dislocation, persons rest their weight more on the heel than on the anterior part of the foot, for the fore part of the foot cannot be bent forward equally well when the rest of the limb is extended as when it is in a state of flexion; neither, again, can the foot be arched to the same degree the limb is bent as when it is extended. The natural state of matters is such as has been now described; and in an unreduced dislocation, persons walk in the manner described, for the reasons which have been stated. The limb, moreover, is less fleshy than the other, at the nates, the calf of the leg, and the whole of its posterior part. When this dislocation occurs in infancy, and is not reduced, or when it is congenital, in these cases the bone of the thigh is more atrophied than those of the leg and foot; but the atrophy of the thigh-bone is least of all in this form of dislocation. The fleshy parts, however, are everywhere attenuated, more especially behind, as has been stated above. If properly trained, such persons, when they grow up, can use the limb, which is only a little shorter than the other, and yet they support themselves on a staff at the affected side. For, not being able to use properly the ball of the foot without the heel, nor to put it down as some can in the other varieties of dislocation (the cause of which has been just now stated), on this

account they require a staff. But those who are neglected, and are not in the practice of putting their foot to the ground, but keep the limb up, have the bones more atrophied than those who use the limb; and, at the articulations, the limb is more maimed in the direct line than in the other forms of dislocation.

Artic. 61

In a word, luxations and subluxations take place in different degrees, being sometimes greater and sometimes less; and those cases in which the bone has slipped or been displaced to a much greater extent, are in general more difficult to rectify than otherwise; and if not reduced, such cases have greater and more striking impairment and lesion of the bones, fleshy parts, and attitudes; but when the bone has slipped, or been displaced to a less extent, it is easier to reduce such cases than the other; and if the attempts at reduction have failed, or have been neglected, the impairment in such cases is less, and proves less injurious than in the cases just mentioned. The other joints present great differences as to the extent of the displacements which they are subject to. But the heads of the femur and humerus are very similar to one another as to their dislocations. For the heads of the bones are rounded and smooth, and the sockets which receive the heads are also circular, and adapted to the heads; they do not admit then of being dislocated in any intermediate degree, but, notwithstanding, from their rounded shape, the bones slip either outward or inward. In the case we are now treating of, then, there is either a complete dislocation or none at all, and yet these bones admit of being displaced to a greater or less extent; and the thigh is more subject to these differences than the arm.

Artic. 62

Wherefore, then, some of these congenital displacements, if to a small extent, may be reduced to their natural condition, and especially those at the ankle-joint. Most cases of congenital club-foot are remediable, unless the declination be very great, or when the affection occurs at an advanced period of youth. The best plan, then,

is to treat such cases at as early a period as possible, before the deficiency of the bones of the foot is very great, and before there is any great wasting of the flesh of the leg. There is more than one variety of club-foot, the most of them being not complete dislocations, but impairments connected with the habitual maintenance of the limb in a certain position. In conducting the treatment, attention must be paid to the following points: to push back and rectify the bone of the leg at the ankle from without inward, and to make counter-pressure on the bone of the heel in an outward direction, so as to bring it into line, in order that the displaced bones may meet at the middle and side of the foot; and the mass of the toes, with the great toe, are to be inclined inward, and retained so; and the parts are to be secured, with cerate containing a full proportion of resin, with compresses, and soft bandages insufficient quantity, but not applied too tight; and the turns of the bandages should be in the same direction as the rectifying of the foot with the hand, so that the foot may appear to incline a little outward. And a sole made of leather not very hard, or of lead, is to be bound on, and it is not to be applied to the skin but when you are about to make the last turns of the bandages. And when it is all bandaged, you must attach the end of one of the bandages that are used to the bandages applied to the inferior part of the foot on the line of the little toe; and then this bandage is to be rolled upward in what is considered to be a sufficient degree, to above the calf of the leg, so that it may remain firm when thus arranged. In a word, as if moulding a wax model, you must bring to their natural position the parts which were abnormally displaced and contracted together, so rectifying them with your hands, and with the bandaging in like manner, as to bring them into their position, not by force, but gently; and the bandages are to be stitched so as to suit the position in which the limb is to be placed, for different modes of the deformity require different positions. And a small shoe made of lead is to be bound on externally to the bandaging, having the same shape as the Chian slippers had. But there is no necessity for it if the parts be

properly adjusted with the hands, properly secured with the bandages, and properly disposed of afterward. This, then, is the mode of cure, and it neither requires cutting, burning, nor any other complex means, for such cases yield sooner to treatment than one would believe. However, they are to be fairly mastered only by time, and not until the body has grown up in the natural shape; when recourse is had to a shoe, the most suitable are the buskins, which derive their name from being used in traveling through mud; for this sort of shoe does not yield to the foot, but the foot yields to it. A shoe shaped like the Cretan is also suitable.

Artic. 63

In cases of complete dislocation at the ankle-joint, complicated with an external wound, whether the displacement be inward or outward, you are not to reduce the parts, but let any other physician reduce them if he choose. For this you should know for certain, that the patient will die if the parts are allowed to remain reduced, and that he will not survive more than a few days, for few of them pass the seventh day, being cut off by convulsions, and sometimes the leg and foot are seized with gangrene. It should be well known that such will be the results; and it does not appear to me that hellebore will do any good, though administered the same day, and the draught repeated, and yet it is the most likely means, if any such there be; but I am of opinion that not even it will be of service. But if not reduced, nor any attempts at first made to reduce them, most of such cases recover. The leg and foot are to be arranged as the patient wishes, only they must not be put in a dependent position, nor moved about; and they are to be treated with pitched cerate, a few compresses dipped in wine, and not very cold, for cold in such cases induces convulsions; the leaves also of beet, or of colt's foot, of any such, when boiled in dark-colored austere wine, form a suitable application to the wound and the surrounding parts; and the wound may further be anointed with cerate in a tepid state. But if it be the winter season, the part is to be covered with unsoured wool, which is to be sprinkled from above

with tepid wine and oil, but on no account is either bandage or compress to be applied; for this should be known most especially, that whatever compresses, or is heavy, does mischief in such cases. And certain of the dressings used to recent wounds are suitable in such cases; and wool may be laid upon the sore, and sprinkled with wine, and allowed to remain for a considerable time; but those dressings for recent wounds which only last for a few days, and into which resin enters as an ingredient, do not agree with them; for the cleansing of the sores is a slow process, and the sore has a copious discharge for a long time. Certain of these cases it may be advantageous to bandage. It ought also to be well understood, that the patient must necessarily be much maimed and deformed, for the foot is retracted outward, and the bones which have been displaced outward protrude: these bones, in fact, not being generally laid bare, unless to a small extent; neither do they exfoliate, but they heal by thin and feeble cicatrices, provided the patient keeps quiet for a length of time; but otherwise there is danger that a small ulcer may remain incurable. And yet in the case we are treating of, those who are thus treated are saved; whereas, when the parts are reduced and allowed to remain in place, the patients die.

Artic. 64

The same rule applies to dislocations at the wrist, attended with a wound and projection of the bone, whether the bones of the arm be displaced inward or outward. For this should be well understood, that the patient will die in the course of a few days, by the same mode of death as formerly described, if the bone be reduced, and allowed to remain so. But in those cases in which they are not reduced, nor any attempt made to reduce them, the patients, for the most part, recover; and the same mode of treatment as has been described will be applicable; but the deformity and impediment of the limb must necessarily be great, and the fingers of the hand will be weak and useless; for if the bones have slipped inward, they cannot bend the fingers, or if outward, they cannot extend them.

Artic. 65

When the os tibiae, having made a wound at the knee, has protruded through the skin, whether the dislocation be outward or inward, in such a case, if the bone be reduced, death will be even more speedy than in the other cases, although speedy also in them. But the only hope of recovery is if you treat them without reduction. These cases are more dangerous than the others, as being so much higher up, as being so much stronger joints, and displaced from bones which are so much stronger. But if the os femoris form a wound at the knee, and slip through it, provided it be reduced and left so, it will occasion a still more violent and speedy death than in the cases formerly described; but if not reduced, it will be much more dangerous than those cases mentioned before, and yet this is the only hope of recovery.

Artic. 66

The same rule applies to the elbow-joint, and with regard to the bones of the fore-arm and arm. For when these bones protrude through a wound which they have made in the skin, all cases in which they are reduced prove fatal; but if not reduced, there is a chance of recovery; but to those that survive there is certain impediment. And if in any instance the bones of the upper articulations (*shoulder-joint?*), should be dislocated, and project through a wound which they have made in the skin, these, if reduced, are followed by more speedy death; and if not reduced, they are more dangerous than the others. But the mode of treatment which appears to me most suitable has been already described.

Artic. 67

When the joints of the toes or hands are dislocated, and the bones protrude through a wound which they have made, and when there is no fracture of the bone, but merely displacement of the joint, in these cases, if the reduction be made and allowed to remain, there is some danger of spasms (*tetanus?*) if not properly treated, and yet it may be worth while to reduce them, having warned the patient beforehand

that much caution and care will be required. The easiest, the most efficient method, and the one most conformable to art, is that by the lever, as formerly described when treating of bones which have been fractured and protruded; then the patient must be as quiet as possible, lie in a recumbent position, and observe a restricted regimen. And it will be better also that he should get some gentle emetics. The sore is to be treated with dressings for fresh wounds, which permit of allusions, or with the leaves of camomile, or with the applications for fractured bones of the head, but nothing very cold must be applied. The first (*most distant?*) joints are least dangerous, but those still higher, are more so. Reduction should be made the same day, or the next, but by no means on the third or fourth, for it is on the fourth day that exacerbations especially attack. In those cases, then, where immediate reduction cannot be accomplished, we must wait until after the aforesaid days; for whatever you reduce within ten days, may be expected to induce spasm. But if the spasm supervene on its being reduced, the joint should be quickly displaced, and bathed frequently with warm water, and the whole body should be kept in a warm, soft, and easy condition, and more especially about the joints, for the whole body should rather be in a bent than in an extended state. Moreover, it is to be expected, that the articular extremities of the bones of the fingers will for this generally happens, if even the least degree of inflammation take place, so that if it were not that the physician would be exposed to censure, owing to the ignorance of the common people, no reduction should be made at all. The reduction of the bones of joints which have protruded through the skin, is attended with the dangers which have been described.

Artic. 68

When the articular bones of the fingers are fairly chopped off, these cases are mostly unattended with danger, unless deliquium come on in consequence of the injury, and ordinary treatment will be sufficient to such sores. But when resection is made, not at the articulations, but at some other point in the bones, these cases also are free from

danger, and are still more easily cured than the others; and the fractured bones of the fingers which protrude otherwise than at the joint admit of reduction without danger. Complete resections of bones at the joints, whether the foot, the hand, the leg, the ankle, the forearm, the wrist, for the most part, are not unattended with danger, unless one be cut off at once by deliquium animi, or if continual fever supervene on the fourth day.

Artic. 69

With regard to the sphacelus of fleshy parts, it takes place in wounds where there are large blood-vessels, which have been strongly compressed, and in fractures of bones which have been bound too tight, and in other cases of immoderate constriction, when the parts which have been strangulated generally drop off; and the most of such patients recover, even when a portion of the thigh comes away, or of the arm, both bones and flesh, but less so in this case; and when the fore-arm and leg drop off, the patients readily recover. In cases then, of fracture of the bones, when strangulation and blackening of the parts take place at first, the separation of the dead and living parts quickly occurs, and the parts speedily drop off, as the bones have already given way; but when the blackening (*mortification*) takes place while the bones are entire, the fleshy parts, in this case, also quickly die, but the bones are slow in separating at the boundary of the blackening, and where the bones are laid bare. Those parts of the body which are below the boundaries of the blackening are to be removed at the joint, as soon as they are fairly dead and have lost their sensibility; care being taken not to wound any living part; for if the part which is cut off give pain, and if it prove not to be quite dead, there is great danger lest the patient may swoon away from the pain, and such swoonings often are immediately fatal. I have known the thigh-bones, when denuded in this manner, drop off on the eightieth day; but in the case of this patient, the parts below were separated at the knee on the twentieth day, and, as I thought, too early, for it appeared to me that this should be done more guardedly. In a case

which I had of such blackening in the leg, the bones of the leg, as far as they were denuded, separated at its middle on the sixtieth day. But the separation of denuded bones is quicker or slower, according to the mode of treatment; something, too, depends upon whether the compression be stronger or weaker, and whether the nerves, flesh, arteries, and veins are quicker or slower in becoming blackened and in dying; since, when the parts are not strongly compressed, the separation is more superficial, and does not go the length of laying the bones bare, and in some cases it is still more superficial, so as not even to expose the nerves. For the reasons now stated, it is impossible to define accurately the time at which each of these cases will terminate. The treatment of such cases, however, is to be readily undertaken, for they are more formidable to look at than to treat; and a mild treatment is sufficient in all such cases, for they come to a crisis of themselves; only the diet must be attended to, so that it may be as little calculated to create fever as possible, and the body is to be placed in the proper positions: these are, neither raised very high up, nor inclined much downward, but rather upward, until the separation be completed; for at that time there is most danger of hemorrhage; on this account, wounds should not be laid in a declining position, but the contrary. But after a while, and when the sores have become clean, the same positions will no longer be appropriate; but a straight position, and one inclining downward, may be proper; and in the course of time, in some of these cases, abscesses form, and require bandages. One may also expect that such patients will be attacked with dysentery; for dysentery usually supervenes in cases of mortification and of hemorrhage from wounds; it comes on generally when the blackening and hemorrhage have arrived at a crisis, and is profuse and intense, but does not last many days; neither is it of a fatal nature, for such patients do not usually lose their appetite, nor is it proper to put them on a restricted diet.

Dislocation inward at the hip-joint is to be reduced in the following manner: (it is a good, proper, and natural mode of reduction, and has something of display in it, if any one takes delight in such ostentatious modes of procedure). The patient is to be suspended by the feet from a cross-beam with a strong, soft, and broad cord; the feet are to be about four inches or less from one another; and a broad and soft leather collar connected with the cross-beam is to be put on above the knees; and the affected leg should be so extended as to moved be two inches longer than the other; the head should be about two cubits from the ground, or a little more or less; and the arms should be stretched along the sides, and bound with something soft; all these preparations should be made while he is lying on his back, so that he may be suspended for as short a time as possible. But when the patient is suspended, a person properly instructed and not weak, having introduced his arm between his thighs, is to place his fore-arm between the perineum and the dislocated head of the os femoris; and then, having joined the other hand to the one thus passed through the thighs, he is to stand by the side of the suspended patient, and suddenly suspend and swing himself in the air as perpendicularly as possible. This method comprises all the conditions which are natural; for the body being suspended by its weight, produces extension, and the person suspended from him, along with the extension, forces the head of the thigh-bone to rise up above the acetabulum; and at the same time he uses the bone of the fore-arm as a lever, and forces the os femoris to slip into its old seat. The cords should be properly prepared, and care should be taken that the person suspended along with the patient have a sufficiently strong hold.

Artic. 71

Wherefore, as formerly stated, men's constitutions differ much from one another as to the facility or difficulty with which dislocations are reduced; and the cause of this was also stated formerly in treating of the shoulder. In some the thigh is reduced with no preparation, with slight extension, directed by the hands, and with slight movement;

and in some the reduction is effected by bending the limb at the joint, and making rotation. But much more frequently it does not yield to any ordinary apparatus, and therefore one should be acquainted with the most powerful means which can be applied in each case, and use whatever maybe judged most proper under all circumstances. The modes of extension have been described in the former parts of the work, so that one may make use of whatever may happen to be at hand. For, extension and counter-extension are to be made in the direction of the limb and the body; and if this be properly effected, the head of the thigh-bone will be raised above its ancient seat; and if thus raised, it will not be easy to prevent it from settling in its place, so that any ordinary impulse with the lever and adjustment will be quite sufficient; but some apply insufficient extension, and hence the reduction gives much trouble. The bands then should be fastened, not only at the foot, but also above the knee, so that the force of the extension may not be expended on the knee-joint more than upon the hip-joint. The extension in the direction of the foot is to be thus contrived. But the counter-extension is not only to be managed by means of something carried round the chest and armpits, but also by a long, double, strong, and supple thong applied to the perineum, and carried behind along the spine, and in front along the collar-bone and fixed to the point from which counter-extension is made; and then force is to be so applied, by means of this extension and counter-extension, that the thong at the perineum may not pass over the head of the thigh-bone, but between it and the perineum; and during the extension one should strike the head of the femur with the fist, so as to drive it outward. And when the patient is raised up by the stretching, you should pass a hand through (*between the legs?*) and grasp it with the other hand, so as at the same time to make extension, and force the dislocated limb outward; while some other person sitting by the knee quietly directs it inward.

It has been formerly stated by us that it will be of importance for any person who practices medicine in a populous city to get prepared a quadrangular board, about six cubits or a little more in length, and about two cubits in breadth; a fathom will be sufficient thickness for it; and then along it from the one end to the other, an excavation must be made, so that the working of the levers may not be higher than is proper; then at both sides we are to raise short, strong, and strongly-fixed posts, having axles; and in the middle of the bench five or six long grooves are to be scooped out about four inches distant from one another, three inches will be a sufficient breadth for them, and the depth in like manner; and although the number of grooves I have mentioned will be sufficient, there is nothing to prevent their being made all over the bench. And the bench should have in its middle a pretty deep hole, of a square shape, and of about three inches in size; and into this hole, when judged necessary, is to be adjusted a corresponding piece of wood, rounded above, which, at the proper time, is to be adjusted between the perineum and the head of the thigh-bone. This upright piece of wood prevents the body from yielding to the force dragging downward by the feet; for sometimes this piece of wood serves the purpose of counter-extension upward; and sometimes, too, when extension and counter-extension are made, this piece of wood, if susceptible of some motion to this side or that, will serve the purpose of a lever for pushing the head of the thigh-bone outward. It is on this account that several grooves are scooped out on the bench, so that this piece of wood, being erected at the one which answers, may act as a lever, either on the sides of the articular heads of bones, or may make pressure direct on the heads along with the extension, according as it may suit to push inward or outward with the lever; and the lever may be either of a round or broad form, as may be judged proper; for sometimes the one form and sometimes the other suits with the articulation. This mode of applying the lever along with extension is applicable in the reduction of all dislocations of the thigh. In the case now on hand, a round lever is proper; but in

dislocations outward a flat lever will be the suitable one. By means of such machines and of such powers, it appears to me that we need never fail in reducing any dislocation at a joint.

Artic. 73

And one might find out other modes of reduction for this joint. If the large bench were to have raised on it two posts about a foot (*in diameter?*), and of a suitable height, on each side near its middle, and if a transverse piece of wood like the step of a ladder, were inserted in the posts, then if the sound leg were carried through between the posts, and the injured limb were brought over the transverse piece of wood, which should be exactly adapted in height to the joint which is dislocated (and it is an easy matter so to adjust it, for the step of the ladder should be made a little higher than required, and a convenient robe, folded several times, is to be laid below the patient's body), then a piece of wood, of suitable breadth and length, is to be laid below the limb, and it should reach from the ankle to beyond the head of the thigh-bone, and should be bound moderately tight to the limb. Then the limb being extended, either by means of the pestle-like piece of wood (formerly described), or by any of the other methods of extension, the limb which is carried over the step with the piece of wood attached to it, is to be forced downward, while somebody grasps the patient above the hip-joint. In this manner the extension will carry the head of the thigh-bone above the acetabulum, while the lever power that is exercised will push the head of the thigh-bone into its natural seat. All the above-mentioned powers are strong, and more than sufficient to rectify the accident, if properly and skillfully applied. For, as formerly stated, in most cases reduction may be effected by much weaker extension, and an inferior apparatus.

Artic. 74

If the head of the bone slip outward, extension and counter-extension must be made as described, or in a similar manner. But along with the extension a broad lever is to be used to force the bone from without inward, the lever being placed at the nates or a little farther up, and

some person is to steady the patient's body, so that it may not yield, either by grasping him at the buttocks with his hands, or this may be effected by means of another similar lever, adjusted to one of the grooves, while the patient has something laid below him, and he is secured, and the dislocated thigh is to be turned gently from within outward at the knee. Suspension will not answer in this form of dislocation, for, in this instance, the arm of the person suspended from him, would push the head of the thigh-bone from the acetabulum. But one might use the piece of wood placed below him as a lever, in such a manner as might suit with this mode of dislocation; it must work from without. But what use is there for more words? For if the extension be well and properly done, and if the lever be properly used, what dislocation of the joint could occur, that might not be thus reduced?

Artic. 75

In dislocation of the thigh, backward, extension and counter-extension should be made as has been described; and having laid on the bench a cloth which has been folded several times, so that the patient may lie soft, he is to be laid on his face, and extension thus made, and, along with the extension, pressure is to be made with a board, as in the case of humpback, the board being placed on the region of the nates, and rather below than above the hip-joint; and the hole made in the wall for the board should not be direct over, but should be inclined a little downward, toward the feet. This mode of reduction is particularly appropriate to this variety of dislocation, and at the same time is very strong. But perhaps, instead of the board, it might be sufficient to have a person sitting (*on the seat of luxation?*), or pressing with his hands, or with his foot, and suddenly raising himself up, along with the extension. None of the other aforementioned modes of reduction are natural in this form of dislocation.

Artic. 76

In dislocation forward, the same mode of extension should be made; but a person who has very strong hands, and is well trained, should

place the palm of the one hand on the groin, and taking hold of this hand with the other, is at the same time to push the dislocated part downward, and at the same time to the fore part of the knee. This method of reduction is most especially conformable to this mode of dislocation. And the mode of suspension is also not far removed from being natural, but the person suspended should be well trained, so that his arm may not act as a lever upon the joint, but that the force of the suspension may act about the middle of the perineum, and at the os sacrum.

Artic. 77

Reduction by the bladder is also celebrated in dislocations at this joint, and I have seen certain persons who, from ignorance, attempted to reduce both dislocations outward and backward therewith, not knowing that they were rather displacing than replacing the parts; it is clear, however, that he who first invented this method intended it for dislocation inward. It is proper, then, to know how the bladder should be used, if it is to be used, and it should be understood that many other methods are more powerful than it. The bladder should be placed between the thighs uninflated, so that it may be carried as far up the perineum as possible, and the thighs beginning at the patella are to be bound together with a swathe, as far up as the middle of the thigh, and then a brass pipe is to be introduced into one of the loose feet of the bladder, and air forced into it, the patient is to lie on his side with the injured limb uppermost. This, then, is the preparation; some, however, do the thing worse than as I have described, for they do not bind the thighs together to any extent, but only at the knees, neither do they make extension, whereas extension should be made, and yet some people by having the good fortune to meet with a favorable case, have succeeded in making reduction. But it is not a convenient method of applying force, for the bladder, when inflated, does not present its most prominent part to the articular extremity of the femur, which is the place that ought to be more especially pressed outward, but its middle, which probably corresponds with the middle

of the thigh, or still lower down, for the thighs are naturally curved, being fleshy, and in contact above, and becoming smaller downward, so that the natural configuration of the parts forces the bladder from the most proper place. And if a small bladder be introduced, its power will be small, and unable to overcome the resistance of the articular bone. But if the bladder must be used, the thighs are to be bound together to a considerable extent, and the bladder is to be inflated along with the extension of the body, and in this method of reduction both legs are to be bound together at their extremity.

Artic. 78

The prime object of the physician in the whole art of medicine should be to cure that which is diseased; and if this can be accomplished in various ways, the least troublesome should be selected; for this is more becoming a good man, and one well skilled in the art, who does not covet popular coin of base alloy. With regard to the subject now on hand, the following are domestic means of making extension of the body, so that it is easy to choose from among the things at hand:—In the first place, when soft and supple thongs are not at hand for ligatures, either iron chains, or cords, or cables of ships, are to be wrapped round with scarfs or pieces of woollen rags, especially at the parts of them which are to be applied, and in this state they are to be used as bands. In the second place, the patient is to be comfortably laid on the strongest and largest couch that is at hand, and the feet of the couch, either those at the (*patient's?*) head, or those at the feet, are to be fastened to the threshold, either within or without, as is most suitable; and a square piece of wood is to be laid across, and extending from the one foot to the other; and if this piece of wood be slender, it should be bound to the feet of the couch, but, not withstanding, if it be thick, there will be no necessity for this; then the heads of the ligatures, both of those at the head and those at the feet, are to be fastened to a pestle, or some such piece of wood, difficult to reduce at either end; the ligatures should run along the line of the body, or be a little elevated above it, and it should be stretched proportionally to

the pestles, so that, standing erect, the one may be fastened to the threshold, and the other to the transverse piece of wood. Extension is then to be made by bending back the ends of the pestles. A ladder, having strong steps, if laid below the bed, will serve the purpose of the threshold and the piece of wood laid along (*the foot of the couch?*), as the pestles can be fastened to the steps at either end, and when drawn back they thus make extension of the ligatures. Dislocation, inward or forward, may be reduced in the following manner: a ladder is to be fastened in the ground, and the man is to be seated upon it, and then the sound leg is to be gently stretched along and bound to it, wherever it is found convenient; and water is to be poured into an earthen vessel, or stones put into a hamper and slung from the injured leg, so as to effect the reduction. Another mode of reduction: a cross-beam is to be fastened between two pillars of moderate height; and at one part of the cross-beam there should be a protuberance proportionate to the size of the nates; and having bound a coverlet round the patient's breast, he is to be seated on the protuberant part of the cross-beam, and afterward the breast is to be fastened to the pillar by some broad ligature; then some one is to hold the sound leg so that he may not fall off, and from the injured limb is to be suspended some convenient weight, as formerly described.

Artic. 79

It should be particularly known that the union of all bones is, for the most part, by a head and socket (*cotyle*); in some of these the place (*socket?*) is cotyloid and oblong, and in some the socket is glenoid (*shallow?*). In all dislocations reduction is to be effected, if possible, immediately, while still warm, but otherwise, as quickly as it can be done; for reduction will be a much easier and quicker process to the operator, and a much less painful one to the patient, if effected before swelling comes on. But all the joints when about to be reduced should be first softened, and gently moved about; for, thus they are more easily reduced. And, in all cases of reduction at joints, the patient must be put on a spare diet, but more especially in the case of the

greatest joints, and those most difficult to reduce, and less so in those which are very small and easily reduced.

Artic. 80

If any joint of the fingers is dislocated, whether the first, second, or the third, the same method of reduction is to be applied, but the largest joints are the most difficult to reduce. There are four modes of displacement—either upward, downward, or to either side; most commonly upward, and most rarely laterally, and in consequence of violent motion. On both sides of its articular cavity there is a sort of raised border. When the dislocation is upward or downward, owing to the articular cavity having smoother edges there than at the sides, if the joint of it be dislocated, it is more easily reduced. This is the mode of reduction:—The end of the finger is to be wrapped round with a fillet, or something such, that, when you lay hold of it and make extension, it will not slip; and when this is done, some person is to grasp the arm at the wrist, and another is to take hold of the finger which is wrapped in the fillet, and then each is to make considerable extension toward himself, and at the same time the projecting bone is to be pushed into its place. But, if the dislocation be lateral, the same mode of reduction is to be used; but when you think that the extremity of the bone has cleared the rim, at the same time that extension is made, the bone is to be pushed direct into its place, while another person on the other side of the finger is to take care and make counter-pressure, so that it may not again slip out there. The twisted nooses formed from palm-shoots are convenient for effecting reduction, if you will make extension and counter-extension by holding the twisted string in the one hand and the wrist in the other. When reduced, you must bind the part as quickly as possible with bandages; these are to be very slender and waxed with cerate, neither very soft nor very hard, but of middle consistence; for that which is hard drops off from the finger, while that which is soft and liquid is melted and lost by the increased heat of the finger. The bandage is to be loosed on the third or fourth day; but on the whole, if inflamed, it is

to be the more frequently loosed, and if otherwise, more rarely; this I say respecting all the joints. The articulation of a finger is restored in fourteen days. The treatment of the fingers and of the toes is the same.

Artic. 81

After all reductions of joints the patient should be confined to a restricted diet and abstinence until the seventh day; and if there be inflammation, the bandages are to be the more frequently loosed, but otherwise, less frequently, and the pained joint is to be kept constantly in a state of rest, and is to be laid in the most convenient position possible.

Artic. 82

Accidents at the knee are more mild than at the elbow, from its being compact, regular, and elegant in its construction; and, therefore, it is more readily dislocated and reduced. It is most frequently dislocated inward, but also outward and backward. The modes of reduction are these: by flexion at the knee, or by sudden calcitration, or having rolled a swathe into a ball, and fixed it in the ham, the patient's body is to be suddenly dropped on its bended knees. Dislocation backward, also, as in the case of the elbow, may be reduced by moderate extension, and to either side, either by flexion or calcitration, but also by moderate extension. The adjustment is the same in all cases. In dislocations backward which are not reduced, the patient cannot bend the joint, but neither can he, to any great extent, in the other varieties; the thigh and leg are wasted in front; but if inward the patients become bow-legged, and the external parts are wasted; but if outward they become more bandy-legged, but the impediment is less, for the body is supported on the larger of the bones, and the inner parts are wasted. When these accidents happen at birth or during adolescence, they follow the rule formerly stated.

Artic. 83

Dislocations at the ankle-joints require strong extension, either with the hands or some such means; and adjustment, which at the same

time effects both purposes, as is common in all cases.

Artic. 84

Injuries of the foot are to be remedied like those of the hand.

Artic. 85

The bones connected with the leg, and which are dislocated, either at birth or during adolescence, follow the same course as those in the hand.

Artic. 86

When persons jumping from a height pitch on the heel, so as to occasion separation (*diastasis*) of the bones, ecchymosis of the veins, and contusion of the nerves; when these symptoms are very violent there is danger of sphacelus, and that the case may give trouble during life, for the bones are so constructed as to slip from one another, and the nerves communicate together. And, indeed, in cases of fracture, either from an injury in the leg or thigh, or in paralysis of the nerves (*tendons*?) connected with these parts, or from neglect during confinement to bed, when the heel gets blackened the most serious consequences result therefrom. Sometimes, in addition to the sphacelus, there come on acute fevers accompanied with hiccup, aberration of intellect, and speedy death, with lividities of the large blood-vessels. With regard to the symptoms attending exacerbations, if the ecchymosed and blackened parts and those around be somewhat hard and red, and if along with the hardness there be lividity, mortification is to be apprehended; but if the parts be slightly livid, or even very livid, and the swelling diffused, or if greenish and soft, these appearances, in such cases, are all favorable. The treatment, if no fever be present, consists in the administration of hellebore, but otherwise (it is not to be given, but *oxyglyky* (*decoction of honeycombs and vinegar*) is to be given for drink, if required. Bandaging as in the other articulations: above all, more especially in contusions, the bandages should be numerous and softer than usual, but the compression should be less; most turns should be made around the heel. Position,

like the bandaging, should be so regulated as not to determine to the heel. Splints are not to be used.

Artic. 87

When the foot is dislocated, either alone or along with its epiphysis, the displacement is, for the most part, to the inside. If not reduced, in the course of time, the hip, the thigh, and the side of the leg opposite the dislocation, become atrophied. Reduction is the same as in the wrist, but the extension requires to be very powerful. Treatment, agreeably to the general rule for joints. Exacerbations do occur, but less frequently than in dislocations at the wrist, provided the parts get rest. While they remain at rest the diet should be restricted. Those which occur at birth, or during adolescence, follow the rule formerly stated.

Mochlicus

Mochl. 1

With regard to the construction of bones⁸, the bones and joints of the fingers are simple, the bones of the hand and foot are numerous, and articulated in various ways; the uppermost are the largest; the heel consists of one bone which is seen to project outward, and the back tendons are attached to it. The leg consists of two bones, united together above and below, but slightly separated in the middle; the external bone (*fibula*), where it comes into proximity with the little toe, is but slightly smaller than the other, more so where they are separated, and at the knee, the outer hamstring arises from it⁹; these bones have a common epiphysis below, with which the foot is moved, and another epiphysis above¹⁰ in which is moved the articular extremity of the femur, which is simple and light in proportion to its length, in the form of a condyle, and having the patella (connected with it?), the femur itself bends outward and forward; its head is a round epiphysis which gives origin to ligament inserted in the acetabulum of the hip-joint¹¹. This bone is articulated somewhat

obliquely, but less so than the humerus. The ischium is united to the great vertebra contiguous to the os sacrum by a cartilaginous ligament¹². The spine, from the os sacrum to the great vertebra, is curved backward; in this quarter are situated the bladder, the organs of generation, and the inclined portion of the rectum; from this to the diaphragm it proceeds in a straight line inclining forward, and the psoae are situated there; from this point, to the great vertebra above the tops of the shoulders, it rises in a line that is curved backward, and the curvature appears greater than it is in reality, for the posterior processes of the spine are there highest; the articulation of the neck inclines forward. The vertebrae on the inside¹³ are regularly placed upon one another, but behind they are connected by a cartilaginous ligament; they are articulated in the form of synarthrosis at the back part of the spinal marrow; behind they have a sharp process having a cartilaginous epiphysis, whence proceeds the roots of nerves running downward, as also muscles extending from the neck to the loins, and filling the space between the ribs and the spine. The ribs are connected to all the intervertebral spaces on the inside, from the neck to the lumbar region, by a small ligament, and before to the sternum, their extremities being spongy and soft; their form is the most arched in man of all animals; for in this part, man is, of all animals, the narrowest in proportion to his bulk. The ribs are united to each vertebra by a small ligament at the place from which the short and broad lateral processes (*transverse processes?*) arise. The sternum is one continuous bone, having lateral pits for the insertion of the ribs; it is of a spongy and cartilaginous structure. The clavicles are rounded in front, having some slight movements at the sternum, but more free at the acromion. The acromion, in man, arises from the scapula differently from most other animals. The scapula is cartilaginous toward the spine, and spongy elsewhere, having an irregular figure externally; its neck and articular cavity cartilaginous; it does not interfere with the movements of the ribs, and is free of all connection with the other bones, except the humerus. The head of the humerus is

articulated with its (*glenoid?*) cavity, by means of a small ligament, and it consists of a rounded epiphysis composed of spongy cartilage, the humerus itself is bent outward and forward, and it is articulated with its (*glenoid?*) cavity by its side, and not in a straight line. At the elbow it is broad, and has condyles and cavities, and is of a solid consistence; behind it is a cavity in which the coronoid process (*olecranon?*) of the ulna is lodged, when the arm is extended; here, too, is inserted the benumbing nerve, which arises from between the two bones of the forearm at their junction, and terminates there.

Mochl. 2

When the nose is fractured, the parts should be modeled instantly, if possible. If the fracture be in its cartilaginous part, introduce into the nostrils a tent formed of caddis, inclosed in the outer skin of a Carthaginian hide, or anything else which does not irritate; the skin is to be glued to the parts displaced, which are to be thus rectified. Bandaging in this case does mischief. The treatment is to consist of flour with manna, or of sulphur with cerate. You will immediately adjust the fragments, and afterward retain them in place with your fingers introduced into the nostrils, and turning the parts into place; then the Carthaginian skin is to be used. Callus forms even when there is a wound; and the same things are to be done, even when there is to be exfoliation of the bones, for this is not of a serious nature.

Mochl. 3

In fractures of the ears, neither bandages nor cataplasms should be used; or, if any bandage be used, it should be put on very tight; the cerate and sulphur should be applied to agglutinate the bandages. When matter forms in the ears, it is found to be more deeply seated than might be supposed, for all parts that are pulpy, and consist of juicy flesh, prove deceptive in such a case. But no harm will result from making an opening, for the parts are lean, watery, and full of mucus. No mention is here made of the places and circumstances which render it fatal to make an opening. The cure is soonest effected by transfixing the ear with a cautery; but the ear is maimed and

diminished in size, if burned across. If opened, one of the gentle medicines for flesh wounds should be used as a dressing.

Mochl. 4

The jaw-bone is often slightly displaced (*subluxated?*), and is restored again; it is dislocated but rarely, especially in gaping; in fact, the bone is never dislocated unless it slips while the mouth is opened wide. It slips, however, the more readily from its ligaments being oblique, supple, and of a yielding nature. The symptoms are: the lower jaw protrudes, it is distorted to the side opposite the dislocation, and the patient cannot shut his mouth; when both sides are dislocated, the jaw projects more, the mouth can be less shut, but there is no distortion; this is shown by the rows of the teeth in the upper and lower jaw corresponding with one another. If, then, both sides be dislocated, and not immediately reduced, the patient for the most part dies on the tenth day, with symptoms of continued fever, stupor, and coma, for the muscles there induce such effects; there is disorder of the bowels attended with scanty and unmixed dejection; and the vomitings, if any, are of the same character. The other variety is less troublesome. The method of reduction is the same in both:—The patient being laid down or seated, the physician is to take hold of his head, and grasping both sides of the jaw-bone with both hands, within and without, he must perform three manoeuvres at once,—rectify the position of the jaw, push it backward, and shut the mouth. The treatment should consist of soothing applications, position, and applying a suitable bandage to support the jaw-bone, so as to cooperate with the reduction.

Mochl. 5

The bone of the shoulder is dislocated downward. I have never heard of any other mode. The parts put on the appearance of dislocation forward, when the flesh about the joint is wasted during consumption, as also seems to be the case with cattle when in a state of leanness after winter. Those persons are most liable to dislocations who are thin, slender, and have humidities about their joints without

inflammation, for it knits the joints. Those who attempt to reduce and rectify dislocations in oxen, commit a blunder, as forgetting that the symptoms arise from the manner in which the ox uses the limb, and that the appearance is the same in a man who is in a similar condition, and forgetting also that Homer has said, that oxen are most lean at that season. In this dislocation, then, when not reduced, the patient cannot perform any of those acts which others do, by raising the arm from the side. I have thus stated who are the persons most subject to this dislocation, and how they are affected. In congenital dislocations the nearest bones are most shortened, as is the case with persons who are *weasel-armed*; the fore-arm less so, and the hand still less; the bones above are not affected. And the parts (near the seat of the injury) are most wasted in flesh; and this happens more especially on the side of the arm opposite the dislocation, and that during adolescence, yet in a somewhat less degree than in congenital cases. The deep-seated suppurations occur most frequently to new-born infants about the joint of the shoulder, and these produce the same consequences as dislocations. In adults, the bones are not so diminished in size, and justly, seeing that the others will not increase as in the former case; but wasting of the flesh takes place, for it is increased, and is diminished every day, and at all ages. And attention should be paid to the force of habit, and to the symptom produced by the tearing away of the acromion, whereby a void is left, which makes people suppose that the humerus is dislocated. The head of the humerus is felt in the armpit, and the patient cannot raise his arm, nor swing it to this side and that, as formerly. The other shoulder shows the difference. Modes of reduction:—The patient himself having placed his fist in the arm pit, pushes up the head of the humerus with it, and brings the hand forward to the breast. Another:—Force it backward, so that you may turn it round. Another:—Apply your head to the acromion, and your hands to the armpit, separate the head of the humerus (*from the side?*), and push the elbow in the opposite direction; or, instead of your knees, another person may turn

aside the elbow, as formerly directed. Or, place the patient on your shoulder, with the shoulder in his armpit. Or, with the heel, something being introduced to fill up the hollow of the armpit, and using the right foot to the right shoulder. Or, with a pestle. Or, with the step of a ladder. Or, by rotation made with piece of wood stretched below the arm. Treatment:—As to attitude, the arm placed by the side, the hand and shoulder raised; the bandaging and adjustment of the parts while in this attitude. If not reduced, the top of the shoulder becomes attenuated.

Mochl. 6

When the acromion is torn away, the appearance is the same as in dislocation of the shoulder; but there is no impediment, except that the bone does not return to its position. The figure should be the same as in dislocation, both as regards bandaging and suspending the limb. The bandaging according to rule.

Mochl. 7

When partial displacement (*sub-luxation?*) takes place at the elbow, either inside or outside, but the sharp point (*olecranon?*) remains in the cavity of the humerus, make extension in a straight line, and push the projecting parts backward and to the sides.

Mochl. 8

In complete dislocations to either side, make extension while the arm is in the position it is put in to be bandaged for a fracture, for thus the rounded part of the elbow will not form an obstacle to it. Dislocation most commonly takes place inward. The parts are to be adjusted by separating the bones as much as possible, so that the end of the humerus may not come in contact with the olecranon, but it is to be carried up and turned round, and not forced in a straight line; at the same time the opposite sides are to be pushed together, and the bones reduced to their place. In these cases rotation of the elbow cooperates; that is to say, turning the arm into a state of supination and pronation; so much for the reduction. With regard to the attitude in which it is to be put,—the hand is to be placed somewhat higher than the elbow,

and the arm by the side; this position suits with it when slung from the neck, is easily borne, is its natural position, and one adapted for ordinary purposes, unless callus form improperly: the callus soon forms. Treatment:—By bandages according to the common rule for articulations, and the point of the elbow is to be included in the bandage.

Mochl. 9

The elbow, when luxated, induces the most serious consequences, fevers, pain, nausea, vomiting of pure bile; and this especially in dislocations backward, from pressure on the nerve which occasions numbness; next to it is dislocation forward. The treatment is the same. The reduction of dislocation backward is by extension and adaptation: the symptom of this variety, loss of the power of extension; of dislocation forward, loss of the power of flexion. In it a hard ball is to be placed in the bend of the elbow, and the fore-arm is to be bent over this while sudden extension is made.

Mochl. 10

Diastasis of the bones may be recognized by examining the part where the vein which runs along the arm divides.

Mochl. 11

In these cases callus is speedily formed. In congenital dislocations, the bones below the seat of the injury are shorter than natural; in this case, the greatest shortening is in the nearest, namely, those of the fore-arm; second, those of the hand; third, those of the fingers. The arm and shoulders are stronger, owing to the nourishment which they receive, and the other arm, from the additional work it has to perform, is still more strong. The wasting of the flesh, if the dislocation was outward, is on the inside; or if otherwise, on the side opposite the dislocation.

Mochl. 12

In dislocation at the elbow, whether outward or inward, extension is to be made with the fore-arm at right angles to the arm; the arm is to be suspended by a shawl passed through the armpit, and a weight is to

be attached to the extremity of the elbow; or force is to be applied with the hands. The articular extremity being properly raised, the parts are to be adjusted with the palms of the hands, as in dislocations of the hands. It is to be bandaged, suspended in a sling, and placed, while in this attitude.

Mochl. 13

Dislocations backward are to be rectified with the palms of the hands along with sudden extension. These two acts are to be performed together, as in other cases of the kind. In dislocation forward, the arm is to bend around a ball of cloth, of proper size, and at the same time replaced.

Mochl. 14

If the displacement be on the other side both these operations are to be performed in effecting the adjustment of the arm. With regard to the treatment,—the position and the bandaging are the same as in the other cases. For all these cases may be reduced by ordinary distention.

Mochl. 15

With regard to the modes of reduction, some act upon the principle of carrying the one piece of bone over the other, some by extension, and some by rotation: these last consist in rapidly turning the arm to this side and that.

Mochl. 16

The joint of the hand is dislocated inward or outward, but most frequently inward. The symptoms are easily recognized; if inward, the patient cannot at all bend his fingers, but if outward, he cannot extend them. Reduction:—By placing the fingers above a table, extension and counter-extension are to be made by assistance, while, with the palm of the hand or the heel on the projecting bone, one presses forward, and from behind, upon the other bone, and lays some soft substance on it; and, if the dislocation be above, the hand is to be turned into a state of pronation; or, if backward, into a state of supination. The treatment is to be conducted with bandages.

Mochl. 17

The whole hand is dislocated either inward, or outward, but especially inward, or to this side or that. Sometimes the epiphysis is displaced, and sometimes there is displacement (*diastasis*) of the one bone from the other. Powerful extension is to be made in this case; and the projecting part is to be pressed upon, and counter-pressure made on the opposite side: both modes being performed at the same time, both backward and laterally, either with the hands on a table, or with the heel. These accidents give rise to serious consequences and deformities; but in time the parts get so strong as to admit of being used. The treatment consists of bandages comprehending the hand and forearm, and splints are to be applied as far as the fingers; when put in splints, they are to be more frequently loosed than in fractures, and more copious allusions of water are to be used.

Mochl. 18

In congenital dislocations the hand becomes shortened, and the atrophy of the flesh is generally on the side opposite the dislocation. In the adult the bones remain of their proper size.

Mochl. 19

The symptoms of dislocation of the finger are obvious, and need not be described. This is the mode of reduction:—By stretching in a straight line, and making pressure on the projecting part, and counter-pressure, at the opposite side, on the other. The proper treatment consists in the application of bandages. When not reduced, the parts unite by callus outside of the joints. In congenital dislocations, and in those which occur during bones below the dislocation are shortened, and the flesh is wasted principally on the side opposite to the dislocation; in the adult the bones remain of their proper size.

Mochl. 20

Dislocation at the hip-joint occurs in four modes, inward most frequently, outward next, the others of equal frequency. The symptoms:—The common, a comparison with the sound leg. The peculiar symptoms of dislocations inward; the head of the bone is felt

at the perineum; the patient cannot bend his leg as formerly; the limb appears elongated, and to a great extent, unless you bring both limbs into the middle space between them in making a comparison of them; and the foot and the knee are inclined outward. If the dislocation has taken place from birth, or during one's growth, the thigh is shortened, the leg less so, and the others according to the same rule; the fleshy parts are atrophied, especially on the outside. Such persons are afraid to stand erect, and crawl along on the sound limb; or, if compelled, they walk with one or two staves, and bear up the affected limb; and the smaller the limb so much the more do they walk. If the accident happens to adults the bones remain of their proper size, but the flesh is wasted, as formerly described; the patients walk in a wriggling manner, like oxen; they are bent toward the flank, and the buttock on the uninjured side is prominent; for the uninjured limb must necessarily come below that it may support the body, whilst the other must be carried out of the way, as it cannot support the body, like those who have an ulcer in the foot. They poise the body by means of a staff on the sound side, and grasp the affected limb with the hand above the knee so as to carry the body in shifting from one place to another. If the parts below the hip-joint be used, the bones below are less atrophied, but the flesh more.

Mochl. 21

The symptoms and attitudes in dislocation outward are the opposite, and the knee and foot incline a little inward. When it is congenital, or occurs during adolescence, the bones do not grow properly; according to the same rule, the bone of the hip-joint is somewhat higher than natural, and does not grow proportionally. In those who have frequent dislocations outward, without inflammation, the limb is of a more humid (flabby?) temperament than natural, like the thumb, for it is the part most frequently dislocated, owing to its configuration; in what persons the dislocation is to a greater or less extent; and in what persons it is more difficultly or easily produced; in what there is reason to hope that it can be speedily reduced, and in what not; and

the remedy for this; and in what cases the dislocation frequently happens, and treatment of this. In dislocation outward from birth, or during adolescence, or from disease, (and it happens most frequently from disease, in which case there is sometimes exfoliation of the bone, but even where there is no exfoliation), the patients experience the same symptoms, but to an inferior degree to those in dislocations inward, if properly managed so that in walking they can put the whole foot to the ground and lean to either side. The younger the patient is, the greater care should be bestowed on him; when neglected, the case gets worse; when attended to, it improves; and, although there be atrophy in all parts of the limb, it is to a less extent.

Mochl. 22

When there is a dislocation on both sides, the affections of the bones are the same; the flesh is well developed, except within, the nates protrude, the thighs are arched, unless there be sphacelus. If there be curvature of the spine above the hip-joint, the patients enjoy good health, but the body does not grow, with the exception of the head.

Mochl. 23

The symptoms of dislocation backward are:—The parts before more empty, behind they protrude, the foot straight, flexion impossible, except with pain, extension least of all: in these the limb is shortened. They can neither extend the limb at the ham, nor at the groin, unless it be much raised, nor can they bend it. The uppermost joint, in most cases, takes the lead: this is common in joints, nerves, muscles, intestines, uteri, and other parts. There the bone of the hip-joint is carried backward to the nates, and on that account it is shortened, and because the patient cannot extend it. The flesh of the whole leg is wasted in all cases, in which most, and to what extent, has been already stated. Every part of the body which performs its functional work is strong, but, notwithstanding, if inactive, it gets into a bad condition, unless its inactivity arise from fatigue, fever, or inflammation. And in dislocations outward, the limb is shortened, because the bone is lodged in flesh which yields; but, not

withstanding, in dislocations inward, it is longer, because the bone is lodged on a projecting bone. Adults, then, who have this dislocation unreduced, are bent at the groins in walking, and the other ham is flexed; they scarcely reach the ground with the ball of the foot; they grasp the limb with the hand, and walk without a staff if they choose; if the staff be too long, their foot cannot reach the grounds—if they wish to reach the ground, they must use a short staff. There is wasting of the flesh in cases attended with pain; and the inclination of the leg is forward, and the sound leg in proportion. In congenital cases, or when in adolescence, or from disease, the bone is dislocated (under what circumstances will be explained afterward), the limb is particularly impaired, owing to the nerves and joints not being exercised, and the knee is impaired for the reasons stated. These persons, keeping the limb bent, walk with one staff or two. But the sound limb is in good flesh from usage.

Mochl. 24

In dislocations forward the symptoms are the opposite: a vacuity behind, a protuberance before; of all motions they can least perform flexion, and extension best; the foot is straight, the limb is of the proper length at the heel; at its extremity the foot a little turned up; they are especially pained at first: of all these dislocations retention of urine occurs most frequently in this variety, because the bone is lodged among important nerves. The fore parts are stretched, do not grow, are diseased, and are obnoxious to premature decay; the back parts are wrinkled. In the case of adults, they walk erect, resting merely on the heel, and this they do decidedly if they can take great steps; but they drag it along; the wasting is least of all in this variety of dislocation, owing to their being able to use the limb, but the wasting is most behind. The whole limb being straighter than natural they stand in need of a staff on the affected side. When the dislocation is congenital, or has occurred during adolescence, if properly managed, the patient has the use of the limb as well as adults (otherwise?) have of it. But, if neglected, it is shortened and extended, for in such cases

the joint is generally in a straight position. The diminution of the bones, and wasting of the fleshy parts, are analogous.

Mochl. 25

In reduction—the extension of the thigh is to be powerful, and the adjustment what is common in all such cases, with the hands, or a board, or a lever, which, in dislocations inward, should be round, and in dislocations outward, flat; but it is mostly applicable in dislocations outward. Dislocations inward are to be remedied by means of bladders, extending to the bare part of the thigh, along with extension and binding together of the limbs. The patient may be suspended, with his feet a little separated from one another, and then a person inserting his arm within the affected limb, is to suspend himself from it, and perform extension and readjustment at the same time; and this method is sufficient in dislocations forward and the others, but least of all in dislocations backward. A board fastened under the limb, like the board fastened below the arm in dislocations at the shoulder, answers in dislocations inward, but less so in the other varieties. Along with extension you will use pressure either with the foot, the hand, or a board, especially in dislocations forward and backward.

Mochl. 26

Dislocations at the knee are of a milder character than those of the elbow, owing to the compactness and regularity of the joint; and hence it is more readily dislocated and reduced. Dislocation generally takes place inward, but also outward and backward. The methods of reduction are—by circumflexion, or by rapid excalcitration, or by rolling a fillet into a ball, placing it in the ham, and then letting the patient's body suddenly drop down on his knees: this mode applies best in dislocations backward. Dislocations backward, like those of the elbows, may also be reduced by moderate extension. Lateral dislocations may be reduced by circumflexion or excalcitration, or by extension (but this is most applicable in dislocation backward), but also by moderate extension. The adjustment is what is common in all. If not reduced, in dislocations backward, they cannot bend the leg and

thigh upon one another, but neither can they do this in the others except to a small extent; and the fore parts of the thigh and leg are wasted. In dislocations inward they are bandy-legged, and the external parts are atrophied. But, in dislocations outward, they incline more outward, but are less lame, for the body is supported on the thicker bone, and the inner parts are wasted. The consequences of a congenital dislocation, or one occurring during adolescence, are analogous to the rule formerly laid down.

Mochl. 27

Dislocations at the ankle-joint require strong extension, either with the hands or some such means, and adjustment, which at the same time effects both acts; this is common in all cases.

Mochl. 28

Dislocations of the bones of the foot are to be treated like those of the hand.

Mochl. 29

Dislocations of the bones connected with the leg, if not reduced, whether occurring at birth or during adolescence, are of the same character as those in the hand.

Mochl. 30

Persons who, in jumping from a height, have pitched on the heel, so as to occasion diastasis (separation) of the bones, ecchymosis of the veins, and contusion of the nerves,—when these symptoms are very violent, there is danger that the parts may sphacelate, and give trouble to the patient during the remainder of his life; for these bones are so constructed as to slip past one another, and the nerves communicate together. And, likewise in cases of fracture, either from an injury in the leg or thigh, or in paralysis of the nerves connected with these parts, or, when in any other case of confinement to bed the heel, from neglect, becomes blackened, in all these cases serious effects result therefrom. Sometimes, in addition to the sphacelus, very acute fevers supervene, attended with hiccup, tumors, aberration of intellect, and speedy death, along with lividity of the large blood-

vessels, and gangrene. The symptoms of the exacerbations are these: if the ecchymosis, the blackened parts, and those around them, be somewhat hard and red, and if lividity be combined with the hardness, there is danger of mortification; but, not withstanding, if the parts are sublivid, or even very livid and diffused, or greenish and soft, these symptoms, in all such cases, are favorable. The treatment consists in the administration of hellebore, if they be free from fever, but otherwise, they are to have *oxyglyky* for drink, if required. Bandaging,—agreeably to the rule in other joints; but this is to be attended to also,—the bandages should be numerous, and softer than usual; compression less; more water than usual to be used in the allusions; to be applied especially to the heel. The same object should be sought after in the position as in the bandaging, namely, that the humors may not be determined to the heel; the limb to be well laid should have the heel higher than the knee. Splints not to be used.

Mochl. 31

When the foot is dislocated, either alone, or with the epiphysis, the displacement is more apt to be inward. If not reduced, in the course of time the parts of the hips, thigh, and leg, opposite the dislocation, become attenuated. Reduction:—As in dislocation at the wrist; but the extension requires to be very powerful. Treatment:—Agreeably to the rule laid down for the other joints. Less apt to be followed by serious consequences than the wrist, if kept quiet. Diet restricted, as being in an inactive state. Those occurring at birth, or during adolescence, observe the rule formerly stated.

Mochl. 32

With regard to slight congenital dislocations, some of them can be rectified, especially club-foot. There is more than one variety of club-foot. The treatment consists in modeling the foot like a piece of wax; applying resinous cerate, and numerous bandages; or a sole, or a piece of lead is to be bound on, but not upon the bare skin; the adjustment and attitudes to correspond.

Mochl. 33

If the dislocated bones cause a wound in the skin, and protrude, it is better to let them alone, provided only they are not allowed to hang, nor are compressed. The treatment consists in applying pitched cerate, or compresses dipped in hot wine (for cold is bad in all such cases), and certain leaves; but in winter unwashed wool may be applied as a cover to the part; neither cataplasms nor bandaging; restricted diet. Cold, great weight, compression, violence, restricted position, all such are to be accounted as fatal measures. When treated moderately (they escape), maimed and deformed; for, if the dislocation be at the ankle, the foot is drawn upward, and, if elsewhere, according to the same rule. The bones do not readily exfoliate; for only small portions of them are denuded, and they heal by narrow cicatrices. The danger is greatest in the greatest joints, and those highest up. The only chance of recovery is, if they are not reduced, except at the fingers and hand, and in these cases the danger should be announced beforehand. Attempts at reduction to be made on the first or second day; or, if not accomplished then, on the tenth, by no means on the fourth. Reduction by levers. Treatment:—As in injuries of the bones of the head, and the part is to be kept hot; and it is better to give hellebore immediately after the parts have been reduced. With regard to the other bones, it should be well known, that, if replaced, death will be the consequence; the more surely and expeditiously, the greater the articulation, and the more high its situation. Dislocation of the foot is attended with spasm (tetanus) and gangrene; and if, upon its being replaced, any of these symptoms come on, the chance of recovery, if there be any chance, is in displacing it anew; for spasms do not arise from relaxation, but from tension of the parts.

Mochl. 34

Excision, either of articular bones or of pieces of bones, when not high up in the body, but about the foot or the hand, is generally followed by recovery, unless the patient die at once from deliquium animi. Treatment:—As in injuries of the head; warmth.

Mochl. 35

Sphacelus of the fleshy parts is produced by the tight compression of bleeding wounds, and by pressure in the fractures of bones, and by blackening, arising from bandages. And in those cases in which a portion of the thigh or arm, both the bones and the flesh drop off, many recover, the case being less dangerous than many others. In cases, then, connected with fracture of the bones, the separation of the flesh quickly takes place, but the separation of the bone, at the boundary of its denuded part, is slower in taking place. But the parts below the seat of the injury, and the sound portion of the body, are to be previously taken away (for they die previously), taking care to avoid producing pain, for *deliquium animi* may occasion death. The bone of the thigh in such a case came away on the eightieth day, but the leg was removed on the twentieth day. The bones of the leg, in a certain case, came away at the middle of the sixtieth day. In these cases the separation is quick or slow, according to the compression applied by the physician. When the compression is gently applied the bones do not drop off at all, neither are they denuded of flesh, but the gangrene is confined in the more superficial parts. The treatment of such cases must be undertaken; for most of them are more formidable in appearance than in reality. The treatment should be mild, but, notwithstanding, with a restricted diet; hemorrhages and cold are to be dreaded; the position, so as that the limb may be inclined upward, and afterward, on account of the purulent abscess, horizontally, or such as may suit with it. In such cases, and in mortifications, there are usually, about the crisis, hemorrhages and crisis, hemorrhages and violent diarrhoeas, which, however, only last for a few days; the patients do not lose their appetite, neither are they feverish, nor should they be put upon a reduced diet.

Mochl. 36

Displacement of the spine, if inward, threatens immediate death, attended with retention of urine and loss of sensibility. Outward, the accident is free from most of these bad effects, much more so than

where there is merely concussion without displacement; the effects in the former case being confined to the spot affected, whereas in the latter they are further communicated to the whole body, and are of a mortal character. In like manner, when the ribs are fractured, whether one or more, provided there be no splinters, there is rarely fever, spitting of blood, and sphacelus, and ordinary treatment without evacuation will suffice, provided there be no fever;—bandaging, according to rule; and the callus forms in twenty days, the bone being of a porous nature. But in cases of contusion, tubercles form, along with cough, suppurating sores, and sphacelus of the ribs, for nerves from all the parts run along each rib. In many of these cases haemoptysis and empyema also take place. The management of this case consists in careful treatment, bandaging according to rule, diet at first restricted, but afterward more liberal, quiet, silence, position, bowels, and venereal matters regulated. Even when there is no spitting of blood, these contusions are more painful than fractures, and are more subject in time to relapses; and when any mucous collection is left in the part, it makes itself be felt in disorders of the body. Treatment:—burning, when the bone is affected, down to the bone, but not touching the bone itself; if in the intercostal space, the burning must not extend through it, nor be too superficial. In sphacelus of the ribs, tents are to be tried, all other particulars will be stated afterward: but they should be learned by sight rather than by words, namely, food, drink, heat, cold, attitude; medicines, dry, liquid, red, dark, white, sour, for the ulcers, and so with regard to the diet.

Mochl. 37

Displacements (*of the vertebrae*) from a fall rarely admit of being rectified, and those above the diaphragm are most difficult to rectify. When the accident happens to children, the body does not grow, with the exception of the legs, the arms, and head. Excurvation, in adults, speedily relieves the individual from the disease he is laboring under, but in time it renews its attack, with the same symptoms as in children, but of a less serious nature. Some individuals have borne this

affection well, and have turned out to be brawny and fat. But few of them have lived to the age of sixty. Lateral curvatures also occur, the proximate cause of which is the attitudes in which these persons lie. These cases have their prognostics accordingly.

Mochl. 38

The rule for the reduction and adjustment:—The axle, the lever, the wedge, pressure above; the axle to separate, the lever to push aside. Reduction and adjustment are to be accomplished by forcible extension, the parts being placed in such a position as will facilitate the conveying of the displaced bone over the extremity of the bone from which it was displaced: this is to be accomplished either with the hands, or by suspension, or axles, or turned round something. With the hands this is to be effected properly, according to the structure of the parts. In the case of the wrist and elbow, the parts are to be forced asunder, at the wrist in the line of the elbow, and the elbow with the fore-arm at a right angle with the arm, as when it is suspended in a sling. When we want to separate the protruding bones, and force them into place, in the case of the fingers, the toes, or the wrist, the proper separation may be made by hands, while the projecting part is forced into its place by pressing down with the heel or the palm of the hand upon some resisting object, while something moderately soft is laid under the projecting part, but nothing such under the other, and then pressure is to be made backward and downward, whether the dislocation be inward or outward. In lateral displacement, pressure and counter-pressure must be made on the opposite sides. Displacements forward can be reduced neither by sneezing, nor coughing, nor by the injection of air, nor by the cupping-instrument; and if anything can do good in such a case, it is extension. People are deceived in fractures of the spinal processes, the pain of which causing the patient to stoop forward, the case is taken for dislocation inward; these fractures heal speedily and easily. Dislocation outward is to be remedied by succussion, when high up, toward the feet; and when situated low down, in the contrary direction; the part is to be

pressed back into its place, either with the foot or a board. Dislocations to either side, if they admit of any remedy, are to be treated by extension, and suitable attitudes, with regimen. The whole apparatus should be broad, soft, and strong; or otherwise, they should be wrapped in rags; before being used, they should all be prepared proportionately to the length, height, and breadth. In applying extension to the thigh, for example, the bands should be fastened at the ankle and above the knee, these stretching in the same direction, another band to be passed by the loins, and around the armpits, and by the perineum and thigh, one end passing up the breast and the other along the back, these all stretching in the same direction and being fastened either to a piece of wood resembling a pestle or to an axle. When this is done on a couch, either of its feet is to be fastened to the threshold, and a strong block of wood is to be laid across the other, and the pieces of wood resembling a pestle are to be raised on these, to make extension and counter-extension; the naves of a wheel are to be fastened in the floor, or a ladder is to be adjusted, so that extension may be made in both directions. The thing commonly used is a bench six cubits long, two cubits broad, one fathom in thickness, having two low axles at this end and that, and having at its middle two moderate sized pillars, to which is to be adjusted a transverse piece of wood like the step of a ladder, which is to receive the piece of wood tied below the limb, as is done in dislocation at the shoulder; and the bench is to have excavations like trays, smooth, four inches in breadth and depth, and at such an interval as to leave room for the lever used to reduce the limb. In the middle of the bench a square hole is to be scooped out to receive a small pillar, which, being adjusted to the perineum, will obviate the tendency of the body to slip downward, and being rather loose may act somewhat as a lever. In certain occasions a piece of wood is required, which is inserted into a hole scooped out of the wall; the other end of it is then to be pressed down, something moderately soft being placed under it.

In those cases where the bone of the palate has exfoliated, the nose sinks in its middle. In contusions of the head without a wound, either from a fall, a fracture, or pressure, in certain of these cases acrid humors descend from the head to the throat, and from the wound in the head to the liver and thigh.

Mochl. 40

The symptoms of subluxations and luxations, and where, and how, and how much these differ from one another. And the cases in which the articular cavity has been broke off, and in which the ligament has been torn, and in which the epiphysis has broken in which, and how, when the limb consists of two bones, one or both are broken: in consequence of these the dangers, chances in which bad, and when the injuries will result in death, and when in recovery. What cases are to be reduced or attempted, and when, and which, and when not; the hopes and dangers in these cases. Which and when congenital dislocations are to be undertaken: the parts in a state of growth, the parts fully grown, and why sooner, or slower: and why a part becomes maimed, and how, and how not: and why a certain part is atrophied, and where, and how, and in what cases to a less extent. And why fractured parts unite sooner or slower, how distortions and callosities form, and the remedy for them. In what cases there are external wounds, either at first or afterwards: in what fractures the bones are shortened, and in what not: in what cases the fractured bones protrude, and when they protrude most: in what cases dislocated bones protrude. That physicians are deceived, and by what means, in what they see, and in what they devise, regarding affections, and regarding cures. Established rules with regard to bandaging: preparation, presentation of the part, extension, adjustment, friction, bandaging, suspension and placing of the limb, attitude, seasons, diet. The more porous parts heal fastest, and vice versa. Distortions, where the bones are crooked. Flesh and tendons wasted on the side of the dislocation. The force used in reduction to be applied at as great a distance as possible from the seat of the displacement. Of nerves

(*ligaments?*), those which are in motion and in humidity (*flabby?*) are of a yielding nature; those that are not, less so. In every dislocation the most speedy reduction is best. Reduction not to be made while the patient is in a febrile state, nor on the fourth or fifth day; and least of all, in those of the elbow, and all cases which induce torpor; the soonest the best, provided the inflammatory stage be avoided. Parts torn asunder, whether nerves, or cartilages, or epiphyses, or parts separated at symphyses, cannot possibly be restored to their former state; but callus is quickly formed in most cases, yet the use of the limb is preserved. Of luxations, those nearest the extremities are least dangerous. Those joints which are most easily dislocated are the least subject to inflammation. Those which have been least inflamed, and have not been subjected to after-treatment, are most liable to be dislocated anew. Extension should be made in the position most calculated to enable the one bone to clear the extremity of the other, attention being paid to configuration and place. Adjustment to be made in the direction of the displacement; to push the displaced limb straight backward and sideways. Parts suddenly drawn aside are to be suddenly drawn back by a rotatory motion. Articulations which have been oftenest dislocated are the most easily reduced; the cause is the conformation of the nerves (*ligaments?*) or of the bones; of the ligaments that they are long and yielding; and of the bones, the shallowness of the articular cavity, and roundness of the head [of the bone that enters it]. Usage, by its friction, forms a new socket. The cause—the disposition, and habit, and age. A part somewhat mucous is not subject to inflammation.

Mochl. 41

In those cases where there are wounds, either at first, or from protrusion of the bones; or afterwards, from pruritus, or irritation; in the latter case you are immediately to unloose the bandages, and having applied pitched cerate to the wound, bandage the limb, placing the head of the roller upon the wound, and proceeding otherwise as if there were no wound in the case; for thus will the swelling be reduced

as much as possible, and the wound will suppurate most quickly, and the diseased parts will separate, and when it becomes clean the wound will most quickly heal. Splints are not to be applied to the place, nor is it to be bound tight. Proceed thus when no large bones exfoliate, but not in the latter case, for then there is great suppuration, and the same treatment is not applicable, but the parts require to be exposed to the air on account of the abscesses. In such cases where the bones protrude, and whether reduced or not, bandaging is not befitting, but distention is to be practiced by means of rolls of cloth, made like those used upon shackles; one of these is to be placed at the ankle, and the other at the knee; they are to be flattened toward the leg, soft, strong, and having rings; and rods made of cornel, and of a proper length and thickness are to be adjusted to them, so as to keep the parts distended; and straps, attached to both extremities, are to be inserted into the rings, so that the extremities being fixed into the rolls, may effect distention. Treatment:—Pitched cerate, in a hot state; the attitudes, position of the foot and hip; regulated diet. The bones which have protruded through the skin are to be replaced the same day, or next; not on the fourth or fifth, but when the swelling has subsided. Reduction is to be performed with levers; when the bone does not present any place upon which the lever can rest, a portion of the part which prevents this is to be sawed off. But the denuded parts will drop off, and the limb become shortened.

Mochl. 42

Dislocations at the joints are to a greater and less extent. Those that are to a less extent are the most easily reduced; those that are to a greater extent occasion lesions of the bones, of the ligaments, of the joints, of the fleshy parts, and of the attitudes. The thigh and arm resemble one another very much in their dislocations.

On Ulcers

Ulc. 1

We must avoid wetting all sorts of ulcers except with wine, unless the ulcer be situated in a joint. For, the dry is nearer to the sound, and the wet to the unsound, since an ulcer is wet, but a sound part is dry. And it is better to leave the part without a bandage unless a cataplasm be applied. Neither do certain ulcers admit of cataplasms, and this is the case with the recent rather than the old, and with those situated in joints. A spare diet and water agree with all ulcers, and with the more recent rather than the older; and with an ulcer which either is inflamed or is about to be so; and where there is danger of gangrene; and with the ulcers an inflammation in joints; and where there is danger of convulsion; and in wounds of the belly; but most especially in fractures of the head and thigh, or any other member in which a fracture may have occurred. In the case of an ulcer, it is not expedient to stand; more especially if the ulcer be situated in the leg; but neither, also, is it proper to sit or walk. But quiet and rest are particularly expedient. Recent ulcers, both the ulcers themselves and the surrounding parts, will be least exposed to inflammation, if one shall bring them to a suppuration as expeditiously as possible, and if the matter is not prevented from escaping by the mouth of the sore; or, if one should restrain the suppuration, so that only a small and necessary quantity of pus may be formed, and the sore may be kept dry by a medicine which does not create irritation. For the part becomes inflamed when rigor and throbbing supervene; for ulcers then get inflamed when suppuration is about to form. A sore suppurates when the blood is changed and becomes heated; so that becoming putrid, it constitutes the pus of such ulcers. When you seem to require a cataplasm, it is not the ulcer itself to which you must apply the cataplasm, but to the surrounding parts, so that the pus may escape and the hardened parts may become soft. Ulcers formed either from the parts having been cut through by a sharp instrument, or excised, admit of medicaments for bloody wounds (ἑναιμα), and which will prevent suppuration by being desiccant to a certain degree. But, when the flesh has been contused and roughly cut by the weapon, it is

to be so treated that it may suppurate as quickly as possible; for thus the inflammation is less, and it is necessary that the pieces of flesh which are bruised and cut should melt away by becoming putrid, being converted into pus, and that new flesh should then grow up.

Ulc. 2

In every recent ulcer, except in the belly, it is expedient to cause blood to flow from it abundantly, and as may seem seasonable; for thus will the wound and the adjacent parts be less attacked with inflammation. And, in like manner, from old ulcers, especially if situated in the leg, in a toe or finger, more than in any other part of the body. For when the blood flows they become drier and less in size, as being thus dried up. It is this (*blood?*) especially which prevents such ulcers from healing, by getting into a state of putrefaction and corruption. But, it is expedient, after the flow of the blood, to bind over the ulcer a thick and soft piece of sponge, rather dry than wet, and to place above the sponge some slender leaves. Oil, and all things of an emollient and oily nature, disagree with such ulcers, unless they are getting nearly well. Neither does oil agree with wounds which have been recently inflicted, nor yet do medicines formed with oil or suet, more especially if the ulcer stands in need of more cleansing. And, in a word, it is in summer and in winter that we are to smear with oil these sores that require such medicines.

Ulc. 3

Gentle purging of the bowels agrees with most ulcers, and in wounds of the head, belly, or joints, where there is danger of gangrene, in such as require sutures, in phagedaenic, spreading and in otherwise inveterate ulcers.

Ulc. 4

And when you want to apply a bandage, no plasters are to be used until you have rendered the sore dry, and then indeed you may apply them. The ulcer is to be frequently cleaned with a sponge, and then a dry and clean piece of cloth is to be frequently applied to it, and in

this way the medicine which it is supposed will agree with it is to be applied, either with or without a bandage.

Ulc. 5

The hot season agrees better than winter with most ulcers, except those situated in the head and belly; but the equinoctial season agrees still better with them.

Ulc. 6

Ulcers which have been properly cleansed and dried as they should be, do not usually get into a the state.

Ulc. 7

When a bone has exfoliated, or has been burned, or sawed, or removed in any other way, the cicatrices of such ulcers become deeper than usual.

Ulc. 8

Ulcers which are not cleansed, are not disposed to unite if brought together, nor do the lips thereof approximate of their own accord. When the points adjoining to an ulcer are inflamed, the ulcer is not disposed to heal until the inflammation subside, nor when the surrounding parts are blackened by mortification, nor when a varix occasions an overflow of blood in the part, is the ulcer disposed to heal, unless you bring the surrounding parts into a healthy condition.

Circular ulcers, if somewhat hollow, you must scarify all along their edges, or to the extent of half the circle, according to the natural stature of the man.

Ulc. 9

When erysipelas supervenes upon any sore, you must purge the body, in the way most suitable to the ulcer, either upward or downward.

Ulc. 10

When swelling arises around an ulcer, and if the ulcer remain free from inflammation, there will be a deposit of matter in process of time. And whatever ulcer gets swelled along with inflammation and does not subside as the other parts subside which became inflamed

and swelled at the same time, there is a danger that such an ulcer may not unite. When from a fall, or in any other way, a part has been torn or bruised, and the parts surrounding the ulcer have become swelled, and, having suppurated, matter flows from the swelling by the ulcer, if in such cases a cataplasm be required, it should not be applied to the sore itself, but to the surrounding parts, so that the pus may have free exit, and the indurated parts may be softened. But when the parts are softened as the inflammation ceases, then the parts which are separated are to be brought toward one another, binding on sponges and applying them, beginning from the sound parts and advancing to the ulcer by degrees. But plenty of leaves are to be bound above the sponge. When the parts are prevented from coming together by a piece of flesh full of humors, it is to be removed. When the ulcer is deep seated in the flesh, it is swelled up, both from the bandaging and the compression. Such an ulcer should be cut up upon a director (specillum) if possible, at the proper time, so as to admit a free discharge of the matter, and then the proper treatment is to be applied as may be needed. For the most part, in every hollow ulcer which can be seen into direct without being any swelling present, if there be putrefaction in it, or if the flesh be flabby and putrid, such an ulcer, and the parts which surround it, will be seen to be black and somewhat livid. And of corroding ulcers, those which are phagedaenic, spread and corrode most powerfully, and, in this case, the parts surrounding the sore will have a black and sub-livid appearance.

Ulc. 11

Cataplasms for swellings and inflammation in the surrounding parts. Boiled mullein, the raw leaves of the trefoil, and the boiled leaves of the epipetrum, and the poley, and if the ulcer stand in need of cleansing, all these things also cleanse; and likewise the leaves of the fig-tree, and of the olive, and the horehound, all these are to be boiled; and more especially the chaste-tree, and the fig, and the olive, and the leaves of the pomegranate are to be boiled in like manner. These are to

be used raw: and the leaves of the mallow pounded with wine, and the leaves of rue, and those of the green origany. With all these, linseed is to be boiled up and mixed by pounding it as a very fine powder. When there is danger of erysipelas seizing the ulcers, the leaves of woad are to be pounded and applied raw in a cataplasm along with linseed, or the linseed is to be moistened with the juice of strychnos or of woad, and applied as a cataplasm. When the ulcer is clean, but both it and the surrounding parts are inflamed, lentil is to be boiled in wine and finely triturated, and, being mixed with a little oil, it is to be applied as a cataplasm; and the leaves of the hip-tree are to be boiled in water and pounded in a fine powder and made into a cataplasm; and apply below a thin, clean piece of cloth wetted in wine and oil; and when you wish to produce contraction, prepare the leaves of the hip-tree like the lentil, and the cress; wine and finely-powdered linseed are to be mixed together. And this is proper: linseed, and raw chaste-tree, and Melian alum, all these things being macerated in vinegar.

Ulc. 12

Having pounded the white unripe grape in a mortar of red bronze, and passed it through the strainer, expose it to the sun during the day, but remove it during the night, that it may not suffer from the dew; rub it constantly during the day, so that it may dry equally, and may contract as much virtue as possible from the bronze: let it be exposed to the sun for as great a length of time as till it acquire the thickness of honey; then put it into a bronze pot with the fresh honey and sweet wine, in which turpentine resin has been previously boiled, boil the resin in the wine until it become hard like boiled honey; then take out the resin and pour off the wine: there should be the greatest proportion of the juice of unripe grape, next of the wine, and third of the honey and myrrh, either the liquid (*stacte*) or otherwise. The finest kind is to be levigated and moistened by having a small quantity of the same wine poured on it; and then the myrrh is to be boiled by itself, stirring it in the wine; and when it appears to have attained the proper degree of thickness, it is to be poured into the juice of the

unripe grape; and the finest natron is to be toasted, and gently added to the medicine, along with a smaller quantity of the flowers of copper (*flos aeris*) than of the natron. When you have mixed these things, boil for not less than three days, on a gentle fire made with fuel of the fig-tree or with coals, lest it catch fire. The applications should all be free from moisture, and the sores should not be wetted when this medicine is applied in the form of liniment. This medicine is to be used for old ulcers, and also for recent wounds of the glans penis, and ulcers on the head and ears. Another medicine for the same ulcers:—The dried gall of an ox, the finest honey, white wine, in which the shavings of the lotus have been boiled, frankincense, of myrrh an equal part, of saffron an equal part, the flowers of copper, in like manner of liquids, the greatest proportion of wine, next of honey, and least of the gall. Another:—Wine, a little cedar honey, of dried things, the flowers of copper, myrrh, dried pomegranate rind. Another:—Of the roasted flower of copper half a drachm, of myrrh two half-drachms, of saffron three drachms, of honey a small quantity, to be boiled with wine. Another:—Of frankincense a drachm, of gall a drachm, of saffron three drachms; let each of these be dried and finely levigated, then, having mixed, triturate in a very strong sun, pouring in the juice of an unripe grape, until it become of a gelatinous consistence, for three days; then let them be allowed to macerate in an austere, dark-colored, fragrant wine, which is gradually poured upon them. Another:—Boil the roots of the holm-oak in sweet white wine; and when it appears to be properly done, having poured off two parts of the wine, and of the lees of wine as free of water as possible one part; then boil, stirring it, so that it may not be burnt, at a gentle fire, until it appear to have attained the proper consistence. Another:—The other things are to be the same; but, notwithstanding, instead of the wine, use the strongest white vinegar, and dip into it wool as greasy as can be procured, and then, moistening it with the lees of oil, boil, and pour in the juice of the wild fig-tree, and add Melian alum, and natron, and the flowers of copper, both toasted. This cleanses the ulcers better than the former,

but the other is no less desiccant. Another:—Dip the wool in a very little water; and then, having added a third part of wine, boil until it attain the proper consistence. By these, recent ulcers are most speedily prevented from getting into a state of suppuration.

Another:—Sprinkle on it dried wakerobin, and add the green bark of the fig-tree, pounding it in the juice: do this with or without wine, and along with honey. Another:—Boiling the shavings of lotus with vinegar (the vinegar should be white); then mix the lees of oil and raw tar-water, and use it as a liniment or wash, and bandage above.

Ulc. 13

These things in powder prevent recent wounds from suppurating, or they may be used for cleansing the sore along with vinegar, or for sponging with wine.

Another:—Sprinkle (*on the sore?*) lead finely triturated with the recement of copper; and sprinkle on it, also, the shavings of lotus, and the scales of copper, and alum, and chalcitis, with copper, both alone, and with the shavings of lotus. And otherwise, when it is wanted to use these in a dry state, do it with the Illyrian spodos triturated with the shavings, and with the shavings alone. And the flowers of silver alone, in the finest powder; and birthwort, when scraped and finely pounded, may be sprinkled on the part.

Ulc. 14

Another, for bloody sores myrrh, frankincense, galls, verdigris the roasted flower of copper, Egyptian alum roasted, vine flowers, grease of wool, plumbago, each of these things is to be diluted, in equal proportions, with wine like the former. And there is another preparation of the same:—The strongest vinegar of a white color, honey, Egyptian alum, the finest natron; having toasted these things gently, pour in a little gall; this cleanses fungous ulcers, renders them hollow, and is not pungent. Another:—The herb with the small leaves, which gets the name of *Parthenium parviflorum*, and is used for removing thymia (*warts?*) from the glans penis, alum, chalcitis, a little

crude Melian alum (?); sprinkle a little dried elaterium, and a little dried pomegranate rind in like manner.

Ulc. 15

The herb which has got the name of lagopyrus, fills up hollow and clean ulcers; (when dried it resembles wheat; it has a small leaf like that of the olive, and more long;) and the leaf of horehound, with oil. Another:—The internal fatty part, resembling honey, of a fig much dried, of water two parts, of linseed not much toasted and finely levigated, one part. Another:—Of the dried fig, of the flower of copper levigated a little, and the juice of the fig. The preparation from dried fig:—The black chamaeleon, the dried gall of an ox, the other things the same. Of the powders:—Of the slender cress in a raw state, of horehound, of each equal parts; of the dried fig, two parts; of linseed, two parts; the juice of the fig. When you use any of these medicines, apply above it compresses wetted in vinegar, apply a sponge about the compresses and make a If the surrounding parts be in an inflamed state, apply to them any medicine which may appear suitable.

Ulc. 16

If you wish to use a liquid application, the medicine called *caricum* may be rubbed in, and the bandages may be applied as formerly described upon the same principle. The medicine is prepared of the following ingredients:—Of black hellebore, of sandarach, of the flakes of copper, of lead washed, with much sulphur, arsenic, and cantharides. This may be compounded so as may be judged most proper, and it is to be diluted with oil of juniper. When enough has been rubbed in, lay aside the medicine, and apply boiled wakerobin in a soft state, either rubbing it in dry, or moistening it with honey. But if you use the *caricum* in a dry state, you must abstain from these things, and sprinkle the medicine on the sore. The powder from hellebore and sandarach alone answers.

Ulc. 17

Another liquid medicine:—The herb, the leaf of which resembles the arum (wakerobin) in nature, but is white, downy, of the size of the ivy-

leaf: this herb is applied with wine, or the substance which forms upon the branches of the ilex, when pounded with wine, is to be applied. Another:—The juice of the grape, the strongest vinegar, the flower of copper, natron, the juice of the wild fig-tree. Alum, the most finely levigated, is to be put into the juice of the wild grape, and it is to be put into a red bronze mortar and stirred in the sun, and removed when it appears to have attained proper consistence.

These are other powders:—Black hellebore, as finely levigated as possible, is to be sprinkled on the sore while any humidity remains about it, and while it continues to spread. The bandaging is the same as when plasters are used. Another, in like manner:—The driest lumps of salt are to be put into a copper, or earthen pot, of equal size, as much as possible, and not large, and the finest honey, of double the size of the salt, as far as can be guessed, is to be poured upon the lumps of salt, then the vessel is to be put upon coals and allowed to sit there until the whole is consumed. Then, having sponged the ulcer and cleansed it, bandage it as before, and compress it a little more. Next day, wherever the medicine has not been taken in, sprinkle it on, press it down, and bandage. But when you wish to remove the medicine, pour in hot vinegar until it separate, and again do the same things, sponging it away, if necessary. Another corrosive powder:—Of the most finely-levigated misy, sprinkle upon the moist and gangrenous parts, and a little of the flower of copper, not altogether levigated. Another powder equally corrosive:—Having sponged the ulcer, burn the most greasy wool upon a shell placed on the fire until the whole be consumed; having reduced this to a fine powder, and sprinkled it on the sore, apply the bandage in the same manner. Another powder for the same ulcers:—The black chamaeleon, when prepared with the juice of the fig. It is to be prepared roasted, and alkanet mixed with it. Or, pimpernel, and Egyptian alum roasted, and sprinkle on them the Orchomenian powder.

For spreading ulcers:—Alum, both the Egyptian roasted, and the Melian; but the part is to be first cleansed with roasted natron and sponged; and the species of alum called chalcitis roasted. It is to be roasted until it catch fire.

Ulc. 19

For old ulcers which occur on the fore part of the legs; they become bloody and black:—Having pounded the flower of the melilot and mixed it with honey, use as a plaster.

Ulc. 20

For nerves (*tendons?*) which have been cut asunder:—Having pounded, sifted, and mixed with oil the roots of the wild myrtle, bind on the part; and the herb cinquefoil (it is white and downy, and more raised above the ground than the black cinquefoil), having pounded this herb in oil bind it on the part, and then remove it on the third day.

Ulc. 21

Emollients (?):—These medicines are to be used in winter rather than in summer. Emollient medicines which make the cicatrices fair:—Pound the inner mucous part of the squill and pitch, with fresh swine's seam, and a little oil, and a little resin, and ceruse. And the grease of a goose, fresh swine's seam, and squill, and a little oil. The whitest wax, fresh clean grease, or squill and white oil, and a little resin. Wax, swine's seam (old and fresh), and oil, and verdigris, and squill and resin. Let there be two parts of the old grease to the fresh, and of the other things, q. s. Having melted the grease that is fresh, pour it into another pot; having levigated plumbago finely and sifted it, and mixed them together, boil and stir at first; boil until when poured upon the ground it concretes; then taking it off the fire, pour it all into another vessel, with the exception of the stony sediment, and add resin and stir, and mix a little oil of juniper, and what has been taken off. In all the emollient medicines to which you add the resin, when you remove the medicine from the fire, pour in and mix the resin while it is still warm. Another:—Old swine's seam, wax, and oil, the dried shavings of the lotus, frankincense, plumbago,—namely, of the frankincense one

part, and of the other one part, and of the shavings of the lotus one part; but let there be two parts of the old grease, one of wax, and of fresh swine's seam one part. Another:—Or old swine's seam along with the fresh grease of a goat; when cleaned, let it retain as little as possible of its membrane: having triturated or pounded it smooth, pour in oil, and sprinkle the lead with the spodium and half the shavings of the lotus. Another:—Swine's seam, spodium, blue chalcitis, oil.

Ulc. 22

For Burns:—You must boil the tender roots of the ilex, and if their bark be very thick and green, it must be cut into small parts, and having poured in white wine, boil upon a gentle fire, until it appear to you to be of the proper consistence, so as to be used for a liniment. And it may be prepared in water after the same manner. Another, not corrosive:—Old swine's seam is to be rubbed in by itself, and it is to be melted along with squill, the root of which is to be divided and applied with a bandage. Next day it is to be fomented; and having melted old swine's seam and wax, and mixed with them oil, frankincense, and the shavings of lotus and vermilion, this is to be used as a liniment. Having boiled the leaves of the wakerobin in wine and oil, apply a bandage. Another:—When you have smeared the parts with old swine's seam let the roots of asphodel be pounded in wine and triturated, and rubbed in. Another:—Having melted old swine's seam, and mixed with resin and bitumen, and having spread it on a piece of cloth and warmed it at the fire, apply a bandage.

Ulc. 23

When an ulcer has formed on the back from stripes or otherwise, let squill, twice boiled, be pounded and spread upon a linen cloth and bound on the place. Afterward the grease of a goat, and fresh swine's seam, spodium, oil, and frankincense are to be rubbed in.

Ulc. 24

Swellings which arise on the feet, either spontaneously or otherwise, when neither the swellings nor the inflammation subside under the

use of cataplasms, and although sponges or wool, or anything else be bound upon the sound part; but the swelling and inflammation return of themselves again, an influx of blood into the veins is the cause, when not occasioned by a bruise. And the same story applies if this happen in any other part of the body. But blood is to be abstracted, especially the from the veins, which are the seat of the influx, if they be conspicuous; but if not, deeper and more numerous scarifications are to be made in the swellings; and whatever part you scarify, this is to be done with the sharpest and most slender instruments of iron. When you have removed the blood, you must not press hard upon the part with the specillum, lest you produce contusion. Bathe with vinegar, and do not allow a clot of blood to remain between the lips of the wounds, and having spread greasy wool with a medicine for bloody wounds, and having carded the woof and made it soft, bind it on, having wetted it with wine and oil. And let the scarified part be so placed that the determination of the blood may be upward and not downward; and do not wet the part at all, and let the patient be put upon a restricted diet and drink water. If upon loosing the bandages you find the scarifications inflamed, apply a cataplasm of the fruit of the chaste-tree and linseed. But if the scarifications become ulcerated and break into one another, we must be regulated by circumstances, and otherwise apply whatever else appears to be proper.

Ulc. 25

When a varix is on the fore part of the leg, and is very superficial, or below the flesh, and the leg is black, and seems to stand in need of having the blood evacuated from it, such swellings are not, by any means, to be cut open; for, generally, large ulcers are the consequence of the incisions, owing to the influx from the varix. But the varix itself is to be punctured in many places, as circumstances may indicate.

Ulc. 26

When you have opened a vein and abstracted blood, and although the fillet be loosed the bleeding does not stop, the member, whether the arm or leg, is to be put into the reverse position to that from which

the blood flows; so that the blood may flow backward, and it is to be allowed to remain in this position for a greater or less space of time. Then bind up the part while matters are so, no clots of blood being allowed to remain in the opening. Then having applied a double compress, and wetted it with wine, apply above it clean wool which has been smeared with oil. For, although the flow of blood be violent, it will be stopped in this way. If a thrombus be formed in the opening, it will inflame and suppurate. Venesection is to be practiced when the person has dined more or less freely and drunk, and when somewhat heated, and rather in hot weather than in cold.

Ulc. 27

When in cupping, the blood continues to flow after the cupping-instrument has been removed, and if the flow of blood, or serum be copious, the instrument is to be applied again before the part is healed up, so as to abstract what is left behind. Otherwise coagula of blood will be retained in the incisions and inflammatory ulcers will arise from them. In all such cases the parts are to be bathed with vinegar, after which they are not to be wetted; neither must the person lie upon the scarifications, but they are to be anointed with some of the medicines for bloody wounds. When the cupping instrument is to be applied below the knee, or at the knee, it should be done, if possible, while the man stands erect.

On Hemorrhoids

Hem. 1

The disease of the hemorrhoids is formed in this way: if bile or phlegm be determined to the veins in the rectum, it heats the blood in the veins; and these veins becoming heated attract blood from the nearest veins, and being gorged the inside of the gut swells outwardly, and the heads of the veins are raised up, and being at the same time bruised by the faeces passing out, and injured by the blood collected in them,

they squirt out blood, most frequently along with the faeces, but sometimes without faeces.

Hem. 2

It is to be cured thus:

In the first place it should be known in what sort of a place they are formed. For cutting, excising, sewing, binding, applying putrefacient means to the anus,—all these appear to be very formidable things, and yet, after all, they are not attended with mischief. I recommend seven or eight small pieces of iron to be prepared, a fathom in size, in thickness like a thick specillum, and bent at the extremity, and a broad piece should be on the extremity, like a small obolus¹⁴. Having on the preceding day first purged the man with medicine, on the day of the operation apply the cautery. Having laid him on his back, and placed a pillow below the breech, force out the anus as much as possible with the fingers, and make the irons red-hot, and burn the pile until it be dried up, and so as that no part may be left behind. And burn so as to leave none of the hemorrhoids unburnt, for you should burn them all up. You will recognize the hemorrhoids without difficulty, for they project on the inside of the gut like dark-colored grapes, and when the anus is forced out they spurt blood. When the cautery is applied the patient's head and hands should be held so that he may not stir, but he himself should cry out, for this will make the rectum project the more. When you have performed the burning, boil lentils and tares, finely triturated in water, and apply as a cataplasm for five or six days. But on the seventh, cut a soft sponge into a very slender slice, its width should be about six inches square. Then a thin smooth piece of cloth, of the same size as the sponge, is to be smeared with honey and applied; and with the index finger of the left hand the middle of the sponge is to be pushed as far up as possible; and afterward wool is to be placed upon the sponge so that it may remain in the anus. And having girded the patient about the loins and fastened a shawl to the girdle, bring up this band from behind between the legs and attach it to the girdle at the navel. Then let the medicine which I formerly said

is calculated to render the skin thick and strong, be bound on. These things should be kept on for not less than twenty days. The patient should once a day take a draught from flour or millet, or bran, and drink water. When the patient goes to stool the part should be washed with hot water. Every third day he should take the bath.

Hem. 3

Another method of cure:—Having got the anus to protrude as much as possible, foment with hot water, and then cut off the extremities of the hemorrhoids. But this medicine should be prepared beforehand, as an application to the wound:—Having put urine into a bronze vessel, sprinkle upon the urine the flower of bronze calcined and finely trituated; then, when it is moistened, shake the vessel and dry in the sun. When it becomes dry, let it be scraped down and levigated, and apply with the finger to the part, and having oiled compresses, apply them, and bind a sponge above.

Hem. 4

Another method:—There grows upon the bleeding condyloma, a protuberance like the fruit of the mulberry, and if the condyloma be far without, an envelope of flesh is adherent to it. Having placed the man over two round stones upon his knees, examine, for you will find the parts near the anus between the buttocks inflated, and blood proceeding from within. If, then, the condyloma below the cover be of a soft nature, bring it away with the finger, for there is no more difficulty in this than in skinning a sheep, to pass the finger between the hide and the flesh. And this should be accomplished without the patient's knowledge, while he is kept in conversation. When the condyloma is taken off, streaks of blood necessarily flow from the whole of the torn part. It must be speedily washed with a decoction of galls, in a dry wine, and the bleeding vein will disappear along with the condyloma, and its cover will be replaced. The older it is, the more easy the cure.

Hem. 5

But if the condyloma be higher up, you must examine it with the speculum, and you should take care not to be deceived by the speculum; for when expanded, it renders the condyloma level with the surrounding parts, but when contracted, it shows the tumor right again. It is to be removed by smearing it with black hellebore on the finger. Then, on the third day, wash it out with a dry wine. You need not be surprised that there is no discharge of blood when you remove the condyloma, for neither, if you cut off the hands or legs at the articulations will there be any flow of blood; but if you cut them off above or below the joints, you will find there hollow veins which pour out blood, and you will have difficulty in stopping the bleeding. In the same manner, the bleeding vein in the anus, if you cut it above or below the point of separation of the condyloma, will pour forth blood; but if you take away the condyloma at its junction (*with the natural parts?*) there will be no flow of blood. If matters then be thus put to rights, it will be well; but otherwise burn it, taking care not to touch the place with the iron, but bringing it close so as to dry it up, and apply the flos aeris in the urine.

Hem. 6

Another method of curing hemorrhoids:—You must prepare a cautery like the *arundo phragmites*, and an iron that exactly fits is to be adapted to it; then the tube being introduced into the anus, the iron, red hot, is to be passed down it, and frequently drawn out, so that the part may bear the more heat, and no sore may result from the heating, and the dried veins may heal up.

Hem. 7

But if you are neither disposed to burn nor excise, having first fomented with plenty hot water and turned out the anus, leavigate myrrh, and having burnt galls and Egyptian alum, in the proportion of one and a half to the other things, and as much of melanteria; these things are all to be used in a dry state. The hemorrhoid will separate under the use of these medicines, like a piece of burnt hide. You are to

proceed thus until the whole are removed, and a half part of burnt chalcitis does the same thing.

Hem. 8

But if you wish to effect the cure by suppositories, take the shell of the part fish a third part of plumbago, bitumen, alum, a little of the flos aeris, galls, a little verdigris; having poured a small quantity of boiled honey on these, and formed an oblong suppository, apply until you remove them.

Hem. 9

An hemorrhoid in a woman may be thus cured. Having fomented with plenty of hot water, boil in the water certain of the fragrant medicines, add pounded tamarisk, roasted litharge and galls, and pour on them white wine, and oil, and the grease of a goose, pounding all together. Give to use after fomenting. In fomenting the anus is to be made to protrude as much as possible.

On Fistulae

Fist. 1

Fistulae are produced by contusions and tubercles, and they are also occasioned by rowing, on horseback, when blood accumulates in the nates near the anus. For, having become putrid, it spreads to the soft parts (the breech being of a humid nature, and the flesh in which it spreads being soft), until the tubercle break and corrupt below at the anus. When this happens, a fistula is formed, having an ichorous discharge, and faeces pass by it, with flatus and much and abomination. It is produced, then, by contusions when any of the parts about the anus are bruised by a blow, or a fall, or a wound, or by riding, or rowing, or any such cause. For blood is collected, and it, becoming corrupted, suppurates; and the from the the same accidents happen, as have been described in the case of tubercles.

Fist. 2

In the first place, then, when you see any such tubercle formed, you must cut it open while still unripe, before it suppurate and burst into the rectum.

Fist. 3

But if a fistula be already formed when you undertake the case, take a stalk of fresh garlic, and having laid the man on his back, and separated his thighs on both sides, push down the stalk as far as it will go, and thereby measure the depth of the fistula. Then, having bruised the root of seseli to a very fine powder, and poured in some water, let it macerate for four days, and, mixing the water with honey, let the patient drink it, fasting, to the amount of three cyathi, and at the same time purge away the ascarides. Those who are left without treatment die.

In the next place, having moistened the strip of cotton cloth, with the juice of the great tithymallus, and sprinkling on it the flos aeris, roasted and triturated, and having made it into a tent equal in length to the fistula, and having passed a thread through the ends of the tent again through the stalk, and having placed the patient in a reclining position, and having examined the ulcerated parts of the rectum with a speculum, pass the stalk by it, and when it reaches the rectum, take hold of it and draw it out until the tent be pushed through, and be brought on a level above and below. When it (*the tent?*) has been pushed inward, introduce a ball of horn into the rectum (the rectum having been previously smeared with Cimolian chalk), and leave it there, and when the patient wants to go to stool, let it be taken out and again replaced, and let this practice be continued for five days. On the sixth day let it be removed, and drawing the tent out of the flesh, and afterwards pounding alum and filling the ball (*pessary*) and introducing it into the rectum, leave it until the alum melts. Anoint the rectum with myrrh until the parts appear to be united.

Fist. 4

Another method of cure:—Taking a very slender thread of raw lint, and uniting it into five folds of the length of a span, and wrapping

them round with a horse hair; then having made a director (specillum) of tin, with an eye at its extremity, and having passed through it the end of raw lint wrapped round as above described, introduce the director into the fistula, and, at the same time, introduce the index finger of the left hand *per anum*; and when the director touches the finger, bring it out with the finger, bending the extremity of the director and the end of the threads in it, and the director is to be withdrawn, but the ends of the threads are to be knotted twice or thrice, and the rest of the raw threads is to be twisted around and fastened into a knot. Then the patient is to be told that he may go and attend to his matters. The rest of the treatment:—Whenever any part of the thread gets loose owing to the fistula becoming putrid, it is to be tightened and twisted every day; and should the raw thread rot before the fistula is eaten through, you must attach another piece of raw thread to the hair, pass it through, and tie it, for it was for this purpose that the hair was rolled round the raw lint, as it is not liable to rot. When the fistula has sloughed through, a soft sponge is to be cut into very slender pieces and applied, and then the flowers of copper, roasted, are to be frequently applied with a director; and the sponge smeared with honey is to be introduced with the index finger of the left hand, and pushed forward; and another bit of added, it is to be bound on in the same manner as in the operation for hemorrhoids. Next day, having loosed the bandages, the fistula is to be washed with hot water, and cleansed, as far as possible, with the finger of the left hand by means of the sponge, and again the flos aeris is to be applied. This is to be done for seven days, for generally the coat of the fistula takes that time to fistula takes that time to slough through. The same mode of bandaging is to be persevered in afterwards, until the cure be completed. For in this way, the fistula being forcibly expanded by the sponge will not fill up and heal unequally, but it will all become whole together. During the treatment, the part should be bathed with plenty of warm water, and the patient kept on a spare diet.

Fist. 5

When the fistula does not get eaten through, having first examined it with a sound, cut down as far as it passes, and sprinkle with the flos aeris, and let it remain for five days. Then pour warm water upon it, and above lay flour mixed with water, and bind on it the leaves of beet. When the flos aeris comes away, and the fistulous sore becomes clean, cure it as before described.

Fist. 6

But if the fistula be in a part which does not admit of this treatment, and if it be deep, syringe it with the flowers of copper, and myrrh, and natron, diluted with urine, and introduce a piece of lead into the orifice of the fistula so that it may not close. Syringe the fistula by means of a quill attached to a bladder, so that the injection may distend the fistula. But it does not heal unless it be cut open.

Fist. 7

If the anus gets inflamed, and there is pain, fever, a frequent desire of going to stool without passing anything, and the anus appears to protrude, owing to the inflammation, and if at times strangury come on, this disease is formed, when phlegm, collected from the whole body, is determined to the rectum. Warm things are beneficial in this case; for these, when applied, can attenuate and dissolve the phlegm, and dilute the acrid and salt particles, so that the heat subsides, and the irritation in the rectum is removed. Wherefore it is to be treated thus: The patient is to be put into a hip-bath of hot water, and sixty grains of the grana gnidia are to be pounded and infused in a hemina of wine, with half a hemina of oil, and injected. This brings away phlegm and faeces. When the patient does not take the hip-bath, boil eggs in dark-colored fragrant wine, and apply to the anus, and spread to the anus, and spread something warm below, either a bladder filled with warm water, or linseed toasted and ground, and its meal stirred up and mixed equally with dark, fragrant wine, and oil, and this applied very warm as a cataplasm; or, having mixed barley and Egyptian alum pulverized, form into an oblong ball (*suppository?*) and

warming it gently at the fire, make it into a cataplasm, foment, form it into shape with the fingers, and then making it quite tepid, introduce it into the anus. The external parts are to be anointed with cerate, and a cataplasm of boiled garlic, with dark wine diluted, is to be applied. But if you remove these things, let him take the hip-bath of hot water, and having mixed together the juice of strychnos, the grease of a goose, swine's seam, chrysocolla, resin, and white wax, and then having melted in the same and mixed together, anoint with these things, and while the inflammation lasts, use the cataplasm of boiled garlic. And if by these means he be freed from the pain, it is enough; but if not, give him the white meconium (*Euphorbia peplus*?), or, if not it, any other phlegmagogue medicine. While the inflammation lasts, the diet should be light.

Fist. 8

The strangury comes on in this way:—The bladder being heated from the rectum, phlegm is attracted by the heat, and by the phlegm (*inflammation*?) the strangury is occasioned. If, then, as is frequently the case, it cease with the disease, well; but, notwithstanding, if not, give any of the medicines for strangury.

Fist. 9

If procidentia ani take place, having fomented the part with a soft sponge, and anointed it with a snail, bind the man's hands together, and suspend him for a short time, and the gut will return. But if it still prolapse, and will not remain up, fasten a girdle round his loins and attach a shawl behind, and having pushed up the anus, apply to it a soft sponge, moistened with hot water in which the shavings of lotus have been boiled; pour of this decoction upon the anus by squeezing the sponge, then, bringing the shawl below between the legs, fasten it at the navel. But if he wish to evacuate the bowels, let him do so upon a very narrow night-stool. Or, if the patient be a child, let him be placed on the feet of a woman, with his back reclined to her knees, and when the bowels are evacuated, let the legs be extended. In this way the anus will be the least disposed to fall out. When a watery and

ichorous discharge flows from the rectum, wash it out with burnt lees of wine, and water from myrtle, and having dried maiden-hair, pound and sift it, and apply as a cataplasm. But if there be a discharge of blood, having washed with the same, and pounded chalcitis, and the shavings of cypress, or of juniper, or of stone-pine, or of turpentine, the in equal proportions the apply as a cataplasm. Anoint the external parts with thick cerate.

When the gut protrudes and will not remain in its place, scrape the finest and most compact silphium (*assafoetida*?) into small pieces and apply as a cataplasm, and apply a sternutatory medicine to the nose and provoke sneezing, and having moistened pomegranate rind with hot water, and having powdered alum in white wine, pour it on the gut, then apply rags, bind the thighs together for three days, and let the patient fast, only he may drink sweet wine. If even thus matters do not proceed properly, having mixed vermillion with honey, anoint.

If procidentia ani be attended with a discharge of blood, pare off the rind of the root of wakerobin, then pound and mix flour with it, and apply it warm as a cataplasm. Another: Having scraped off the rind of the most tender roots of the wild vine, which some call *psilothrion*, boil in a dark austere wine undiluted; then having pounded, apply as a tepid cataplasm; but mix also flour and stir it up with white wine and oil in a tepid state. Another:—Having pounded the seed of hemlock, pour on it a fragrant white wine, and then apply in a tepid state as a cataplasm.

But if it be inflamed, having boiled in water the root of the ivy, finely powdered, and mixing the finest flour, and stirring it up with white wine, apply as a cataplasm, and mix up some fat with these things. Another:—Take the root of the mandrake, especially the green (fresh) root, but otherwise the dried, and having cleaned the green root and cut it down, boil in diluted wine, and apply as a cataplasm; but the dry may be pounded and applied as a cataplasm in the manner. Another:—

Having bruised the inner part of a ripe cucumber to a soft state, apply as a cataplasm.

Fist. 10

If there be pain without inflammation, having roasted red natron, and pounded it to a fine powder, and added alum and roasted salts, finely triturated, mix together in equal proportions; then having mixed it up with the best pitch and spread upon a rag, apply, and bind. Another:— Having pounded the green leaves of capers, put into a bag and bind on the part; and when it appears to burn, take it away and apply it afterward; or, if you have not the leaves of capers, pound the rind of its roots, and having mixed it up with dark-colored wine, bind on the part in the same manner. This is a good application also for pains of the spleen. Of these poultices, those which are cooling, stop the discharge; those which are emollient and heating, discuss; and those which are attractive, dry up and attenuate. This disease is formed when bile and phlegm become seated in the parts. When the anus is inflamed, it should be anointed with the ointment, the ingredients of which are resin, oil, wax, plumbago, and suet, these being all melted and applied quite hot as a cataplasm.

Notes

¹ The practice advocated in this paragraph is alluded to by Paulus Ægineta, in his chapter on Fractures of the Skull. (vi., 90.)

² The operation here described by our author is the more deserving of attention, as it appears to have been peculiar to him. It is not described by Celsus, Paulus Ægineta, Albucasis, nor any one of the ancient authorities, as far as I can find; neither am I aware of its having been attempted in modern times. The object of it, however, seems to be very rational, namely, to avoid doing serious injury to the dura mater by tearing the bone forcibly from it at once.

³ The meaning of the first clause of this sentence, according to Galen, is, that the first thing which the medical practitioner must do is to make himself well acquainted with semeiology, by comparing carefully the condition of disease with

that of health. In all cases of accident, it was the practice of the ancient surgeons to compare carefully the injured part with its fellow or corresponding part on the opposite side.

⁴ By bloody veins is meant veins of a large size, as Galen explains. Contusions of such necessarily produce extravasation and hemorrhage, and the other bad consequences described by our author.

⁵ That gangrene should have often supervened in such a case, as described by our author, need not appear surprising. It shows that Hippocrates had a wonderful talent for original observation when he was able to detect and describe such a case; and it ought to teach our profession a lesson of humility, in comparing our present state of knowledge with that of our forefathers, when we thus find that the "old man of Cos," twenty-two centuries ago, understood the nature of this accident better than many of us did not many years since.

⁶ It is certain that there is no such line in the works of Homer as they have come down to us, and it is singular that Galen takes no notice of it, so that it is impossible to explain how our author came to use it.

⁷ See the Plate at the end of the volume.

⁸ The brief description of the bones given in this paragraph is evidently condensed from a larger work on the subject. A considerable portion of the matter which is found here in an abridged form, is taken from the preceding treatises, *On Fractures* and *On the Articulations*, but not the whole of it.

⁹ The tendon of the biceps.

¹⁰ It will be here perceived that by epiphysis is merely meant a close union of the two bones by means of a ligament. The term in this paragraph is not always used in this sense. Strictly speaking, its signification would appear to be a protuberance of a bone. It is applied to the malleoli, to the head of the tibia, to the head and neck of the femur, to the spinous processes of the vertebrae, to the upper and lower extremities of the humerus and to the lower extremity of the radius.

¹¹ Allusion is evidently made to the ligamentum teres.

¹² It will readily be perceived that the term ischium is not used here exactly as applied by modern anatomists. It is applied in this place to the ilium where it is articulated with the os sacrum. By the great vertebra, as stated in the preceding work, is meant the last vertebra of the loins.

¹³ Meaning before, that is to say, at their anterior part.

¹⁴ I would direct the attention of my surgical readers to the form of the ancient cautery or burning iron; it resembles a small coin, that is to say, it was a disk. I have often thought that modern practitioners in surgery erred in making their cauteries globular, instead of making them flat disks like the ancient.